

Sourccey

Assembly Manual

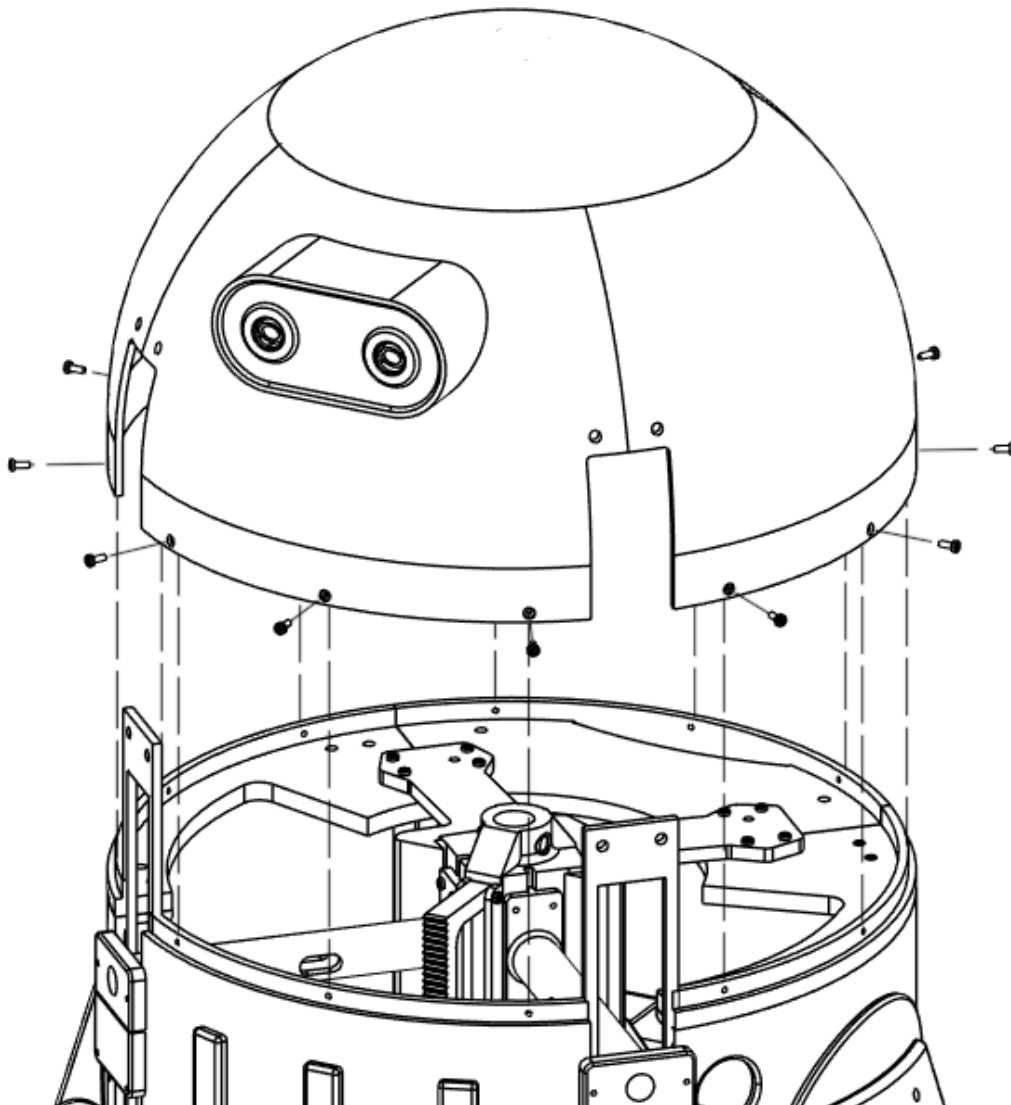
Vulcan Robotics

Revision

Draft 0.1

Date

April 15th, 2026



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Stage 0: Preparation & Body Sub Assemblies

Door Assemblies

Materials

Type	Name	Count
Part	1st Level Left Wall	1
Part	1st Level Right Wall	1
Part	2nd & 3rd Level Rear Wall	2
Part	Door	3
Part	Door Mirrored	1
Part	Hinge Pin	4
Part	Door Handle	4
Fastener	M3x10mm	16

Procedure

1. Fasten Door Handle to Door as shown in Figure 0.1
 - a. (4) M3x10mm
 - b. Repeat for (3) other Doors
2. Assemble hinge mechanism as shown in Figure 0.2
 - a. Repeat for (3) other Wall Assemblies
 - b. **NOTE:** One of the doors is a mirror image of the other three. The mirrored door goes with the 1st Level Right Wall Assembly

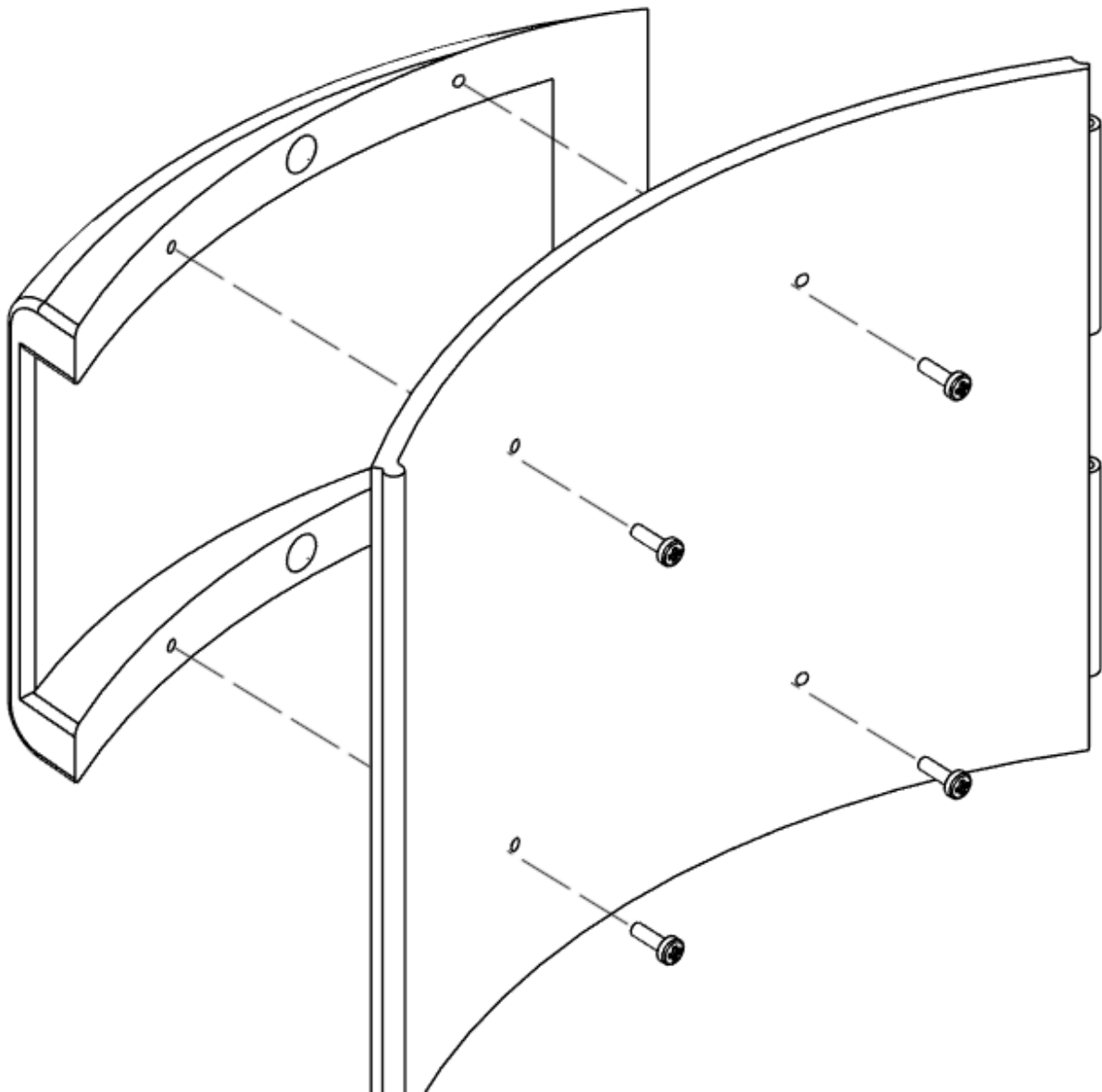


Figure 0.1 - Door Handle Assembly

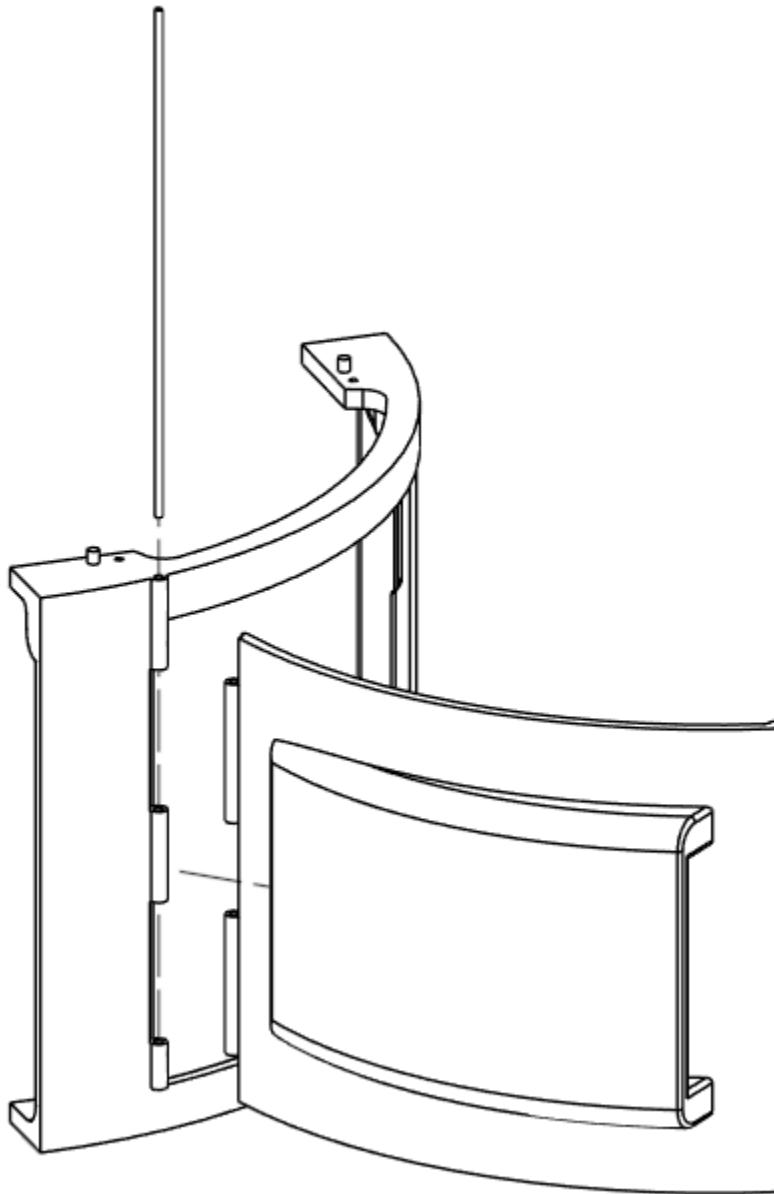


Figure 0.2 - Door Handle Assembly

Potentiometer Assembly

Materials

Type	Name	Count
Electronics	Potentiometer	1
Electronics	SVG Wire 600mm	1
Part	Potentiometer Holder	1
Part	Potentiometer Gear	1
Fastener	Potentiometer Nut	1

Procedure

1. Soldier Potentiometer
 - a. Cut plug terminal off one end of SVG wire, removing minimal wire length
 - b. Soldier SVG Wire ends to Potentiometer terminals as shown in Figure 0.3
2. Fasten Potentiometer to Potentiometer Holder as shown in Figure 0.4
3. Assemble Potentiometer Gear on Potentiometer shaft as shown in Figure 0.4
 - a. **NOTE:** This is a very tight fit, using a clamp or vise is recommended

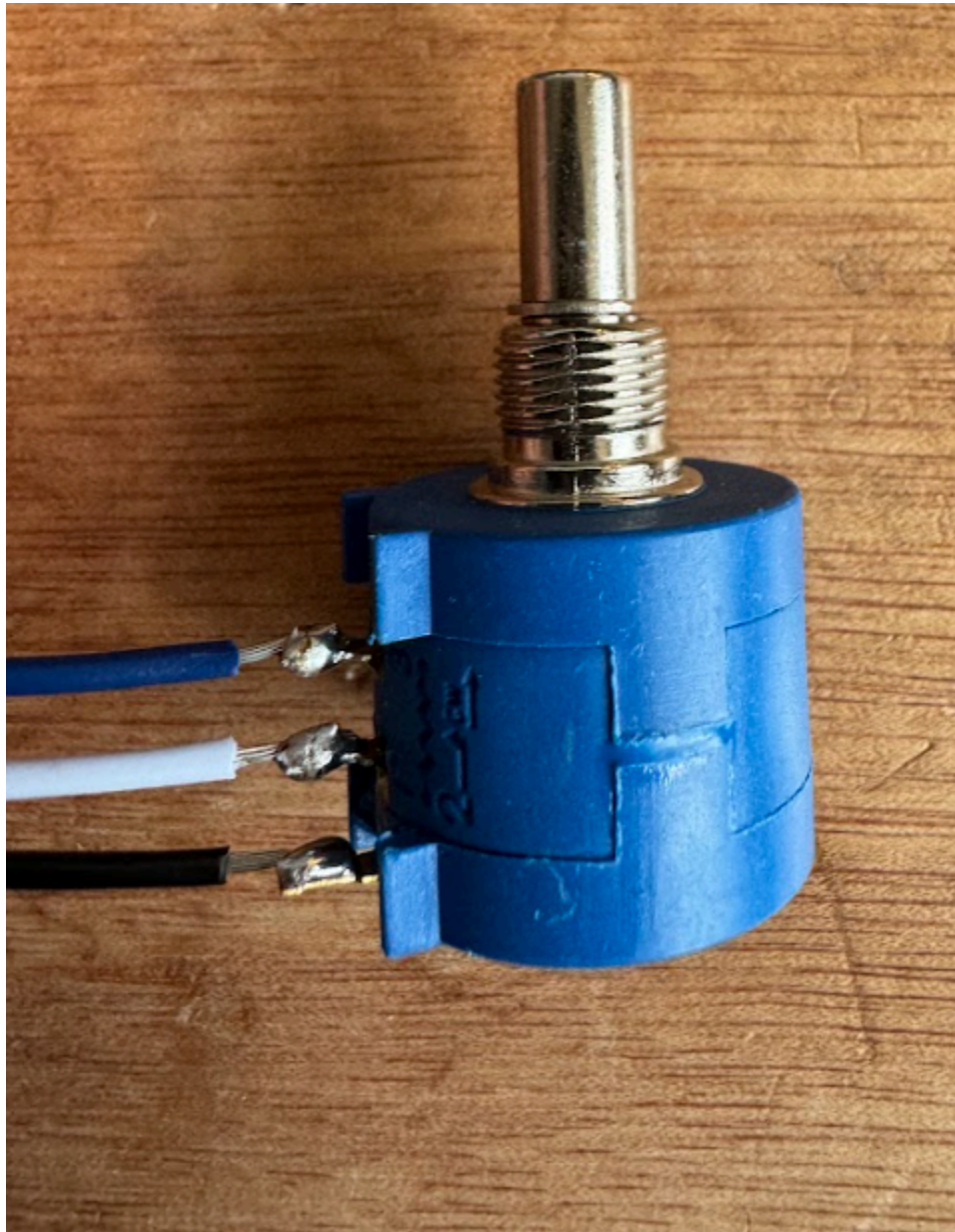


Figure 0.3 - Potentiometer Wire Soldering

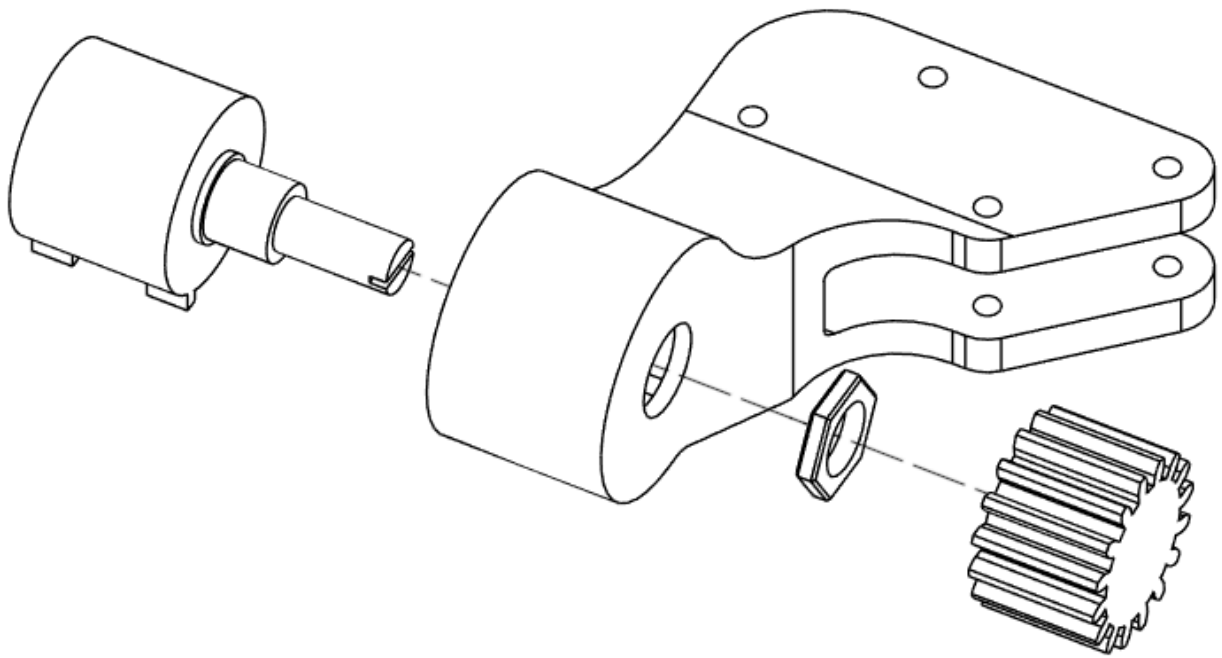


Figure 0.4 - Potentiometer Assembly

Wheel Housing Assemblies

Materials

Type	Name	Count
Actuator	Wheel Motor	4
Electronics	18 gauge wire ~1.8m	
Electronics	Heat Shrink 20mm	8
Part	Mecanum Wheel Set	1
Part	Wheel Holder FL & RR	2
Part	Wheel Holder FR & RL	2
Fastener	M3x5mm	24
Fastener	Structural Methacrylate Adhesive Epoxy	NA

Procedure

SAFETY NOTE: Uncured epoxy releases fumes that can be harmful to your health, ensure you are wearing nitrile gloves, face mask rated for organic fumes, and are in a well ventilated area before using the epoxy

NOTE: The mecanum wheels look identical at first glance but they are not. Pay close attention to Figure XX to ensure the correct wheel goes with the correct Wheel Holder

NOTE: The set screw is the shorter of the 2 screws provided with Mecanum Wheel Kit. The center screw is the longer of the 2

NOTE: Ensure that mecanum wheel rollers are oriented correctly

1. Prepare Wheel Motors

- Solder ~450mm of 18 Gauge Wire to Wheel Motor terminals
- Apply Heat Shrink to each terminal connection
- Repeat for all (4) wheel motors

2. Assemble Wheel Motor Housings

- Fasten Wheel Motor into Wheel Motor Holder as shown in Figure 0.5
 - (6) M3x5mm
- Repeat for all (4) Wheel Motor Housings

3. Fasten Wheels to Wheel Motor Housings as shown in Figures 0.6 and 0.7

- Apply Epoxy to Wheel Motor Shaft
- Fasten Wheel Coupler to Wheel Motor Shaft using set screw provided with Mecanum Wheel Kit

- c. Apply Epoxy to outer surface of Wheel Coupler
 - d. Fasten Mecanum Wheel to Wheel Coupler using center screw provided with Mecanum Wheel Kit
 - e. Repeat for all (4) Wheel Housing Assemblies
4. Wait at least 1 hour for epoxy to cure before continuing

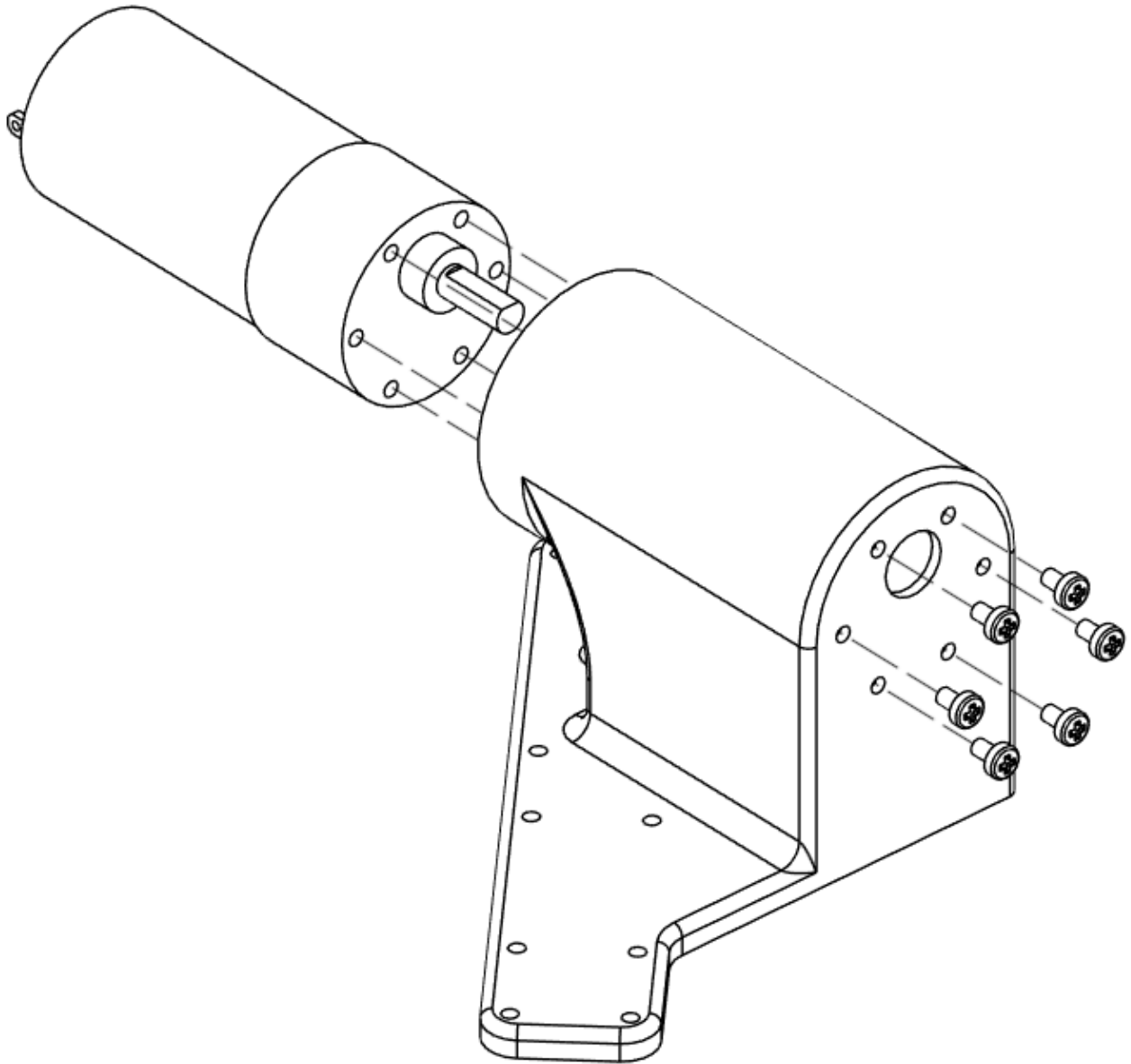


Figure 0.5 - Wheel Housing Assembly

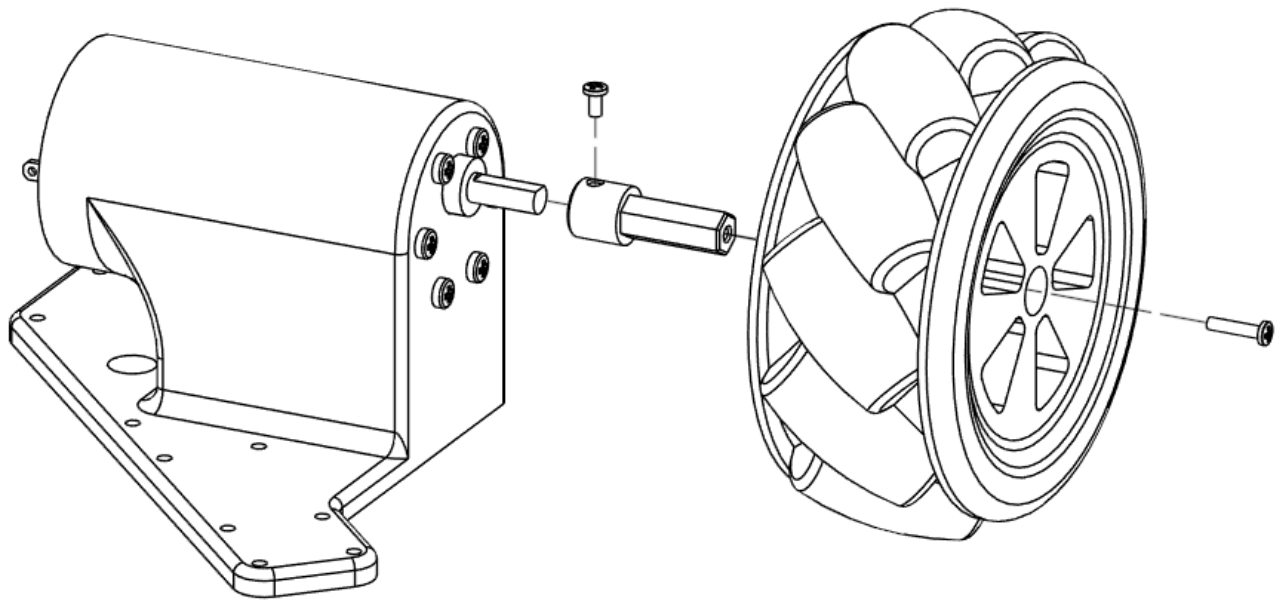


Figure 0.6 - Wheel & Coupler Assembly

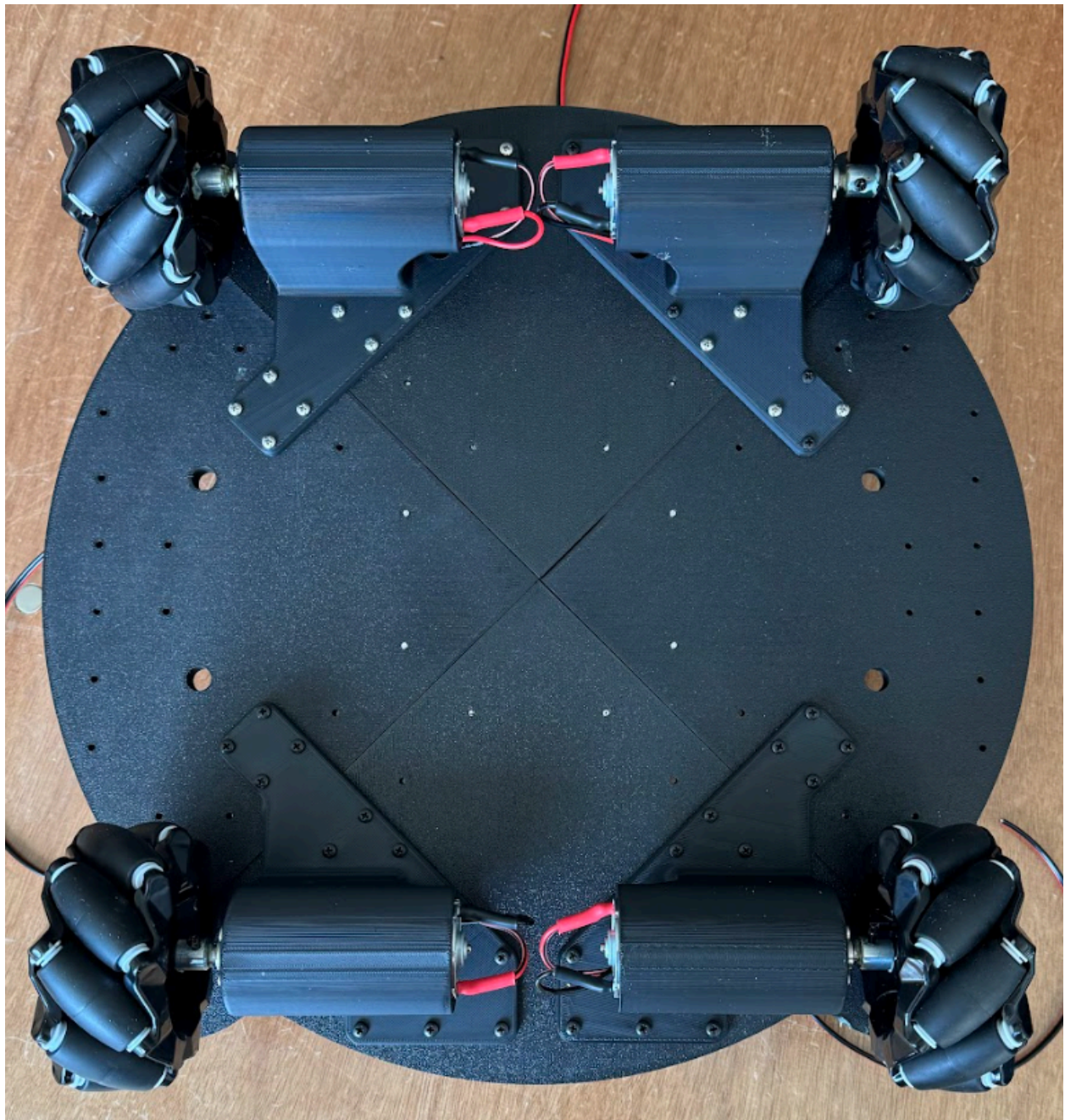


Figure 0.7 - Mecanum Wheel Roller Orientation

Raspberry Pi Assembly

Materials

Type	Name	Count
Electronics	Raspberry Pi 5	1
Electronics	Raspberry Pi USB HAT	1
Electronics	Raspberry Pi Power Board	1
Electronics	Raspberry Pi GPIO Ribbon Cable	1
Electronics	Raspberry Pi USB-C Power Cable	1
Part	Raspberry Pi Holder	1
Part	Raspberry Pi Holder Lid	1
Fastener	Raspberry Pi Connecting Screw	8
Fastener	Raspberry Pi Connecting Standoff: Male-Female	2
Fastener	Raspberry Pi Connecting Standoff: Female-Female	4
Fastener	M3x8mm	4

Procedure

1. Fasten standoffs to Raspberry Pi as shown in Figure 0.8
 - a. (4) Raspberry Pi Connecting Screw
 - b. (4) Raspberry Pi Connecting Standoff: Female-Female
2. Fasten USB HAT to standoffs as shown in Figure 0.8
 - a. (2) Raspberry Pi Connecting Screw
 - b. (2) Raspberry Pi Connecting Standoff: Male-Female
3. Connect GPIO Ribbon Cable to USB HAT
 - a. **NOTE:** Not pictured in Figure 0.8
4. Fasten Power Board to standoffs as shown in Figure 0.8
 - a. (2) Raspberry Pi Connecting Screw
5. Connect USB HAT to Power Board using USB-C Power Cable
 - a. **NOTE:** Not pictured in Figure 0.8
6. Insert Raspberry Pi Assembly into Raspberry Pi Holder and Fasten Lid as shown in Figure 0.9
 - a. (4) M3x8mm

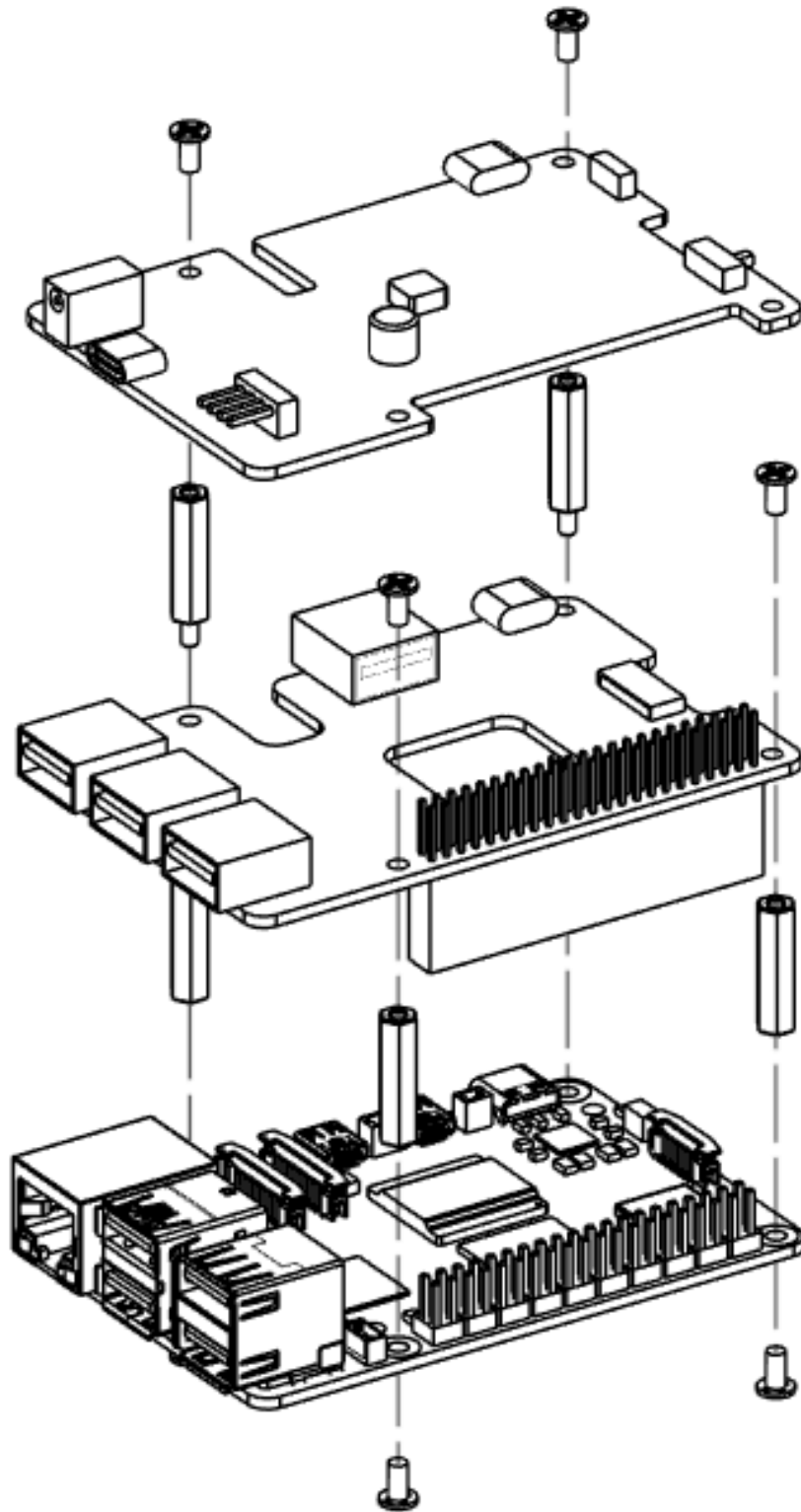


Figure 0.8 - Raspberry Pi Assembly

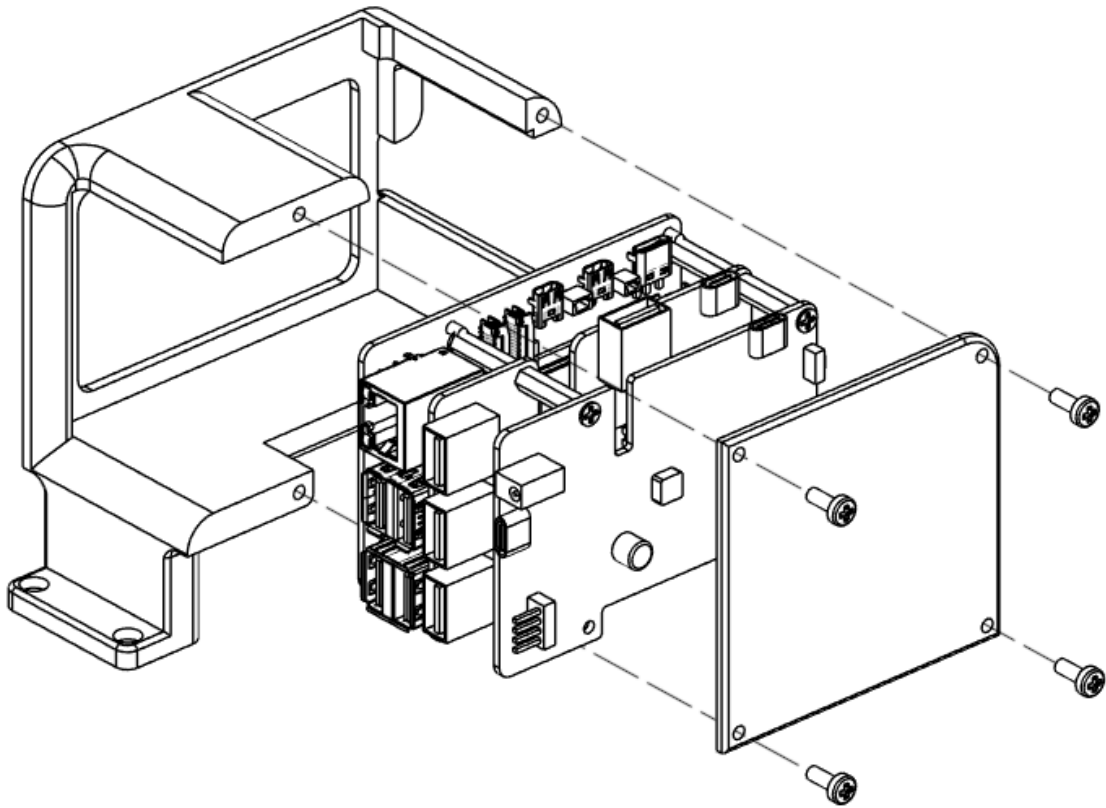


Figure 0.9 - Raspberry Pi Holder Assembly

Stage 1: Teleoperator Stand

Teleoperator Stand

Materials

Type	Name	Count
Electronics	Motor Driver Board	2
Electronics	Barrel Jack Power Cable - Male	2
Electronics	Barrel Jack Power Cable - Female	1
Electronics	Sourccey Teleop Power Cable	1
Electronics	Wire Splitter	1
Electronics	USB-C to USB Data Cable	2
Electronics	USB Splitter	1
Electronics	Arm Motor Cable - 400mm	2
Part	Teleop Stand Left	1
Part	Teleop Stand Right	1
Part	Teleop Arm Base Left	1
Part	Teleop Arm Base Right	1
Part	Teleop Stand Lid	1
Fastener	M2.5x8mm	8
Fastener	M3x8mm	2
Fastener	M3x20mm	12
Fastener	M3x30mm	6

Procedure

NOTE: Pay careful attention to the orientation of the parts and motors in this stage. Some parts may seem symmetrical at first, but they are not. The arm illustrated in this section's figures is the left arm

NOTE: Some parts are left/right specific. Ensure the correct part is used

1. Fasten Teleop Stand Right to Teleop Stand Left as shown in Figure 1.1
 - a. (2) M3x8mm
2. Fasten (2) Motor Driver Boards to Teleop Stand Left and Teleop Stand Right as shown in Figure 1.2
 - a. (8) M2.5x8mm
3. Connect cables to Motor Driver Boards as shown in Figure 1.2
 - a. (2) Barrel Jack Power Cable - Male (~60mm in length)
 - b. (1) Barrel Jack Power Cable - Female (~25mm in length)
 - c. (1) Sourcsey Teleop Power Cable
 - d. (1) Wire Splitter
 - e. (2) USB-C to USB Data Cable
4. Connect USB Splitter to the USB Data Cables from Step 3
5. Fasten Teleop Arm Base to Teleop Stand as shown in Figure 1.3
 - a. (6) M3x20mm
 - b. Repeat on other side of the stand
6. Connect (1) 400mm Arm Motor Cable to Motor Driver Board and thread through channels up to the top of Arm Base
 - a. Repeat on other side of the stand
7. Fasten Teleop Stand Lid to Teleop Stand as shown in Figure 1.4
 - a. (6) M3x30mm

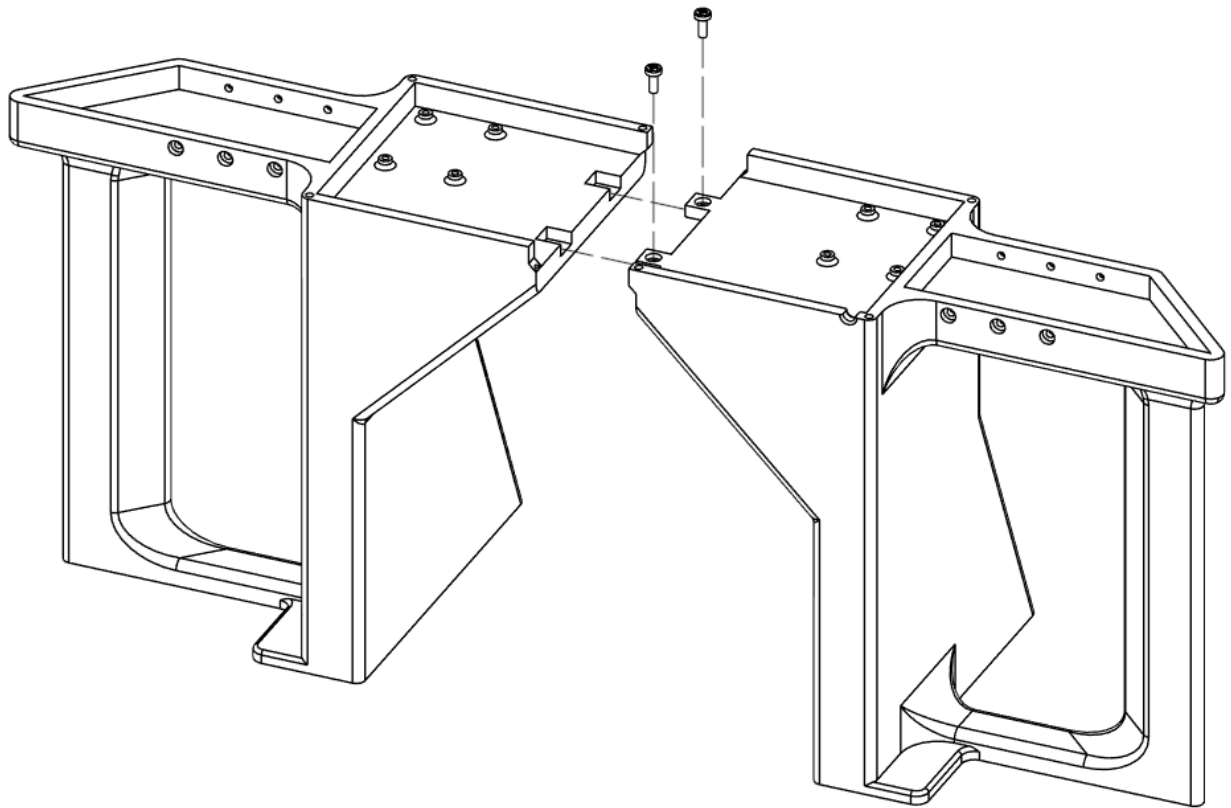


Figure 1.1 - Teleop Stand Assembly

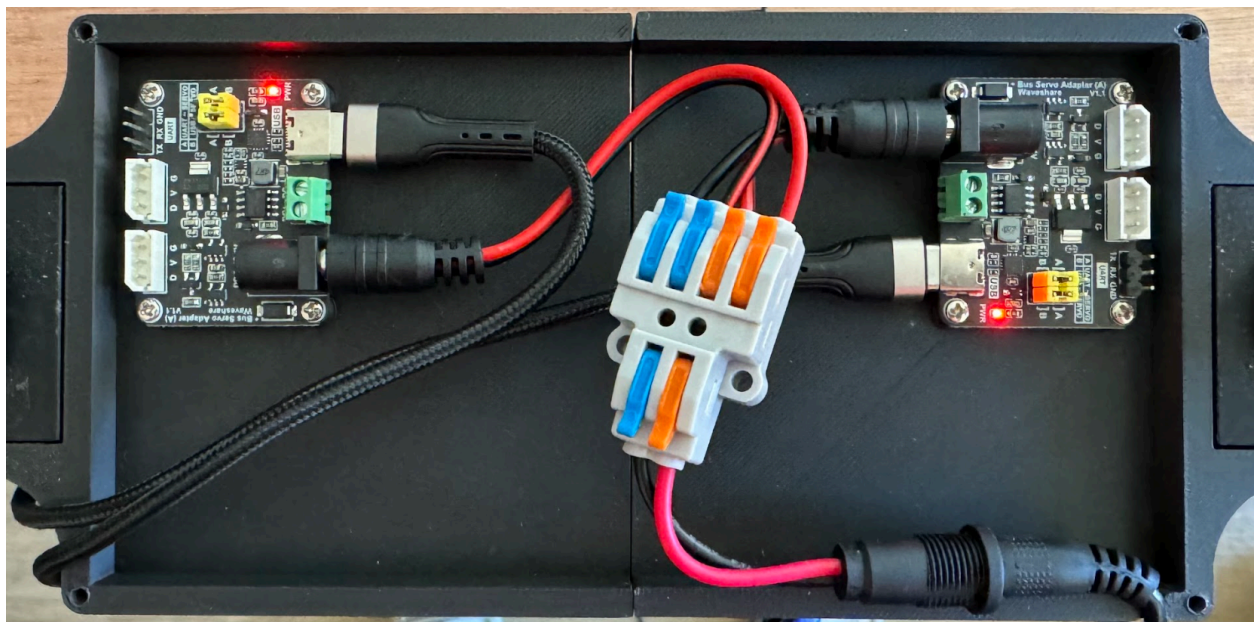


Figure 1.2 - Teleop Stand Electronics Assembly

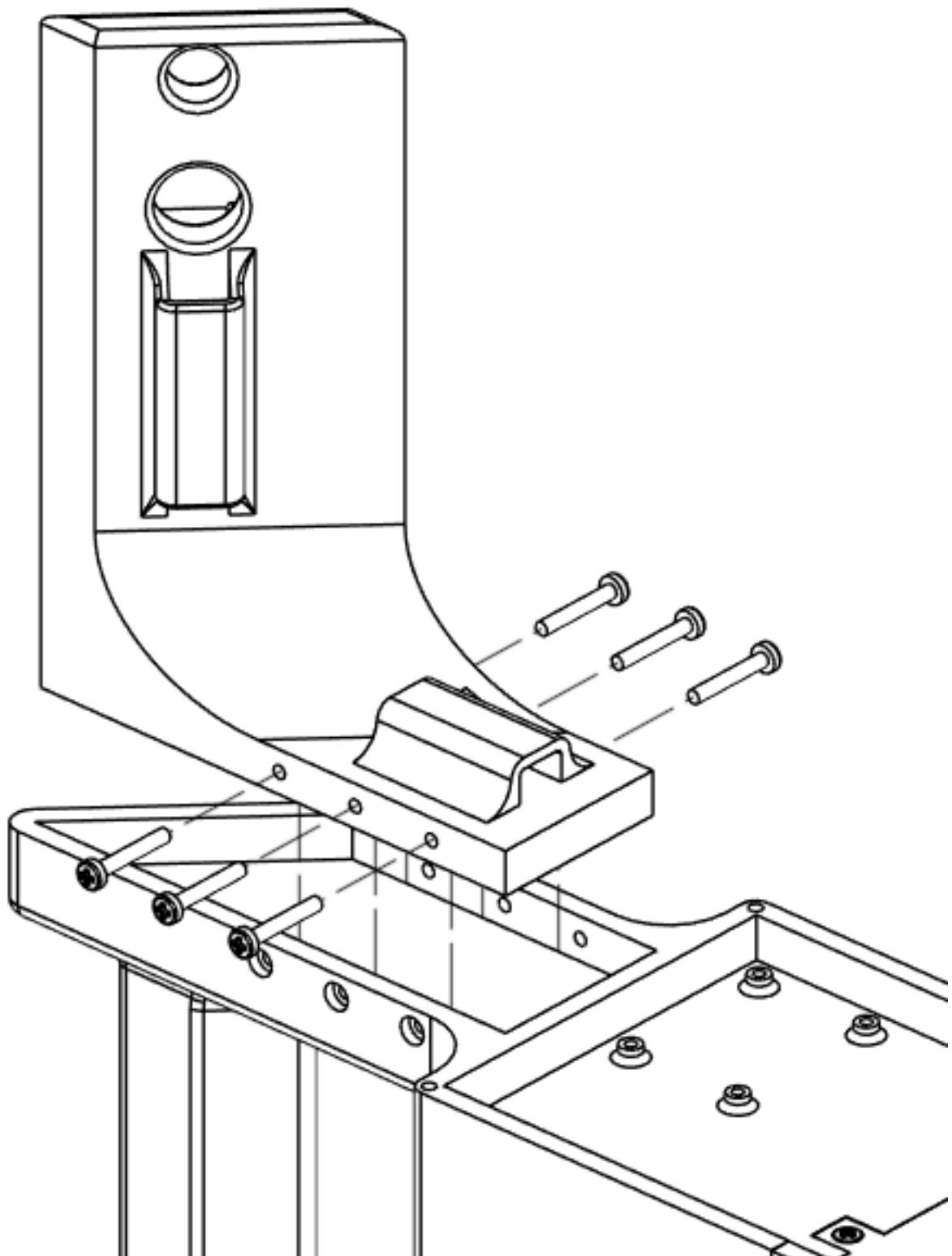


Figure 1.3 - Teleop Arm Base Assembly

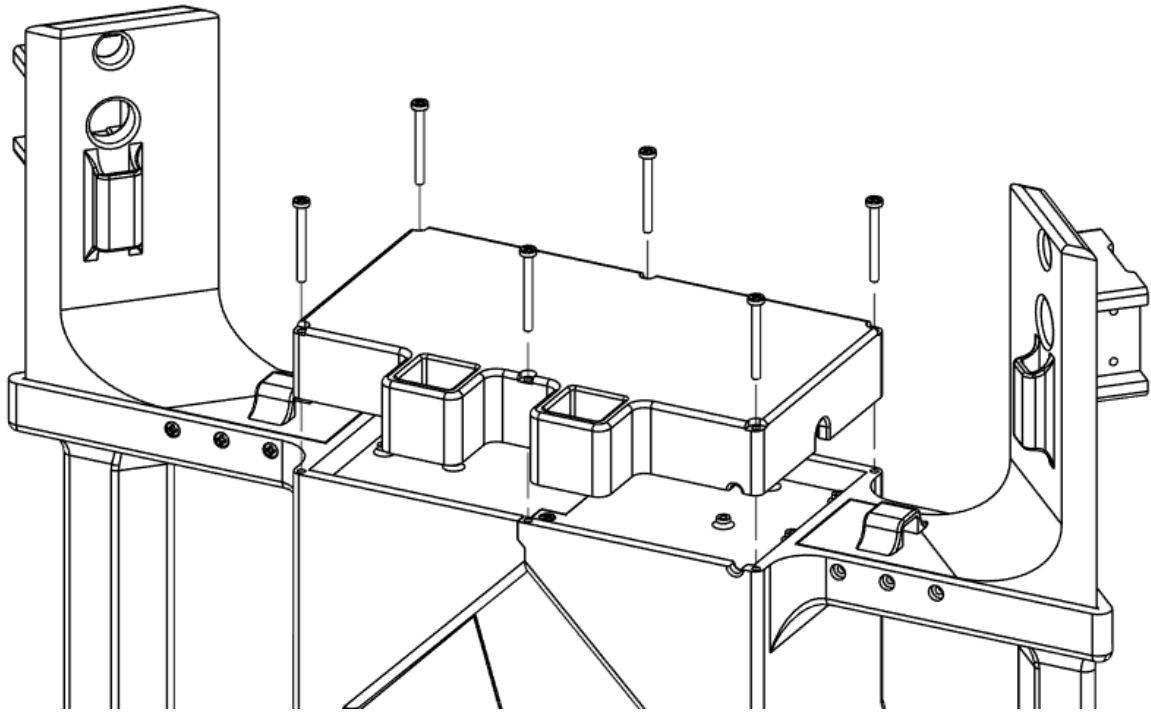


Figure 1.4 - Teleop Stand Lid Assembly

Stage 2: Motor ID & Software Preparation

Motor ID & Software Preparation

Materials

Type	Name	Count
Actuator	Feetech Servo Motor STS3215 - TELEOP	12
Actuator	Feetech Servo Motor STS3215	8
Actuator	Feetech Servo Motor STS3250	4
Electronics	SD Card	1

Procedure

NOTE: Motor models and IDs are very important. Take care to keep track of which motor is which after IDs have been assigned.

1. Power on the Teleop Stand by plugging in the Power Cord
2. Connect the USB data cable from Teleop Stand into your computer
3. Download FD Feetech Manager Software
 - a. <https://www.feetechrc.com/software.html>
 - b. FD 1.9.8.3 (or latest version)
4. Prepare the FD Software by defining the COM Settings as shown in Figure 2.1. Ensure there are 2 COM ports listed in the COM dropdown menu, one for each of the motor driver boards. Select either of these ports. Set the Baud Rate to 1000000. Open the COM port by selecting "Open" at the bottom of the COM settings.
 - a. **NOTE:** The COM port numbers may be different from the Figure
5. Plug in a Feetech Servo Motor to either of the Motor Arm Cables on the Teleop Stand
6. Select "Search" at the top of the Servo List as shown in Figure 2.2. A motor should appear as shown in the figure. If a motor does not appear, the selected COM port is incorrect. Change this in the COM Settings
7. Select the motor from the Servo List and select the "Programming" Tab as shown in Figure 2.3. Then select the ID in address 5 and change it to the desired ID in the bottom right corner as shown in Figure 2.3
8. Repeat this step for all Motors in the Robot and Teleoperator ensuring that each Motor is the correct model
 - a. Robot Motor 1: STS3215
 - b. Robot Motor 2-3: STS3250
 - c. Robot Motor 4-6: STS3215
 - d. Robot Motor 7: STS3215
 - e. Robot Motor 8-9: STS3250
 - f. Robot Motor 10-12: STS3215
 - g. Teleop Motor 1-12: STS3215 - TELEOP
9. Download and open the Raspberry Pi Imager Software
 - a. <https://www.raspberrypi.com/software/>
10. Download latest Vulcan Pi OS from Vulcan Robots website
11. For Device, select Raspberry Pi 5 and click "next"
12. For OS, select "Use Custom". Select the Vulcan Pi OS and click "next"
13. For Storage, select the SD card and click "next"
14. Select "Write" and wait for the flash to finish

15. Insert the SD Card into the Raspberry Pi

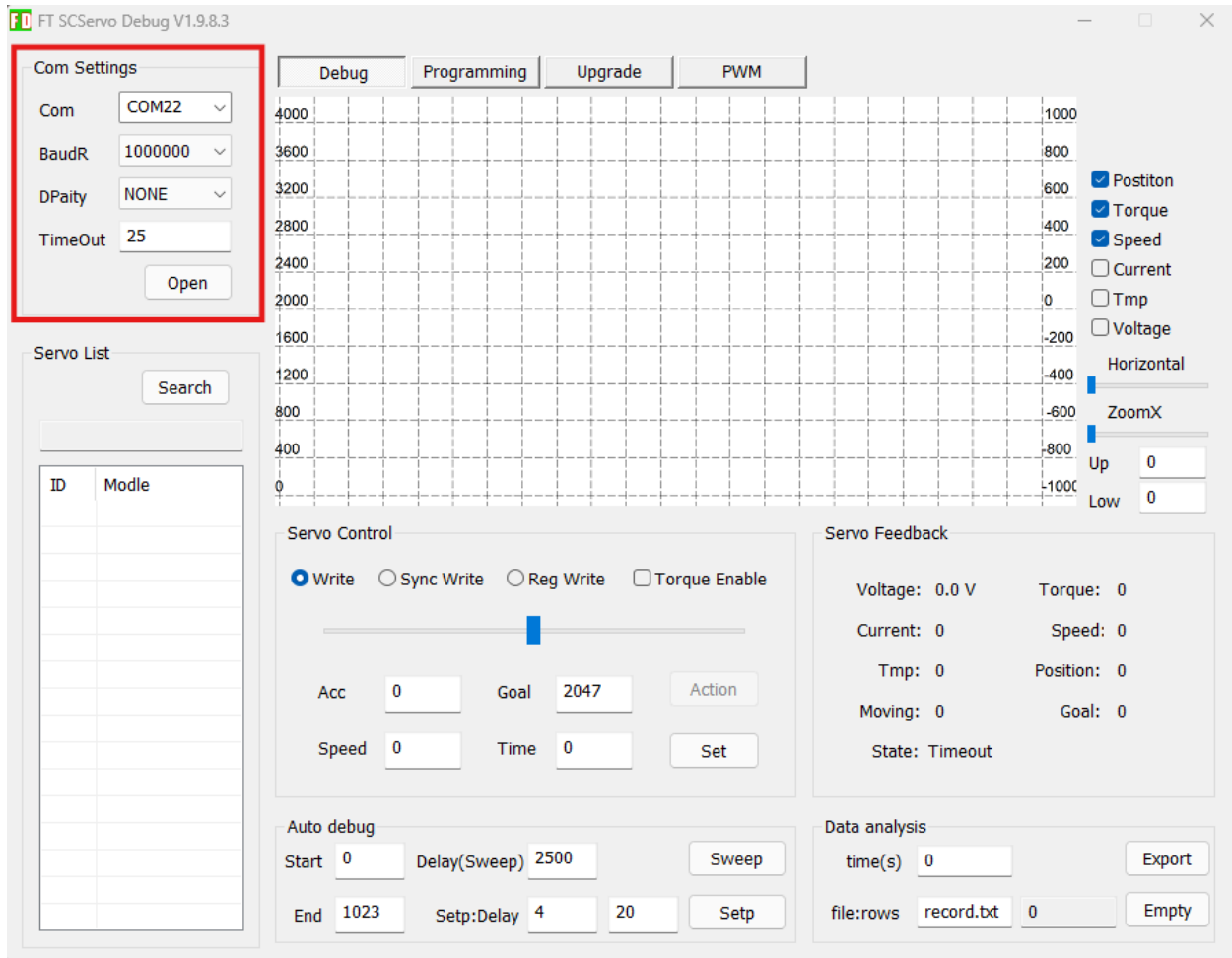


Figure 2.1 - Select COM Settings

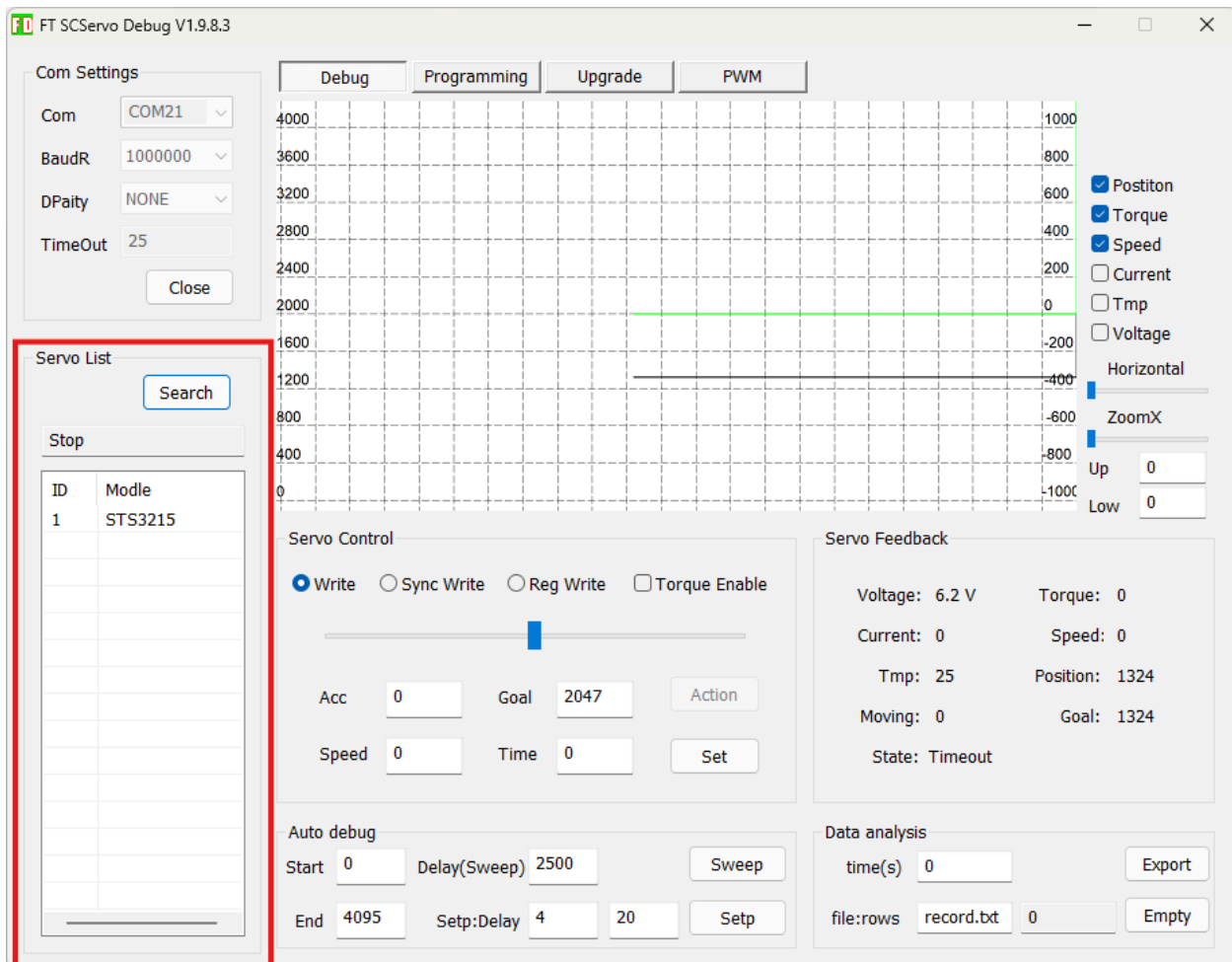


Figure 2.2 - Servo Search List

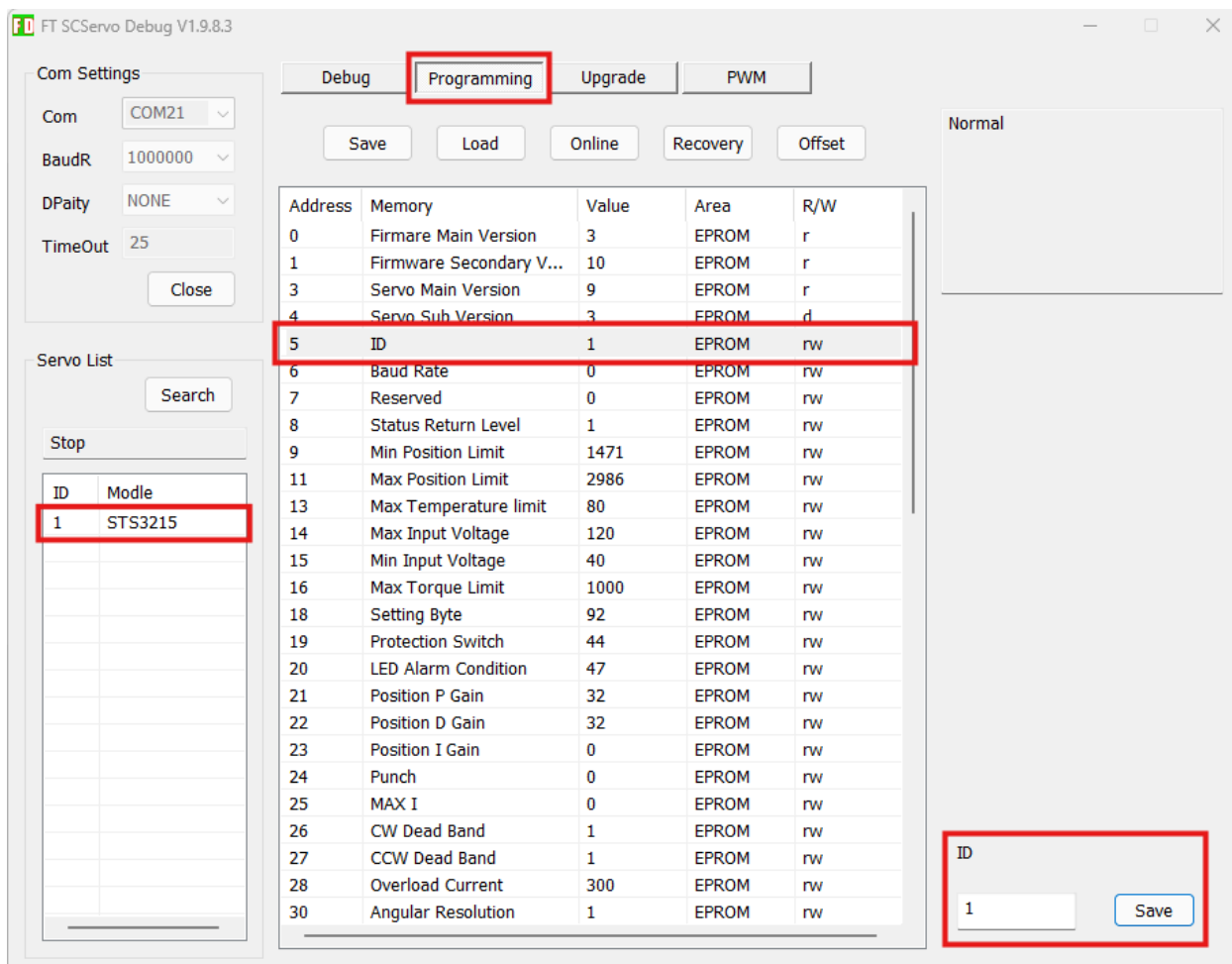


Figure 2.3 - Servo Programming Tab

Stage 3: Teleoperator Arms

Teleoperator Arms

Materials

Type	Name	Count
Actuator	Feetech Servo Motor STS3215-C046 - TELEOP	12
Electronics	Arm Motor Wire - 100mm	4
Electronics	Arm Motor Wire - 170mm	2
Electronics	Arm Motor Wire - 200mm	4
Part	Teleop Shoulder	2
Part	Teleop Motor 2 Support	2
Part	Teleop Forearm	4
Part	Teleop Motor 3/4 Support	4
Part	Teleop Wrist	2
Part	Teleop Motor 5 Support	2
Part	Teleop Motor 5 Wire Clip	2
Part	Teleop Motor 6 Casing Left	1
Part	Teleop Motor 6 Casing Right	1
Part	Teleop Motor 6 Support	2
Part	Teleop Handle	2
Part	Teleop Trigger Left	1
Part	Teleop Trigger Right	1
Part	Teleop Motor 6 Wire Clip	2
Part	Teleop Motor 3/4 Wire Clip	4
Part	Teleop Motor 2 Wire Clip	2
Part	Teleop Motor 1 Wire Clip	2
Fastener	M2x5mm	10

Type	Name	Count
Fastener	M2x8mm	9
Fastener	M3x5mm	14
Fastener	M3x8mm	56
Fastener	M3x10mm	12
Fastener	M3x12mm	8
Fastener	M3x16mm	32

Procedure

NOTE: Pay careful attention to the orientation of the parts and motors in this stage. Some parts may seem symmetrical at first, but they are not. The arm illustrated in this section's figures is the left arm

NOTE: Some parts are left/right specific. Ensure the correct part is used

NOTE: Each Servo Motor comes with 2 horns that attach to either side of the motor. The one with the teeth on the inside is referred to as the "Driven Motor Horn". The one without teeth is referred to as the "Idle Motor Horn". Both of these horns are to be placed on the motors prior to installation unless otherwise specified

1. Insert Teleop Motor 1 into Teleop Arm Base as shown in Figure 3.1
2. Insert Teleop Motor 2 into Teleop Shoulder and Fasten Teleop Motor 2 Support to Teleop Shoulder as shown in Figure 3.2
 - a. (4) M3x8mm
3. Fasten Teleop Shoulder Assembly to Teleop Motor 1 as shown in Figure 3.3
 - a. (8) M3x8mm
4. Fasten Teleop Motor 1 Support to Teleop Arm Base as shown in Figure 3.4
 - a. (4) M3x12mm
5. Fasten Teleop Forearm to Teleop Motor 2 as shown in Figure 3.5
 - a. (8) M3x16mm
6. Fasten Teleop Motor 3 and Teleop Motor 3/4 Support to Teleop Forearm as shown in Figure 3.6
 - a. (2) M3x5mm
7. Fasten Teleop Forearm to Teleop Motor 3 as shown in Figure 3.7
 - a. (8) M3x16mm
8. Fasten Teleop Motor 4 and Teleop Motor 3/4 Support to Teleop Forearm as shown in Figure 3.6
 - a. (2) M3x5mm
9. Fasten Teleop Motor 5 and Teleop Motor 5 Support to Teleop Wrist as shown in Figure 3.8
 - a. (3) M3x5mm
 - b. **NOTE:** Do NOT attach the Driven Motor Horn to Teleop Motor 5
10. Connect a 100mm Arm Motor Wire and 170mm Arm Motor Wire to Motor 5
11. Fasten Teleop Motor 5 Wire Clip to Teleop Wrist as shown in Figure 3.9
 - a. (1) M2x5mm

- b. NOTE:** Arm Motor Wires not pictured in Figure 3.9
12. Fasten Teleop Wrist Assembly to Teleop Motor 5 as shown in Figure 3.10
 - a. (8) M3x8mm
 13. Fasten Driven Motor Horn to Motor 6 Casing as shown in Figure 3.11
 - a. (4) M3x10mm
 14. Fasten Motor 6 Casing Assembly to Teleop Motor 5 as shown in Figure 3.12
 - a. (1) M3x10mm
 15. Insert Teleop Motor 6 into Teleop Motor 6 Casing as shown in Figure 3.13
 - a. **NOTE:** Do NOT attach the Idle Motor Horn to Teleop Motor 6
 16. Fasten Teleop Motor 6 Support to Teleop Motor 6 Casing as shown in Figure 3.14
 - a. (3) M3x8mm
 17. Fasten Teleop Handle to Teleop Motor 6 Casing as shown in Figure 3.15
 - a. (2) M3x8mm
 18. Fasten Teleop Trigger to Teleop Motor 6 as shown in Figure 3.16
 - a. (4) M3x8mm
 - b. (1) M3x10mm
 19. Connect the 170mm Arm Motor Wire from Teleop Motor 5 to Teleop Motor 6
 20. Fasten Teleop Motor 6 Wire Clip to Teleop Motor 6 Casing as shown in Figure 3.17
 - a. (2) M2x5mm
 - b. **NOTE:** Arm Motor Wires not shown in Figure 3.17
 21. Connect the 100mm Arm Motor Wire from Teleop Motor 5 to Teleop Motor 4
 22. Connect a 200mm Arm Motor Wire from Teleop Motor 4 to Teleop Motor 3
 23. Fasten Teleop Motor 3/4 Wire Clip to Teleop Forearm as shown in Figure 3.18
 - a. (1) M2x5mm
 - b. **NOTE:** Arm Motor Wires not shown in Figure 3.18
 24. Connect a 200mm Arm Motor Wire from Teleop Motor 3 to Teleop Motor 2
 25. Fasten Teleop Motor 3/4 Wire Clip to Teleop Forearm as shown in Figure 3.19
 - a. (1) M2x5mm
 - b. **NOTE:** Arm Motor Wires not shown in Figure 3.19
 26. Connect a 100mm Arm Motor Wire from Teleop Motor 2 to Teleop Motor 1
 27. Fasten Teleop Motor 2 Wire Clip to Teleop Motor 2 Support as shown in Figure 3.20
 - a. (2) M2x8mm
 - b. **NOTE:** Arm Motor Wires not shown in Figure 3.20
 28. Connect the 400mm Arm Motor Wire from Teleop Stand to Teleop Motor 1
 29. Fasten Teleop Motor 1 Wire Clip to Teleop Motor 1 Support as shown in Figure 3.21
 - a. (2) M2x8mm
 - b. **NOTE:** Arm Motor Wires not shown in Figure 3.21
 30. Repeat Steps 1-29 for the Right Arm
 - a. **NOTE:** Teleop Motor IDs will be different for the right arm. It will start with Teleop Motor 7 in place of Teleop Motor 1, and continue to increment down the arm

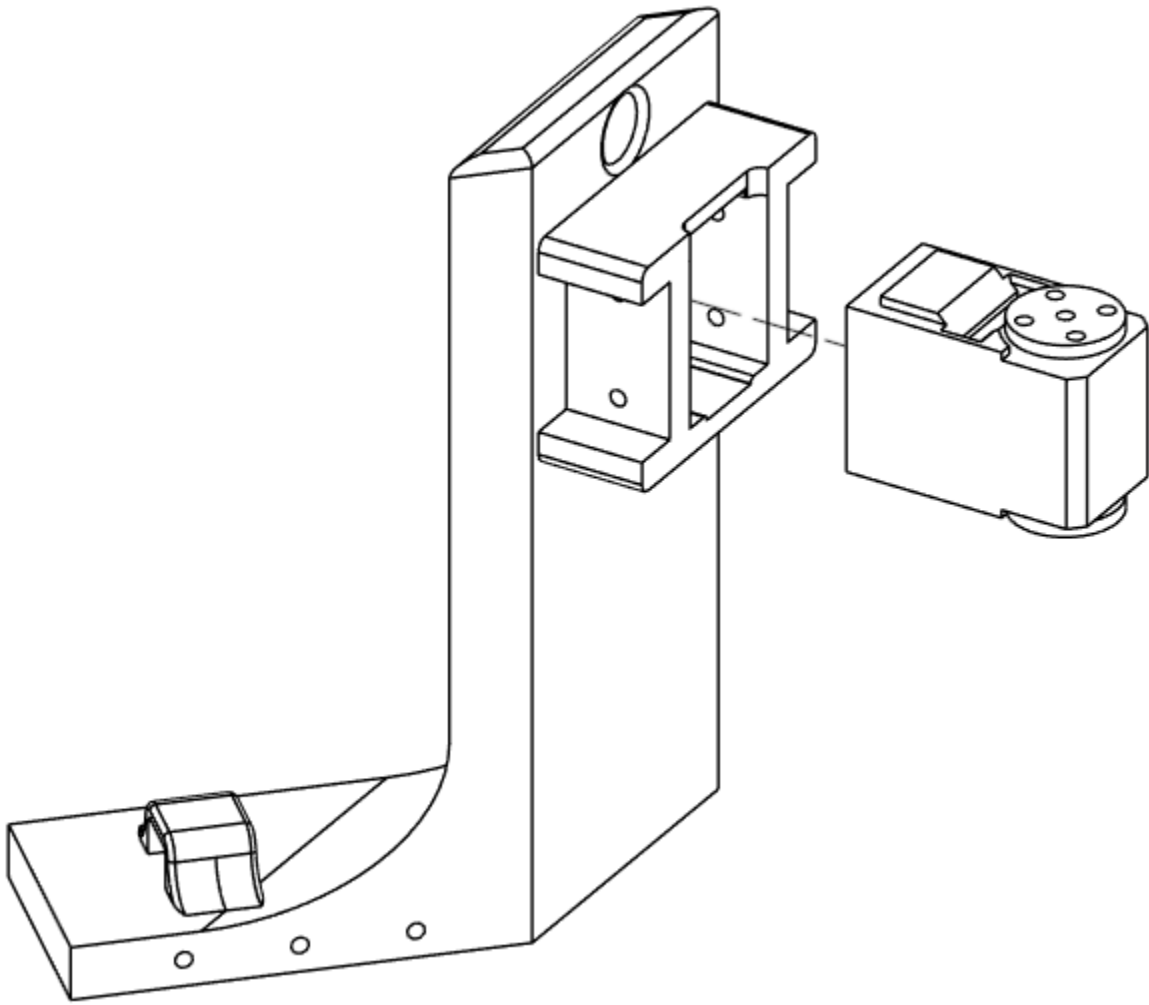


Figure 3.1 - Teleop Motor 1 Assembly

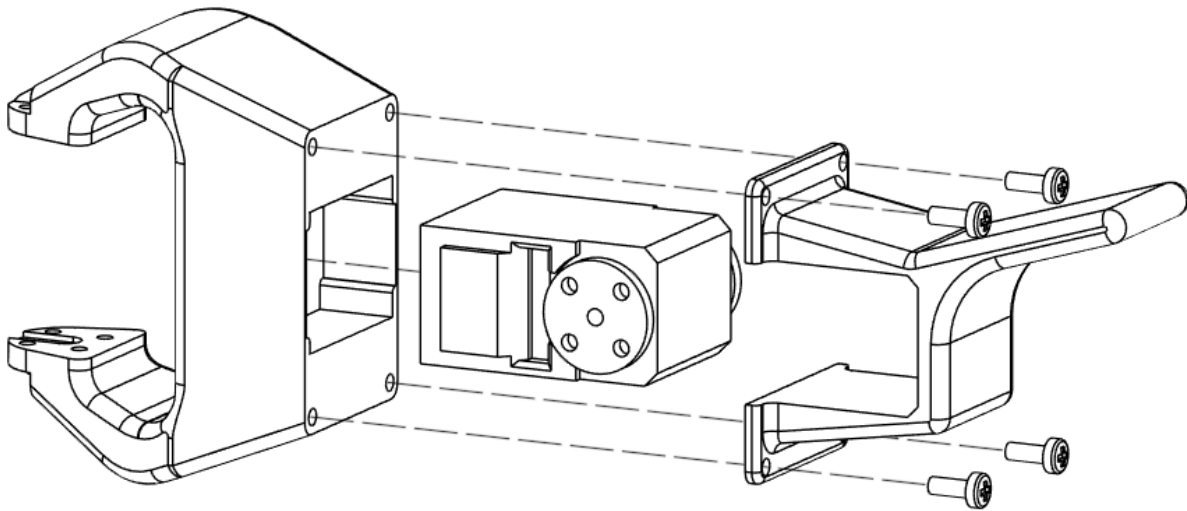


Figure 3.2 - Teleop Shoulder Assembly

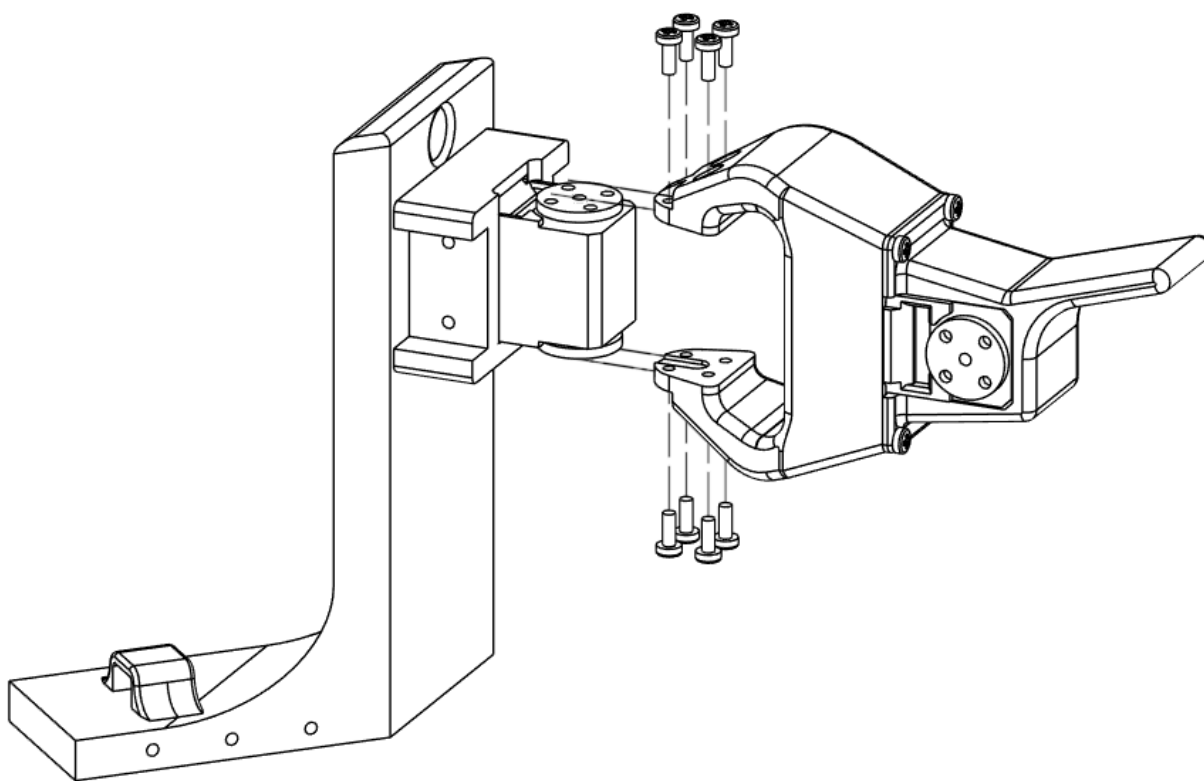


Figure 3.3 - Teleop Shoulder & Teleop Motor 1 Assembly

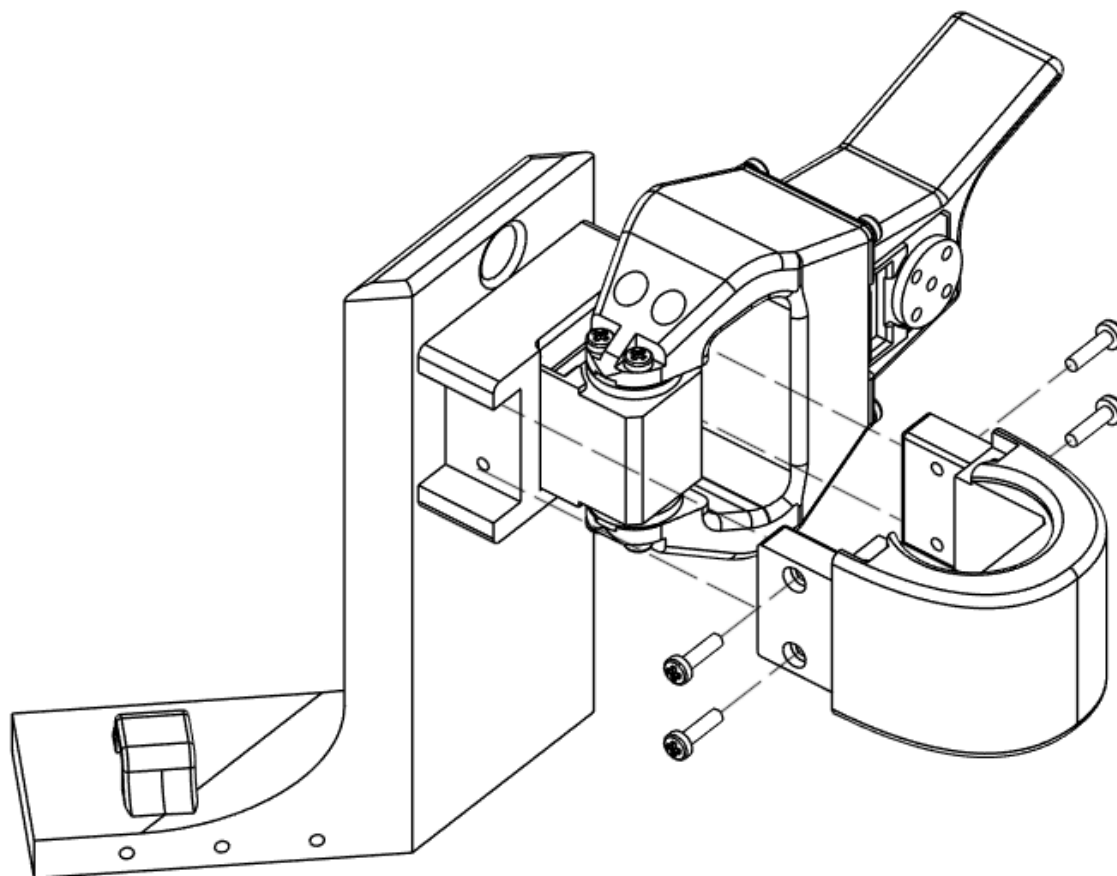


Figure 3.4 - Teleop Motor 1 Support Assembly

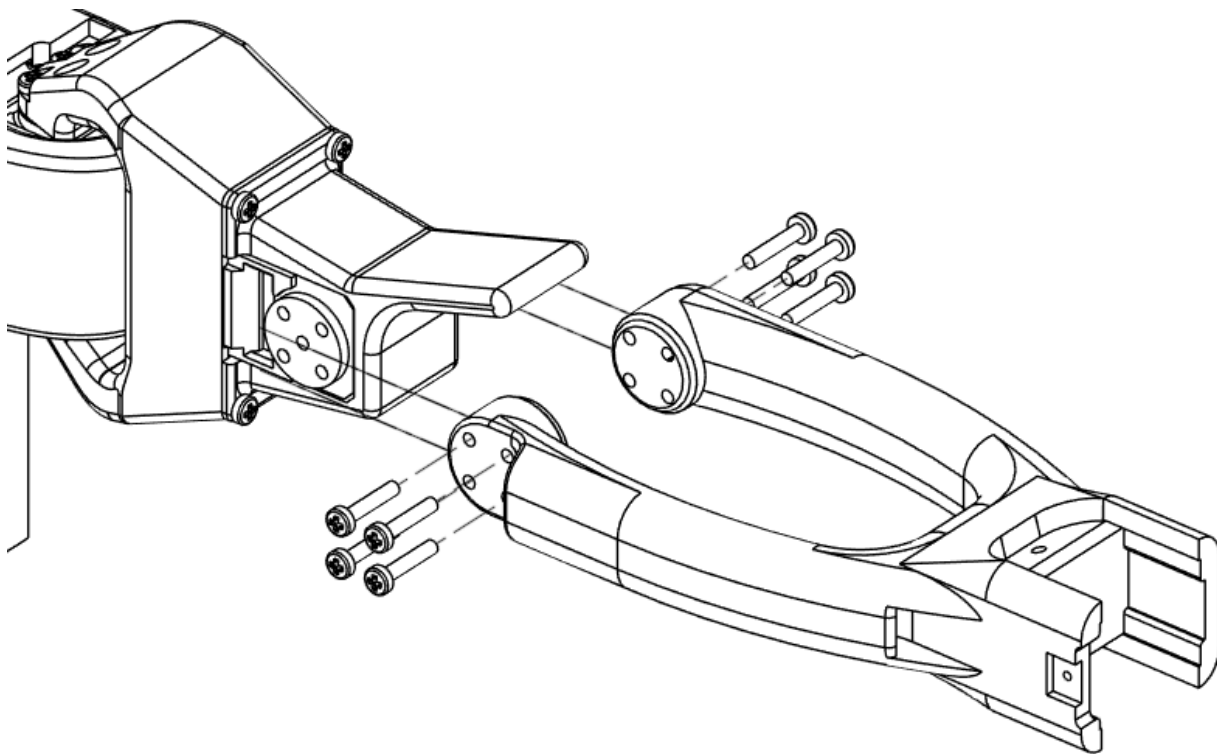


Figure 3.5 - Teleop Forearm & Teleop Motor 2 Assembly

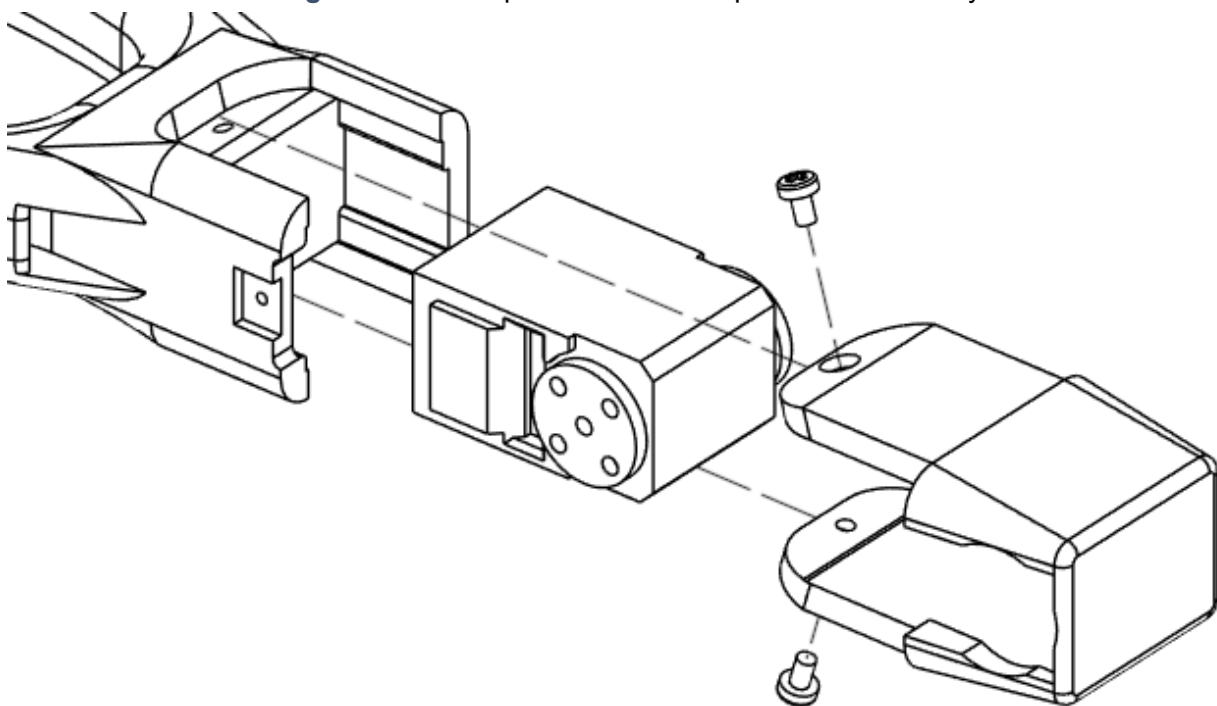


Figure 3.6 - Teleop Motor & Teleop Motor Support Assembly

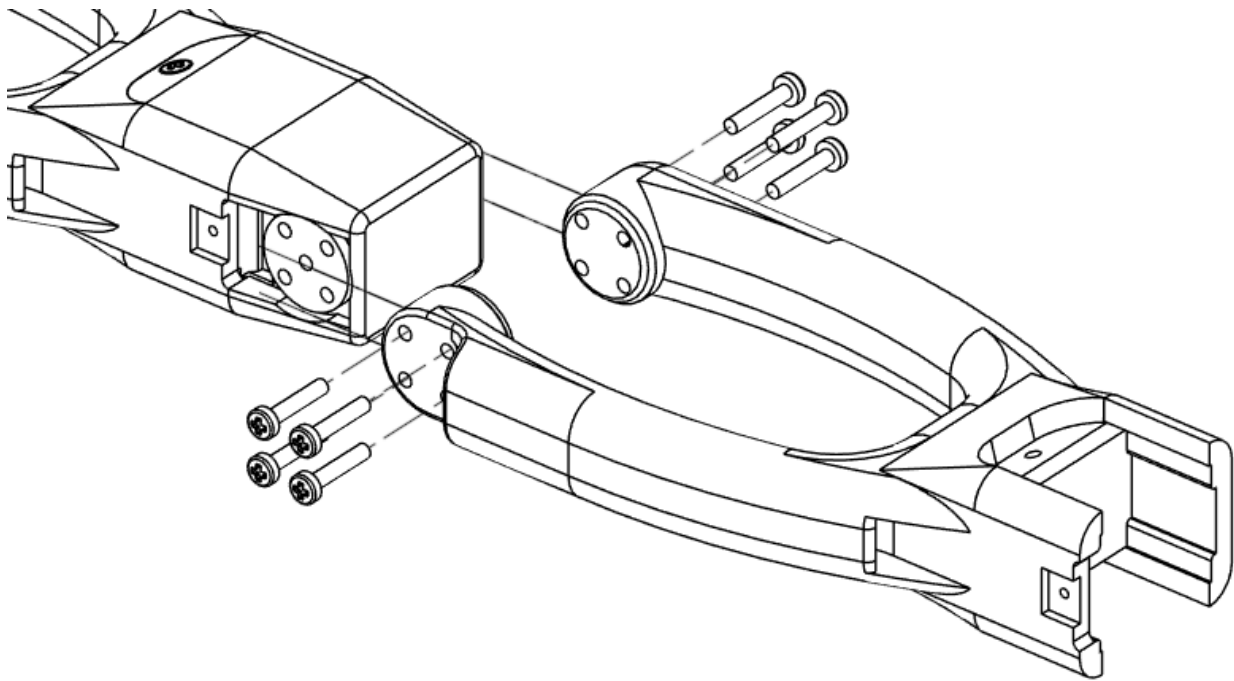


Figure 3.7 - Teleop Forearm & Teleop Motor 3 Assembly

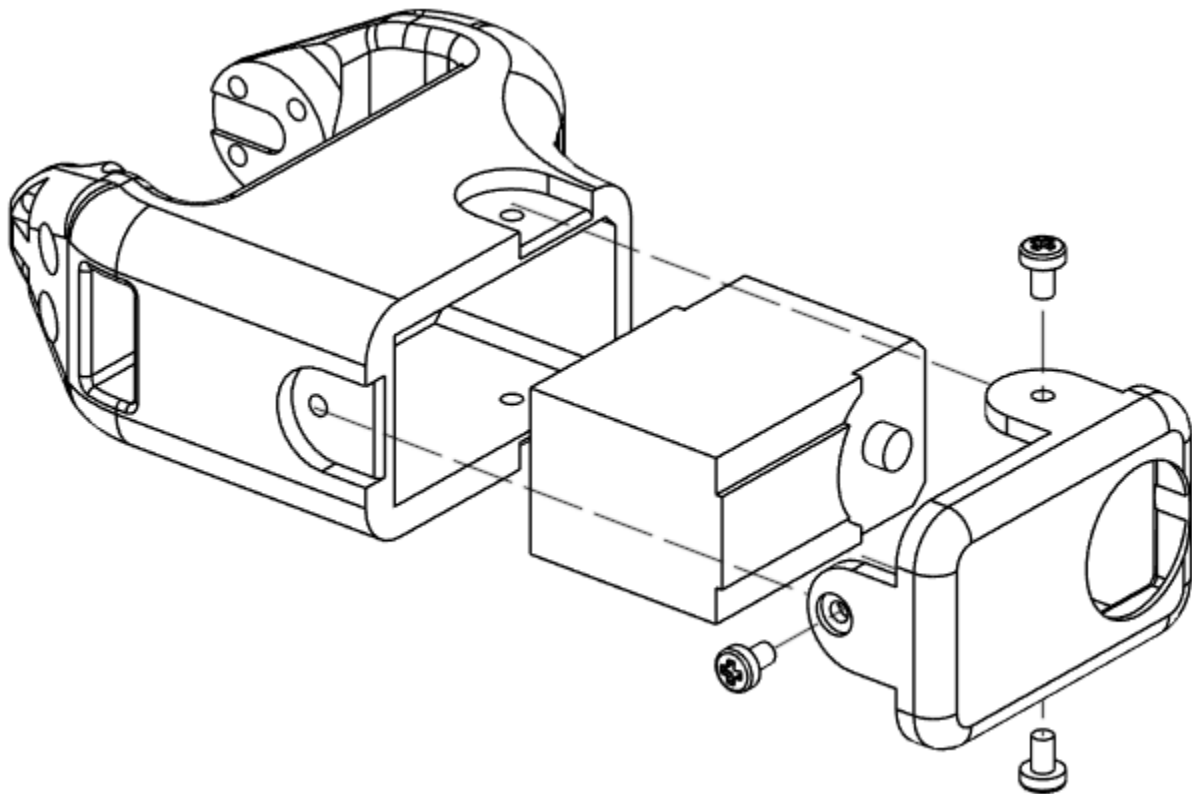


Figure 3.8 - Teleop Wrist Assembly

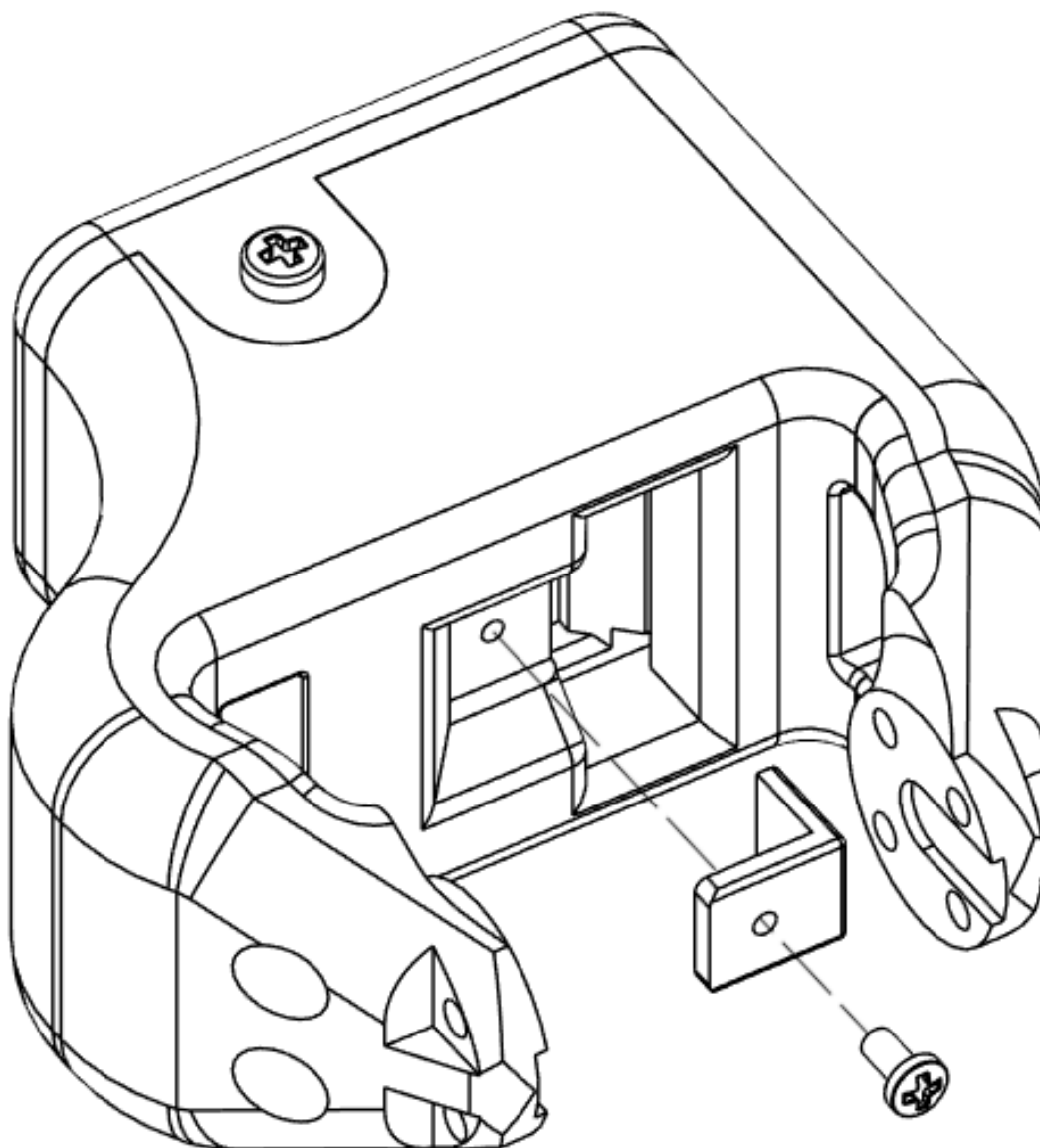


Figure 3.9 - Teleop Motor 5 Wire Clip Assembly

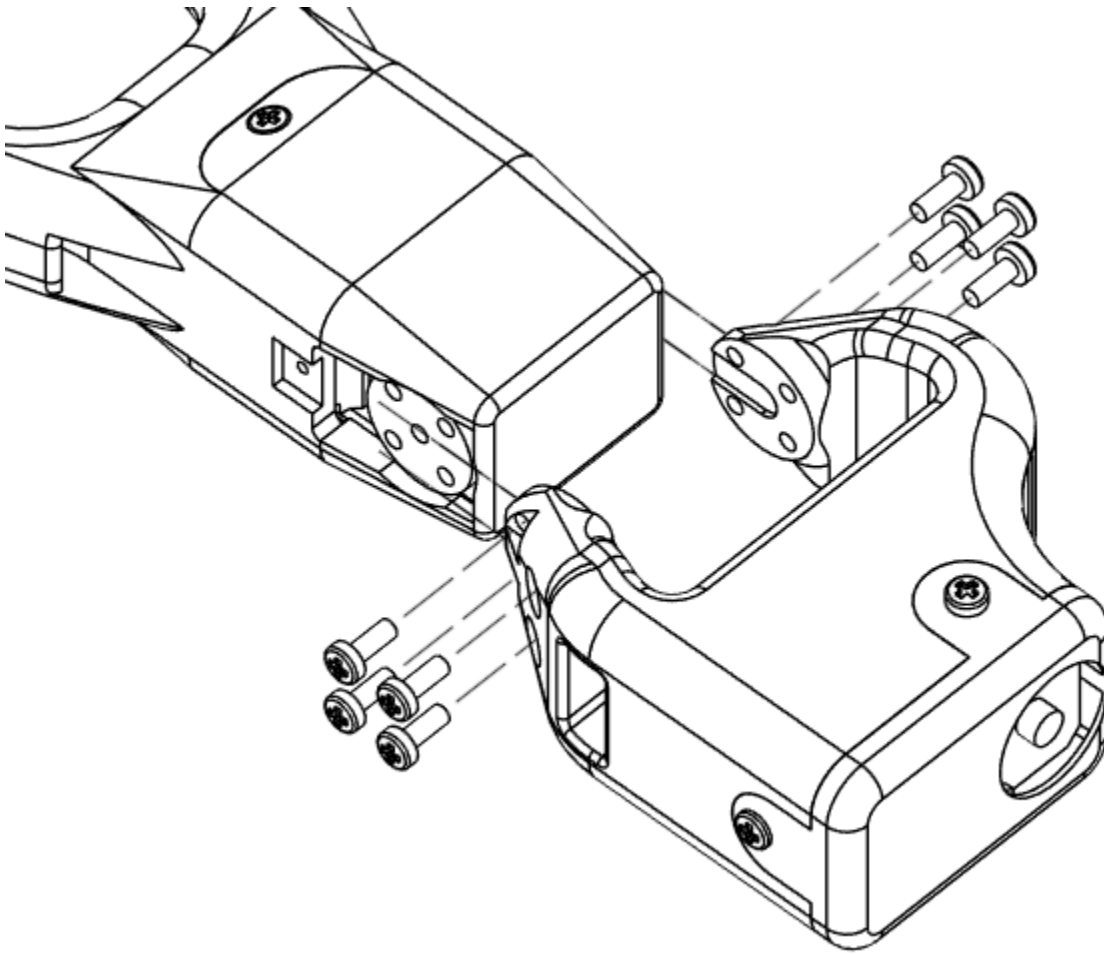


Figure 3.10 - Teleop Wrist & Teleop Motor 5 Assembly

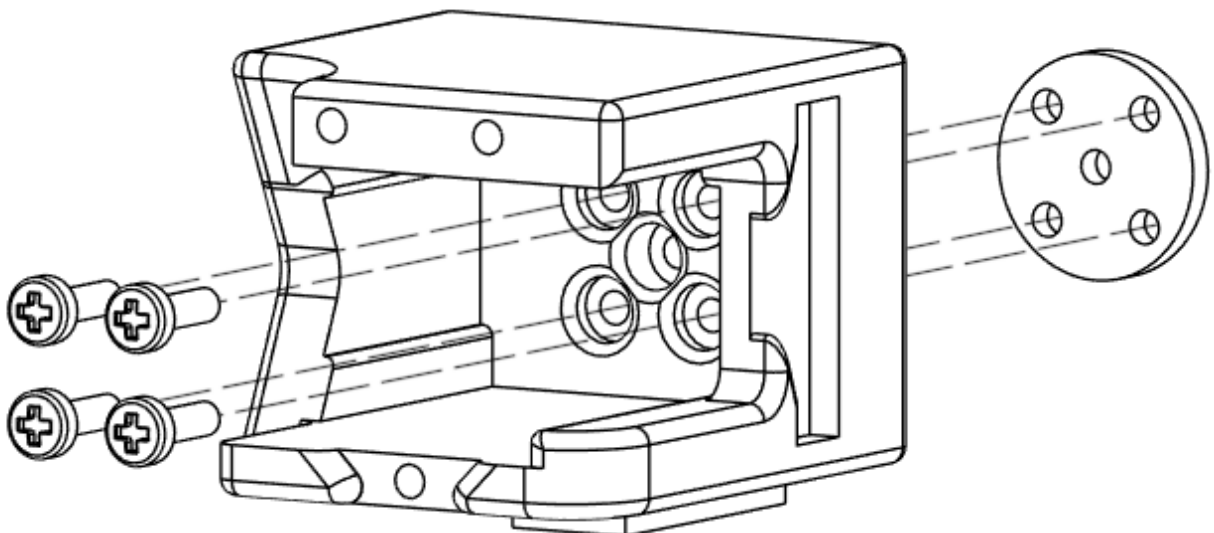


Figure 3.11 - Teleop Motor 6 Casing & Driven Motor Horn Assembly

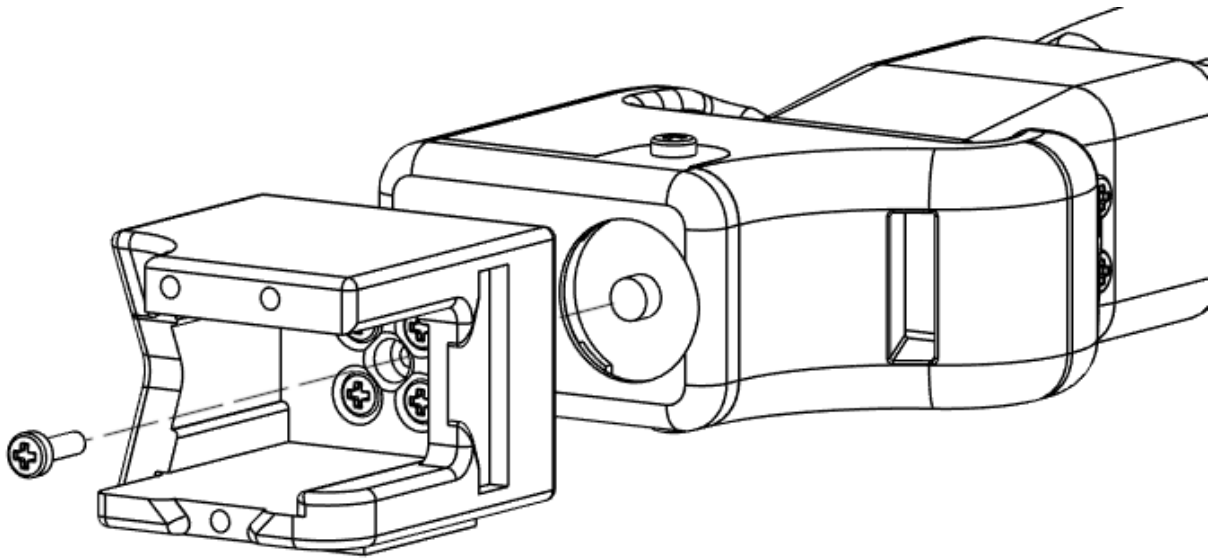


Figure 3.12 - Teleop Motor 6 Casing & Teleop Motor 5 Assembly

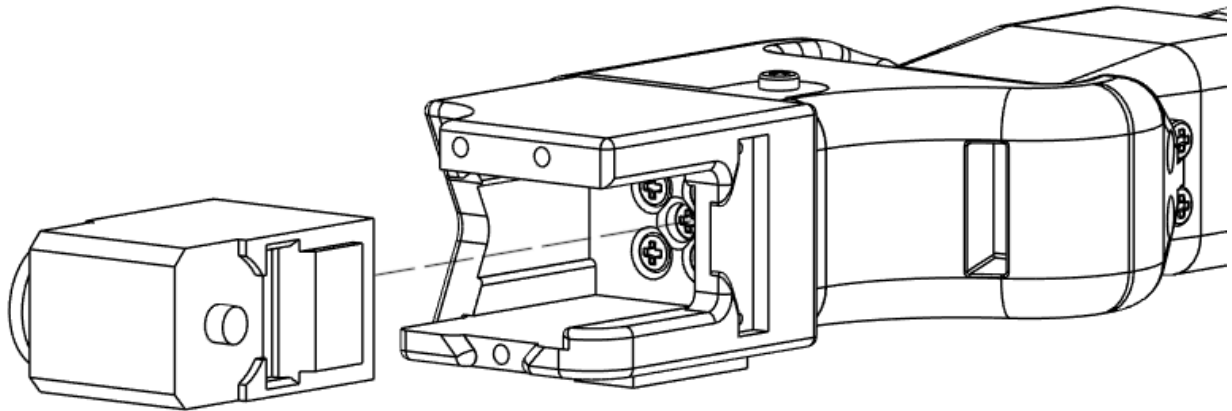


Figure 3.13 - Teleop Motor 6 Assembly

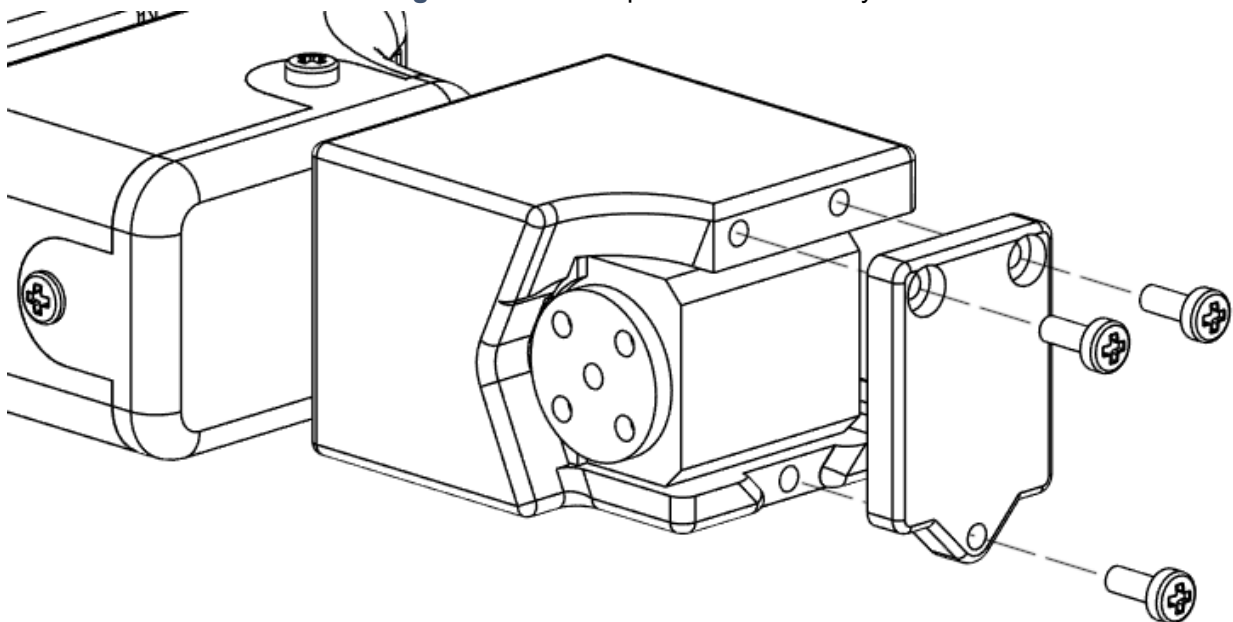


Figure 3.14 - Teleop Motor 6 Support Assembly

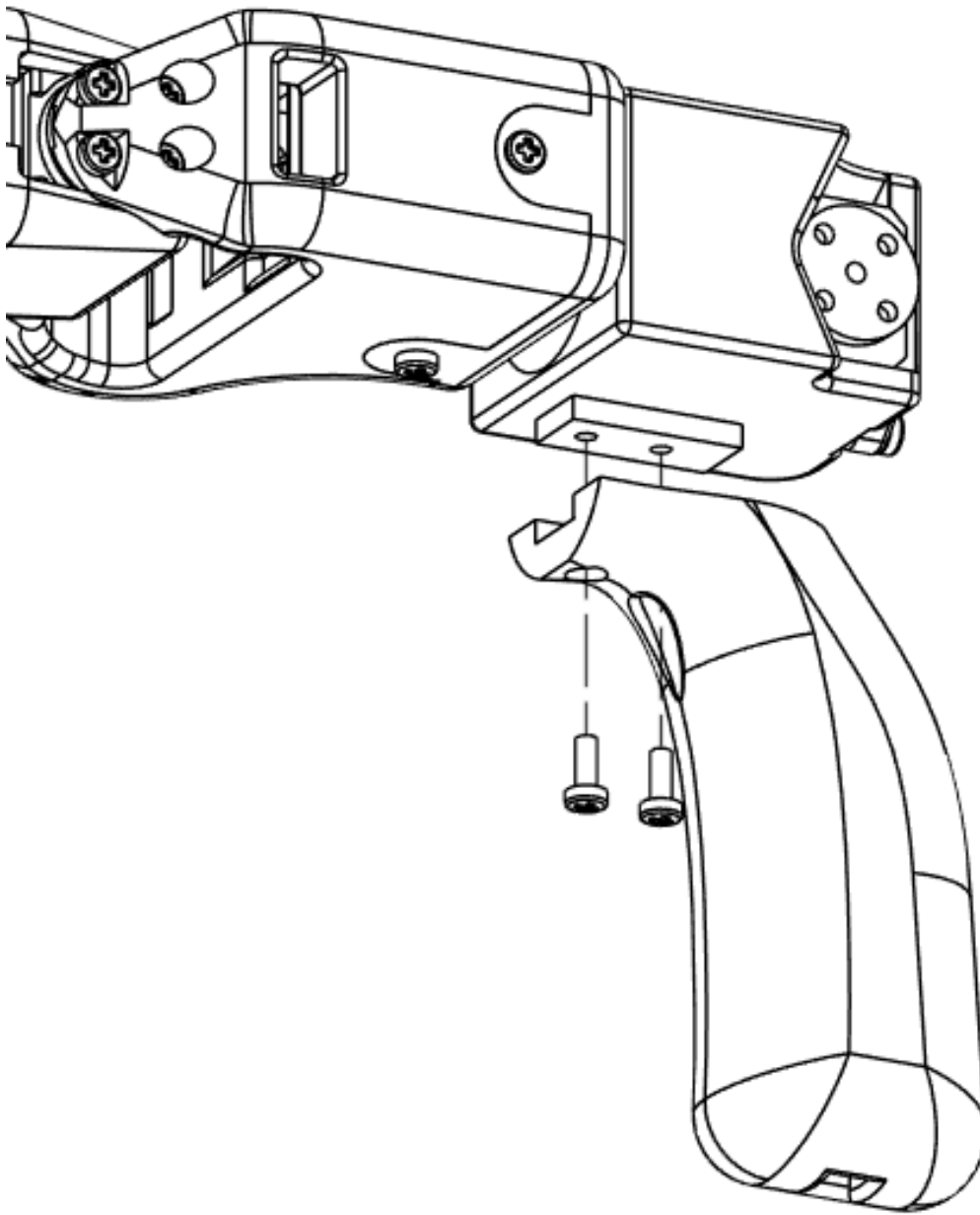


Figure 3.15 - Teleop Handle Assembly

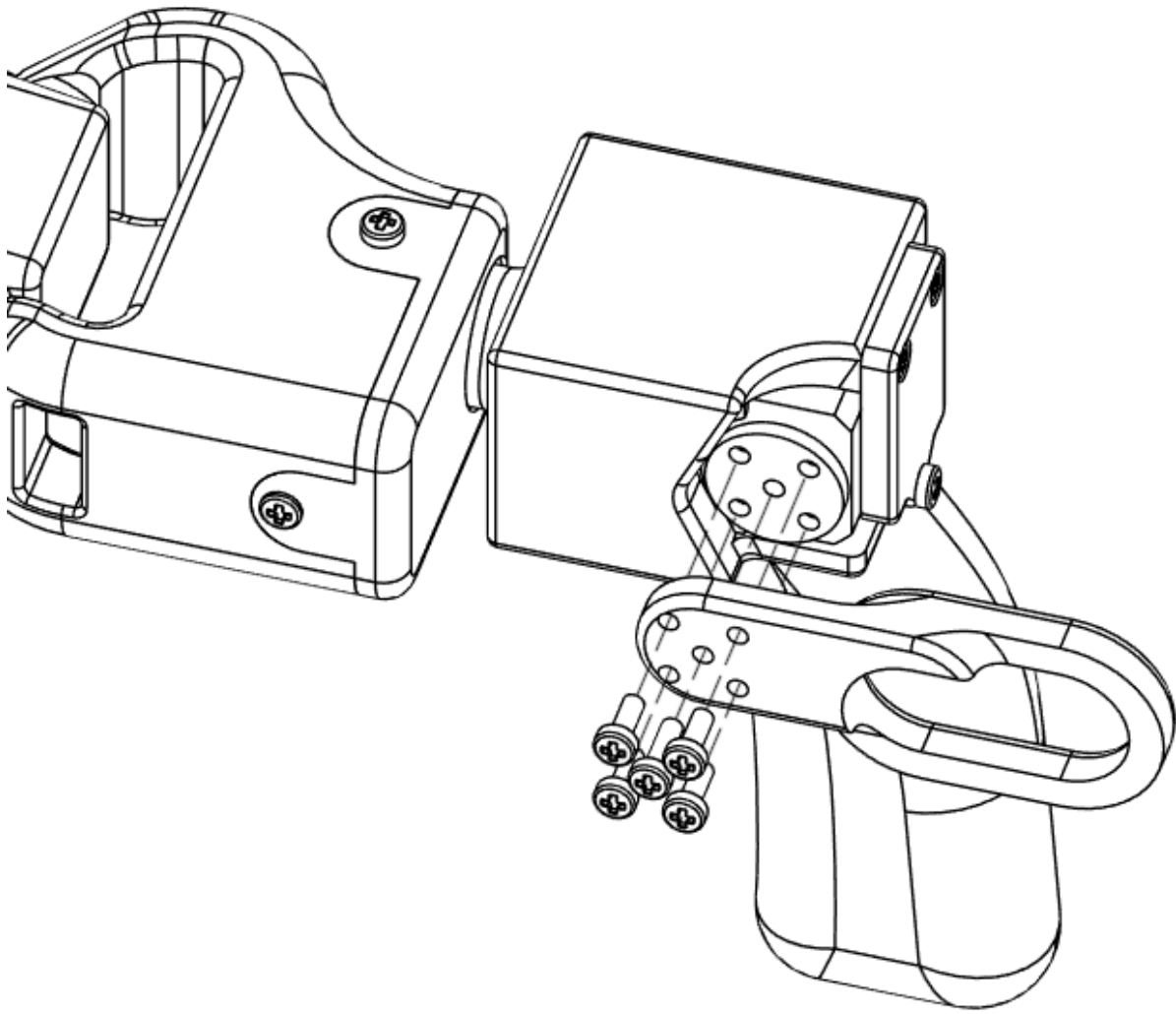


Figure 3.16 - Teleop Trigger Assembly

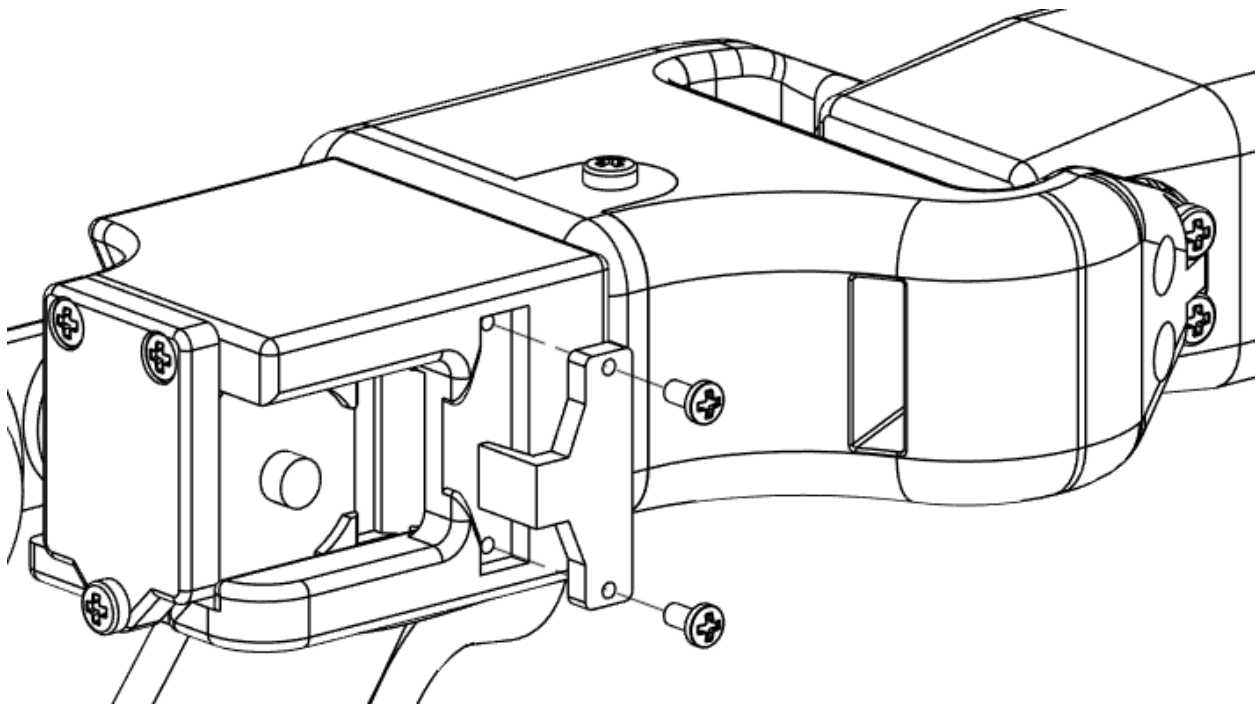


Figure 3.17 - Teleop Motor 6 Wire Clip Assembly

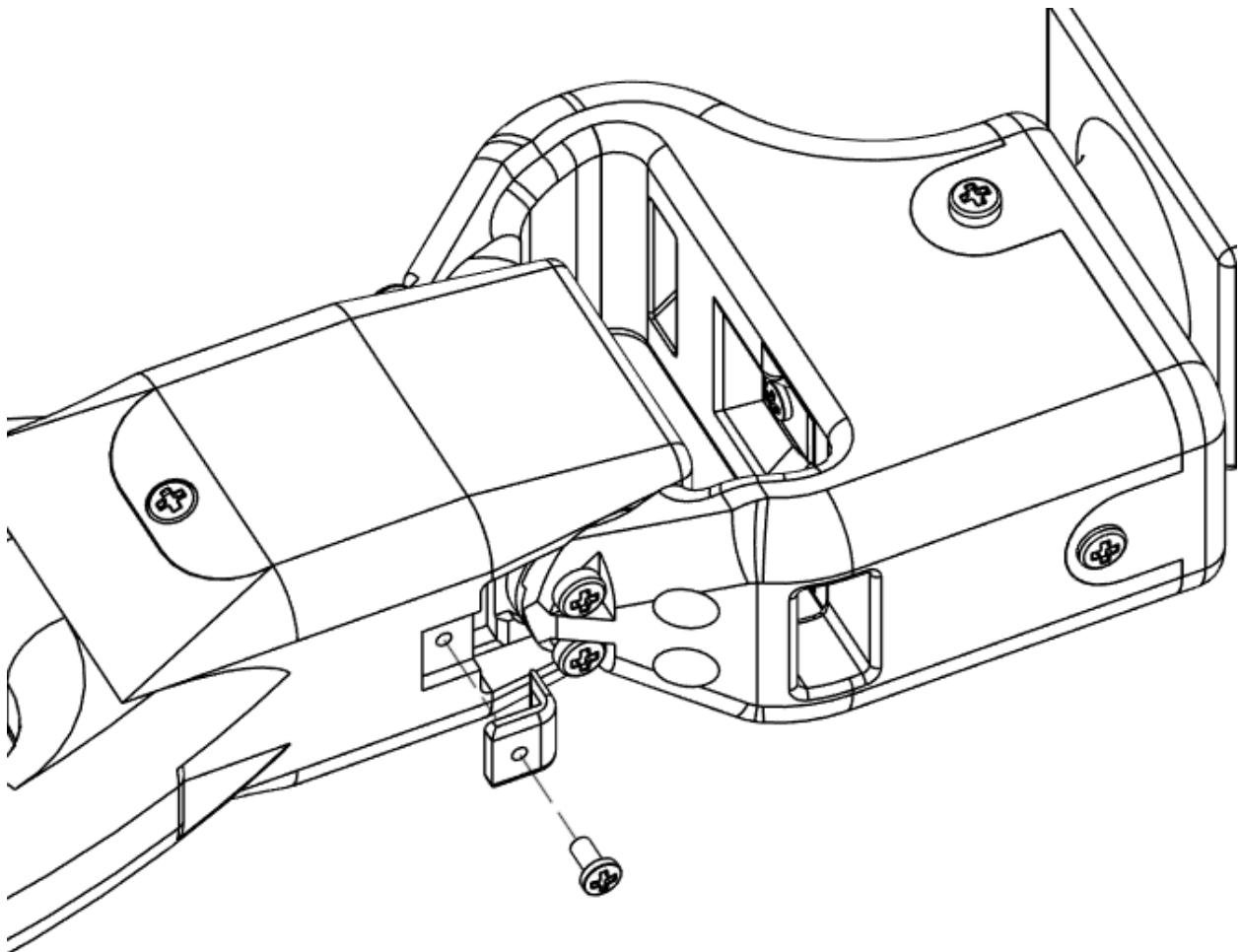


Figure 3.18 - Teleop Motor 4 Wire Clip Assembly

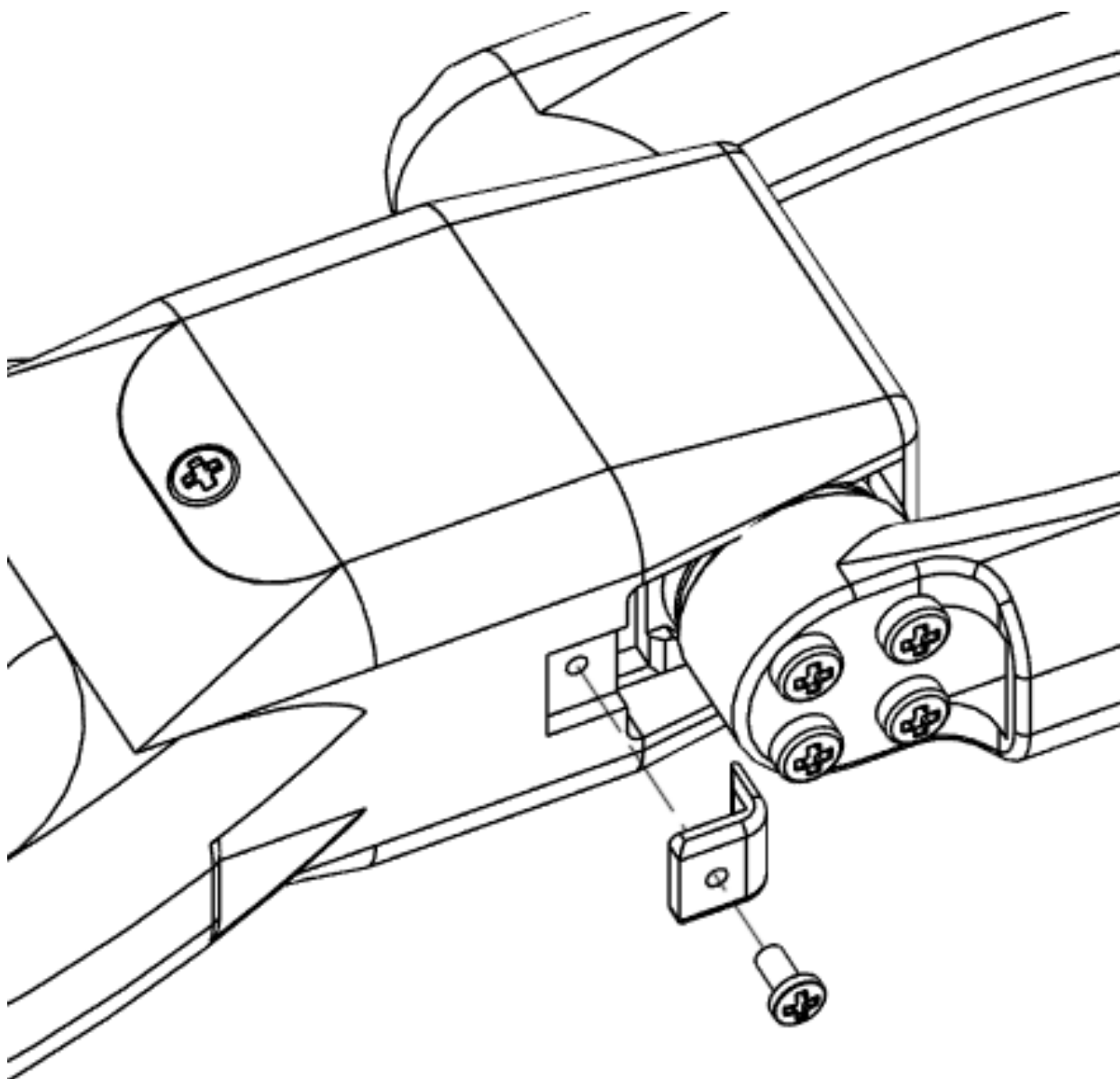


Figure 3.19 - Teleop Motor 3 Wire Clip Assembly

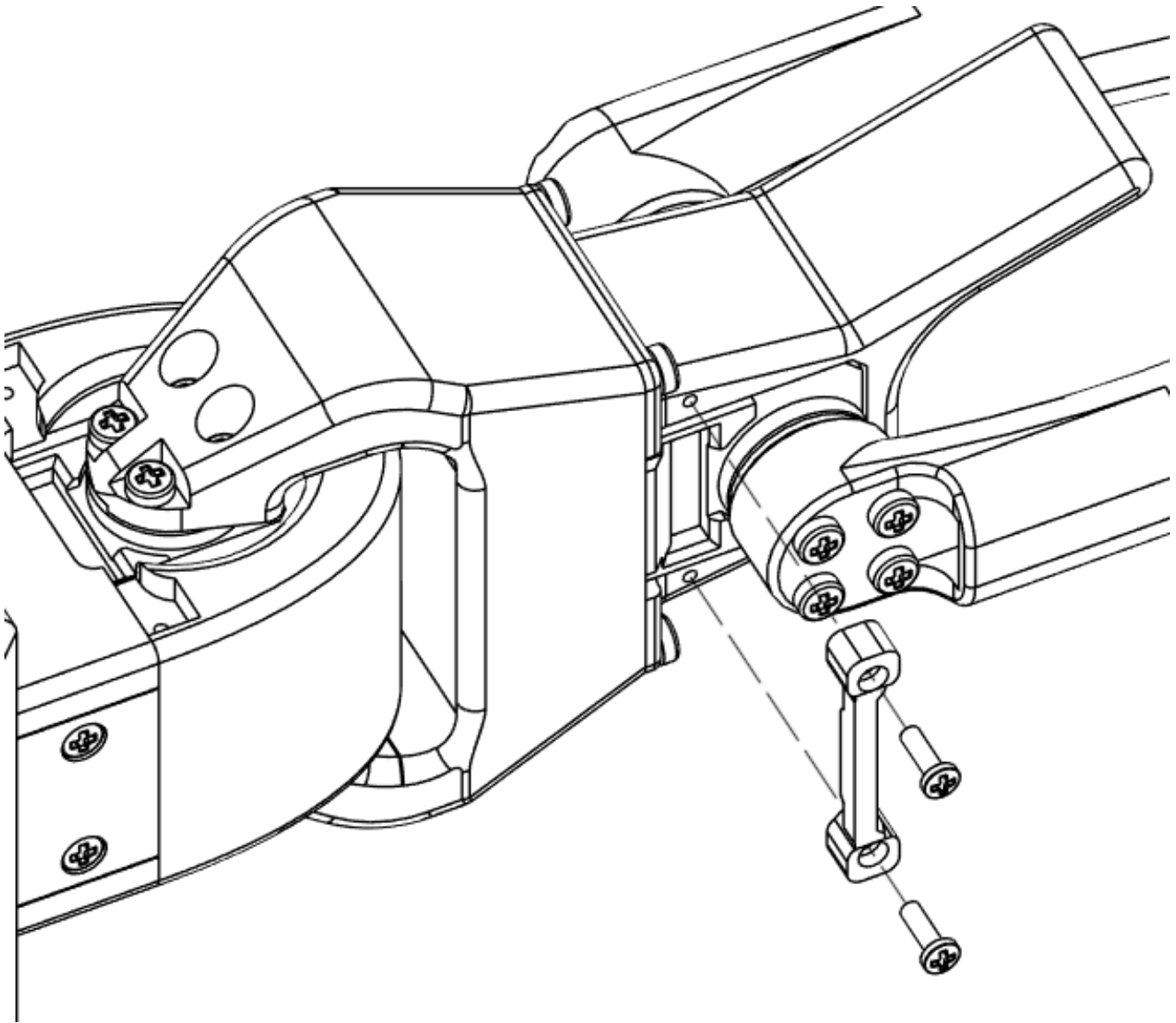


Figure 3.20 - Teleop Motor 2 Wire Clip Assembly

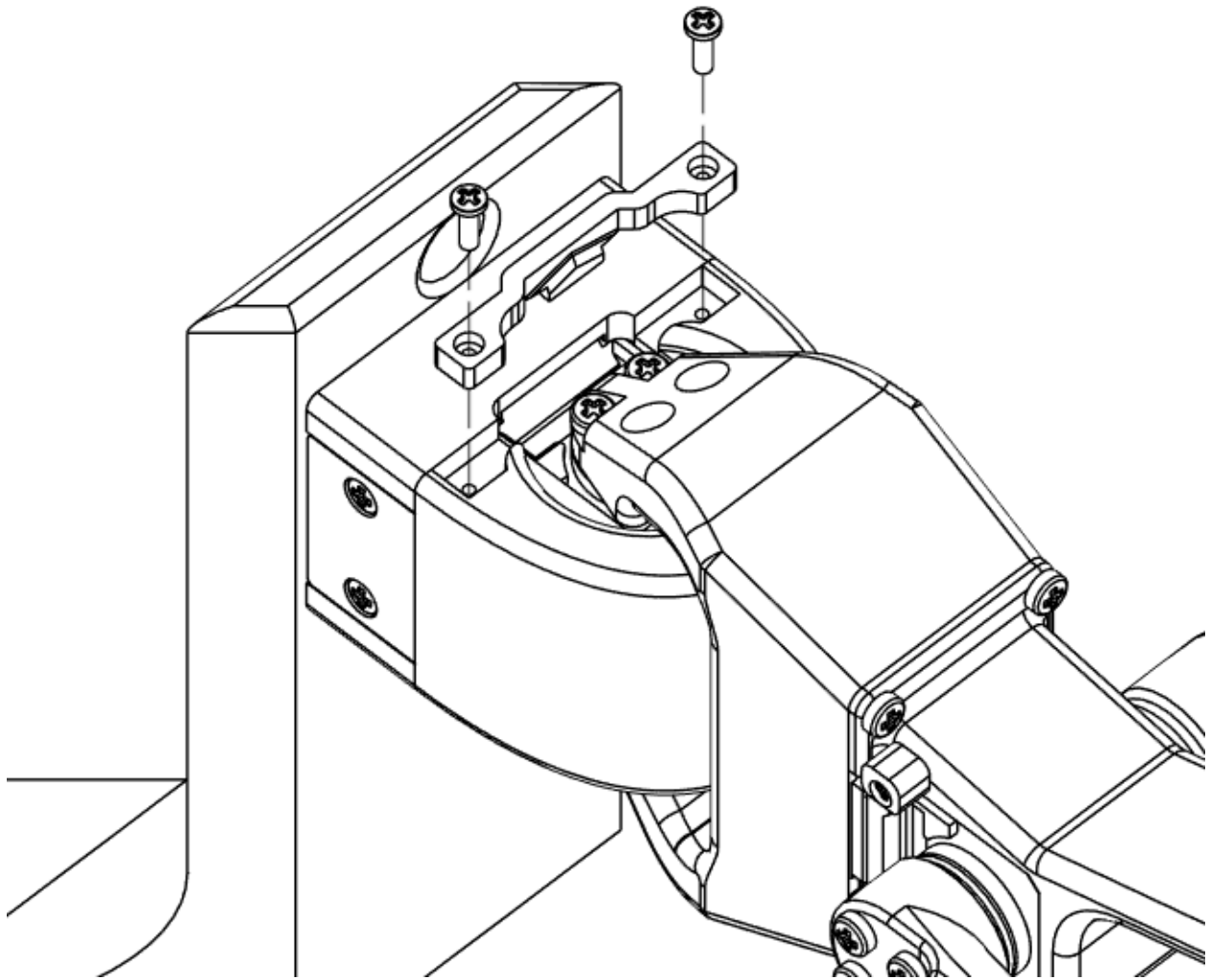


Figure 3.21 - Teleop Motor 1 Wire Clip Assembly

Stage 4: Base

Base

Materials

Type	Name	Count
Sub Assembly	Wheel Housing Assembly FL & RR	2
Sub Assembly	Wheel Housing Assembly FR & RL	2
Part	1st Level Base Plate	4
Part	Linear Actuator Base	1
Part	Battery Holder	1
Part	1st Level Rear Wall	1
Fastener	M3x16mm	48
Fastener	M3x8mm	8
Fastener	M3x12mm	4

Procedure

1. Assemble 1st level base plate

- a. Fasten wheel motor housing assemblies to Level 1 Base Plates as shown in Figure 4.1
 - i. (44) M3x16mm
- b. Fasten Linear Actuator Base to Level 1 Base Plates as shown in Figure 4.2
 - i. (8) M3x8mm
- c. Thread all wheel motor wires through

2. Assemble Rear Wall

- a. Fasten Battery Holder to rear Level 1 Base Plate as shown in Figure 4.3
 - i. (4) M3x12mm
- b. Fasten Level 1 Rear Wall to rear Level 1 Base Plate as shown in Figure 4.4
 - i. (4) M3x16mm

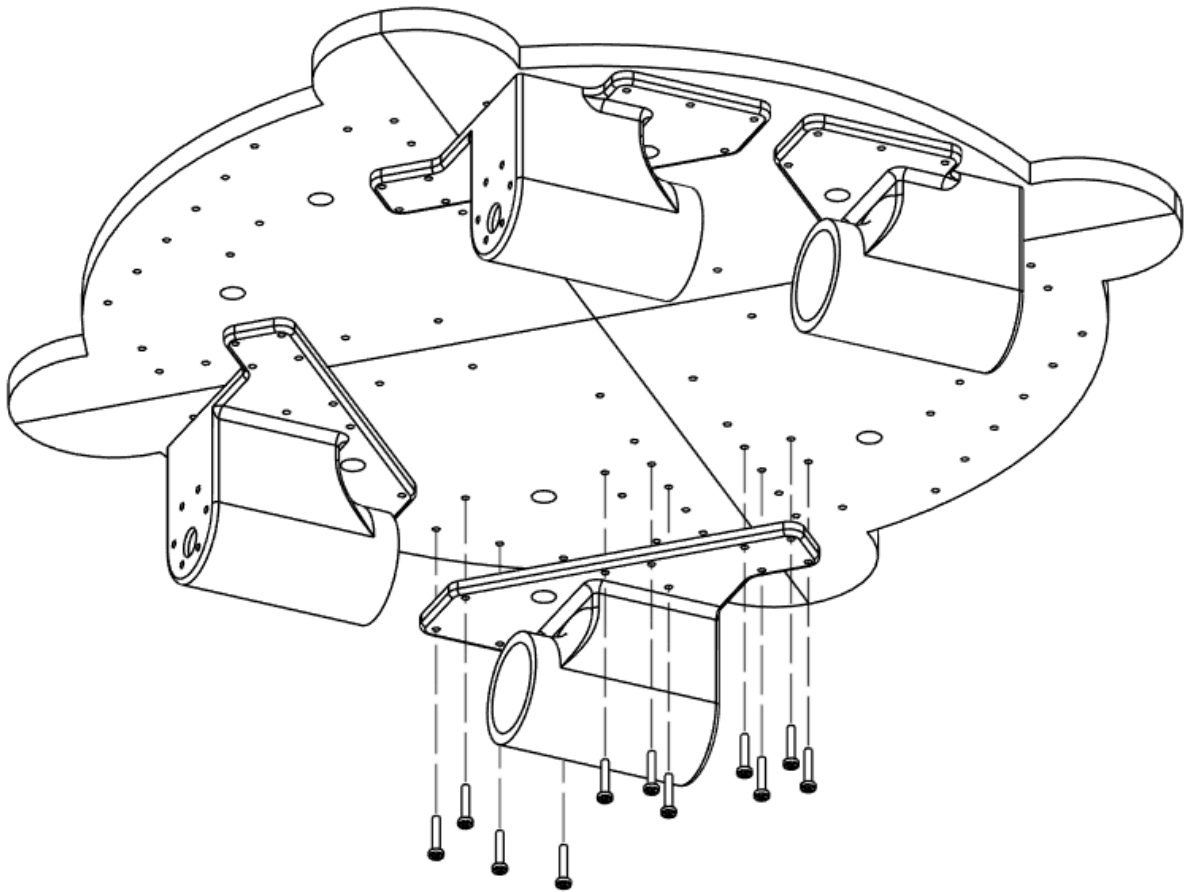


Figure 4.1 - Base Plate Undercarriage Assembly

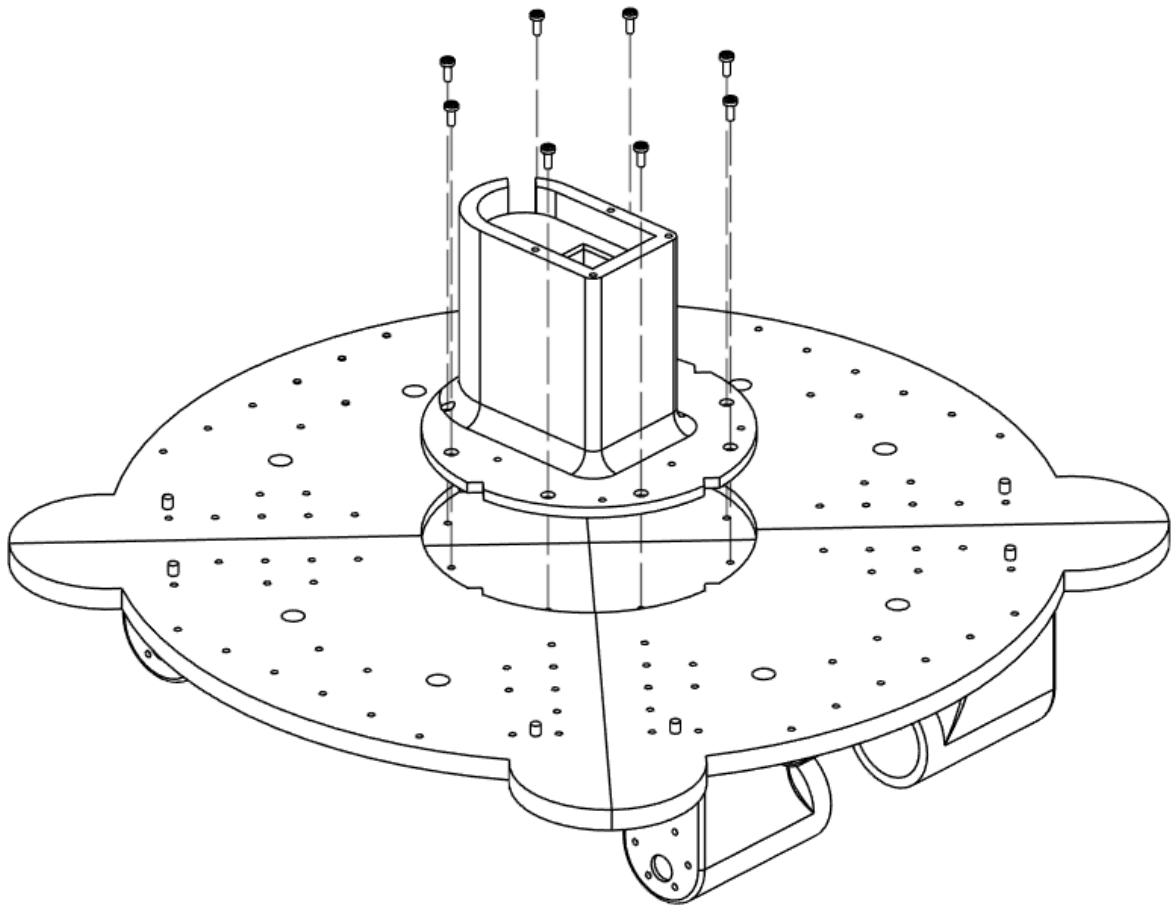


Figure 4.2 - Linear Actuator Base Assembly

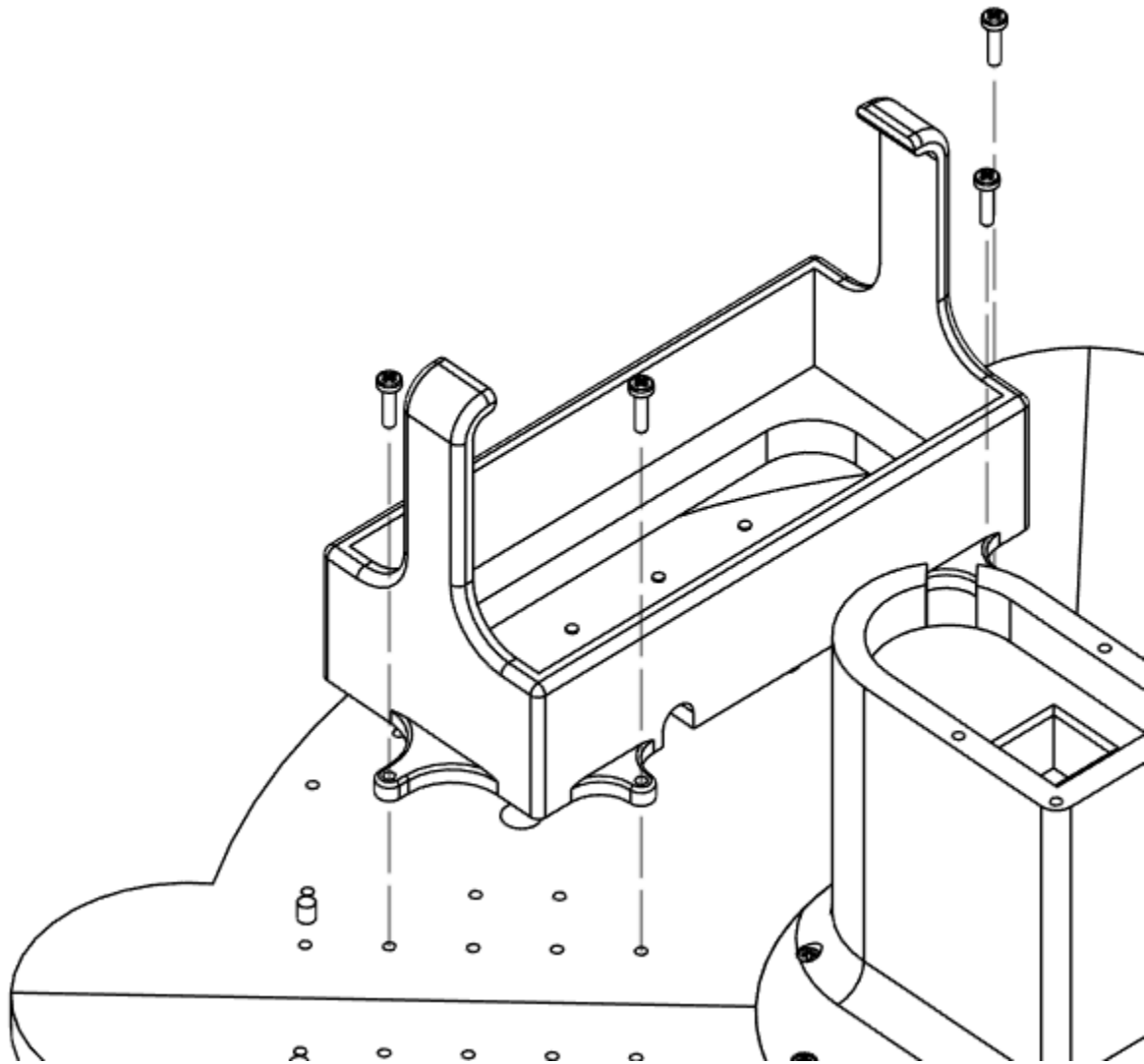


Figure 4.3 - Battery Holder Assembly

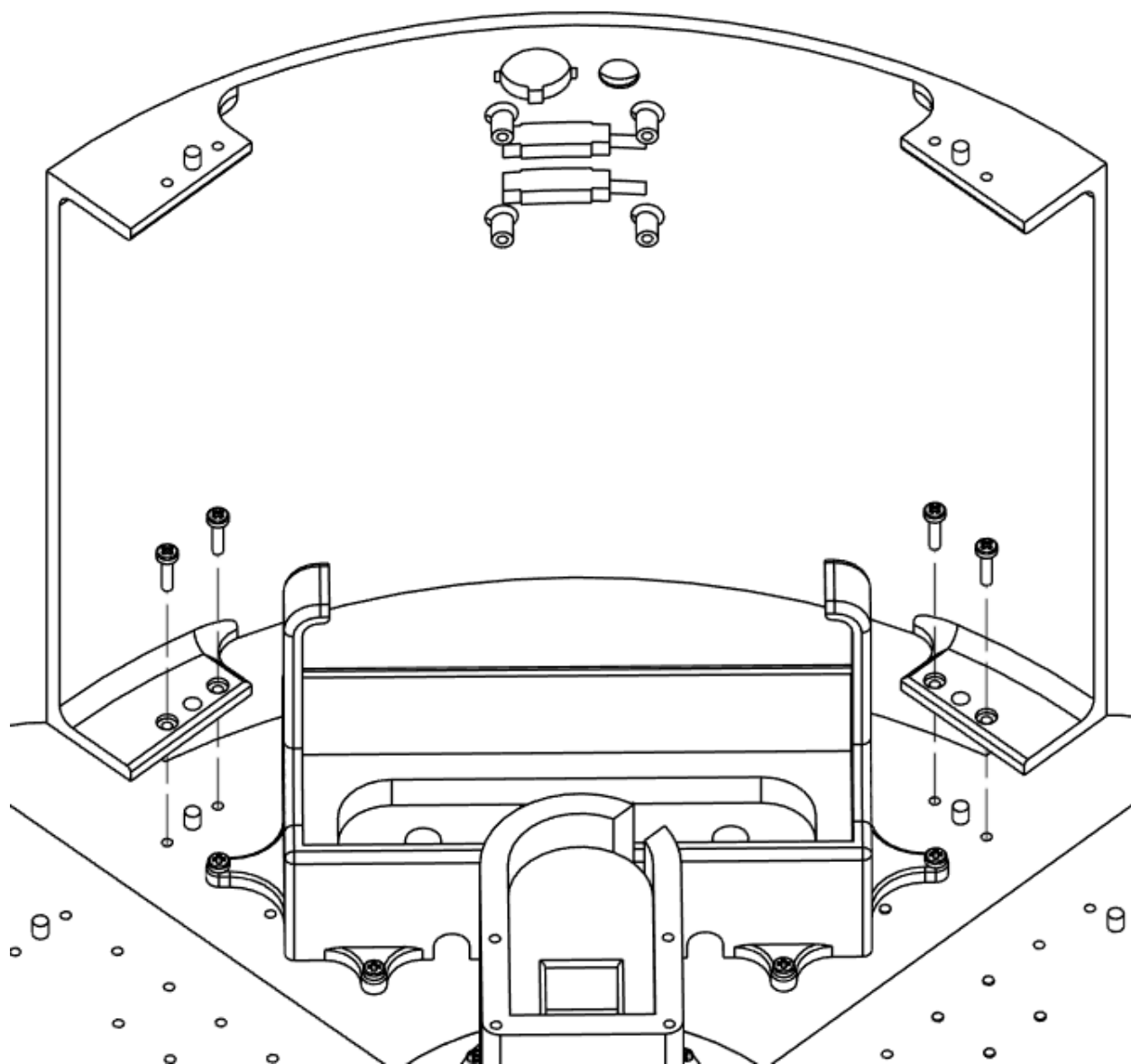


Figure 4.4 - First Level Rear Wall Assembly

Stage 5: Main Electrical

Main Electrical

Materials

Type	Name	Count
Actuator	Linear Actuator	1
Electronics	Fuse Holder	2
Electronics	Power Switch	1
Electronics	Charging Port	1
Electronics	Battery	1
Electronics	Spade Connector	1
Electronics	Sourcney PCB	1
Electronics	Buck Converter	1
Electronics	XT30 Wire - 150mm	7
Electronics	XT30 Wire - 350mm	4
Electronics	XT30 Wire - 700mm (Connected to Motor)	4
Electronics	Thermistor	1
Electronics	LED light strips	4
Electronics	Lever Wire Connectors - Single Orange	12
Electronics	Lever Wire Connectors - Single Blue	12
Electronics	Lever Wire Connectors - Double Orange	1
Electronics	Lever Wire Connectors - Double Orange/Blue	1
Electronics	Lever Wire Connectors - 5-Prong Orange	1
Electronics	Lever Wire Connectors - 5-Prong Blue	1

Type	Name	Count
Electronics	USBC to Red Black Wire	1
Electronics	5.5mm x 2.5mm Barrel Jack to Red Black Wire	2
Part	Linear Actuator Stabilizer	1
Part	Fuse Holder	1
Part	2nd Level Base Plate	1
Part	PCB Holder	1
Fastener	M3x8mm	10
Fastener	M3x12mm	9
Fastener	M3x40mm	4

Procedure

SAFETY NOTE: Uncured epoxy releases fumes that can be harmful to your health, ensure you are wearing nitrile gloves, face mask rated for organic fumes, and are in a well ventilated area before using the epoxy

1. Assemble rear electrical area

- a. Install battery in Battery Holder as shown in Figure 5.1
- b. Install (2) fuse holders in Level 1 Rear Wall as shown in Figure 5.2
- c. Fasten Fuse Holder to Level 1 Rear Wall as shown in Figure 5.2
 - i. (4) M3x8mm
- d. Install power switch in Level 1 Rear Wall as shown in Figure 5.2
- e. Install charging port in Level 1 Rear Wall as shown in Figure 5.2
- f. Attach main power fuse to power switch as shown in Figure 5.2
 - i. (1) Spade connector

2. Install PCB

- a. Fasten Level 2 Base Plate to Level 1 Rear Wall as shown in Figure 5.3
 - i. (2) M3x12mm
- b. Fasten PCB Holder to Level 2 Base Plate as shown in Figure 5.3
 - i. (2) M3x12mm
- c. Fasten Buck Converter to PCB Holder as shown in Figure 5.4
 - i. (2) M3x8mm
- d. Install PCB in PCB Holder
 - i. (4) M3x8mm

3. Wire main electrical system

- a. Wire Battery and 1st Level Rear Wall
 - i. Crimp Spade Connectors to red and black 14AWG wire (~125mm in length)
 - ii. Connect wires to Battery terminals as shown in Figure 5.2
 - iii. Connect red & black wires from Battery to Double Lever Wire Connector as shown in Figure 5.5
 - iv. Connect Fuse Holder wires to Double Lever Wire Connector as shown in Figure 5.5
 - v. Connect Double Orange Lever Wire Connector to upper fuse holder as shown in Figure 5.2
 - vi. Crimp Spade Connector to red 14 AWG Wire (~50mm in length). Connect to Double Orange Lever Wire

Connector and left terminal on Power Switch as shown in Figure 5.2

vii. Crimp Spade Connector to red 18 AWG Wire (~100mm in length) and connect to middle terminal of Power Switch as shown in Figure 5.2. Connect other end of wire to Single Orange Lever Wire Connector.

viii. Connect Black Wire to right terminal on Power Switch as shown in Figure 5.2. Connect other end of wire to Blue Lever Wire Connector. Connect Male-Female Jumper Wire to Blue Lever Wire Connector and plug into PCB (Blue-1 Port) as shown in Figure 5.2

ix. Connect Charging Port wires to Lever Wire Connectors

x. Clip all Lever Wire Connectors together

b. PCB Wiring

i. Connect red side of 350mm XT30 wire to Single Orange Lever Wire Connector from step 3-a-7. Connect black side of XT30 wire to Single Blue Lever Wire Connector as shown in Figure 5.6. Connect XT30 plug to main power port on PCB (Port Red-1) as shown in Figure 5.9

ii. Connect red side of 350mm XT30 wire to Double Orange Lever Wire Connector from step 3-a-5 as shown in Figure 5.5. Connect black side to Double Lever Connector from step 3-a-4 as shown in Figure 5.5. Connect XT30 plug to fuel gauge port on PCB (Port Red-2) as shown in Figure 5.9

iii. Connect red and black sides of 350mm XT30 wire to Lever Wire Connectors from step 3-a-9. Connect XT30 plug to charging input port on PCB (Red-3) as shown in Figure 5.9

iv. Connect red side of 350mm XT30 wire to free end of lower Fuse Holder. Connect black side of XT30 wire to Single Blue Lever Wire Connector from step 3-b-1. Connect XT30 plug to charging output port on PCB (Port Red-4) as shown in Figure 5.9

v. Connect red & black 150mm XT30 wires to 5 Prong Lever Wire Connectors. Connect XT30 plug to LED power port on PCB (Port Red-5) as shown in Figure 5.9. Connect LED Strips to 5 Prong Lever Wire Connectors

vi. Connect red & black 150mm XT30 wires to Single Lever Wire Connectors. Connect XT30 plug to raspberry pi power port on PCB (Port Red-6) as shown in Figure 5.9

vii. Connect Buck Converter input wires to Single Lever Wire Connectors. Connect 150mm XT30 wire to Single Lever Wire Connectors. Connect XT30 plug to buck converter input port on PCB (Port Red-7) as shown in Figure 5.9

viii. Connect Buck Converter output wires to Single Lever Wire Connectors. Connect 150mm XT30 wire to Connectors. Connect XT30 plug to buck converter output port on PCB (Port Red-8) as shown in Figure 5.9

ix. Connect (2) 150mm XT30 plugs to motor driver power ports on PCB (Ports Red-9 & Red-10) as shown in Figure 5.9. Connect Single Lever Wire Connectors to ends of XT30 wires. Connect (2) Male Barrel Jacks to Connectors

x. Connect cables from wheel motors to Lever Wire Connectors

xi. Connect (4) 700mm XT30 plugs to DC motor ports on PCB as shown in Figure 5.9

1. Port Red-12: Front Left Wheel
2. Port Red-13: Front Right Wheel
3. Port Red-14: Rear Left Wheel
4. Port Red-15: Rear Right Wheel

xii. Fasten Thermistor to Battery as shown in Figure 5.8. Plug Thermistor Cable into PCB thermistor port (Port Purple-1)

5. Install linear actuator

a. Place linear actuator in Linear Actuator Base

b. Fasten Linear Actuator Stabilizer to Linear Actuator Base as shown in Figure 5.10

i. (5) M3x12mm

ii. (4) M3x40mm

c. Connect Linear Actuator to PCB DC motor port (Port Red-11) as shown in Figure 5.9 with Lever Wire Connectors and 150mm XT30 wire

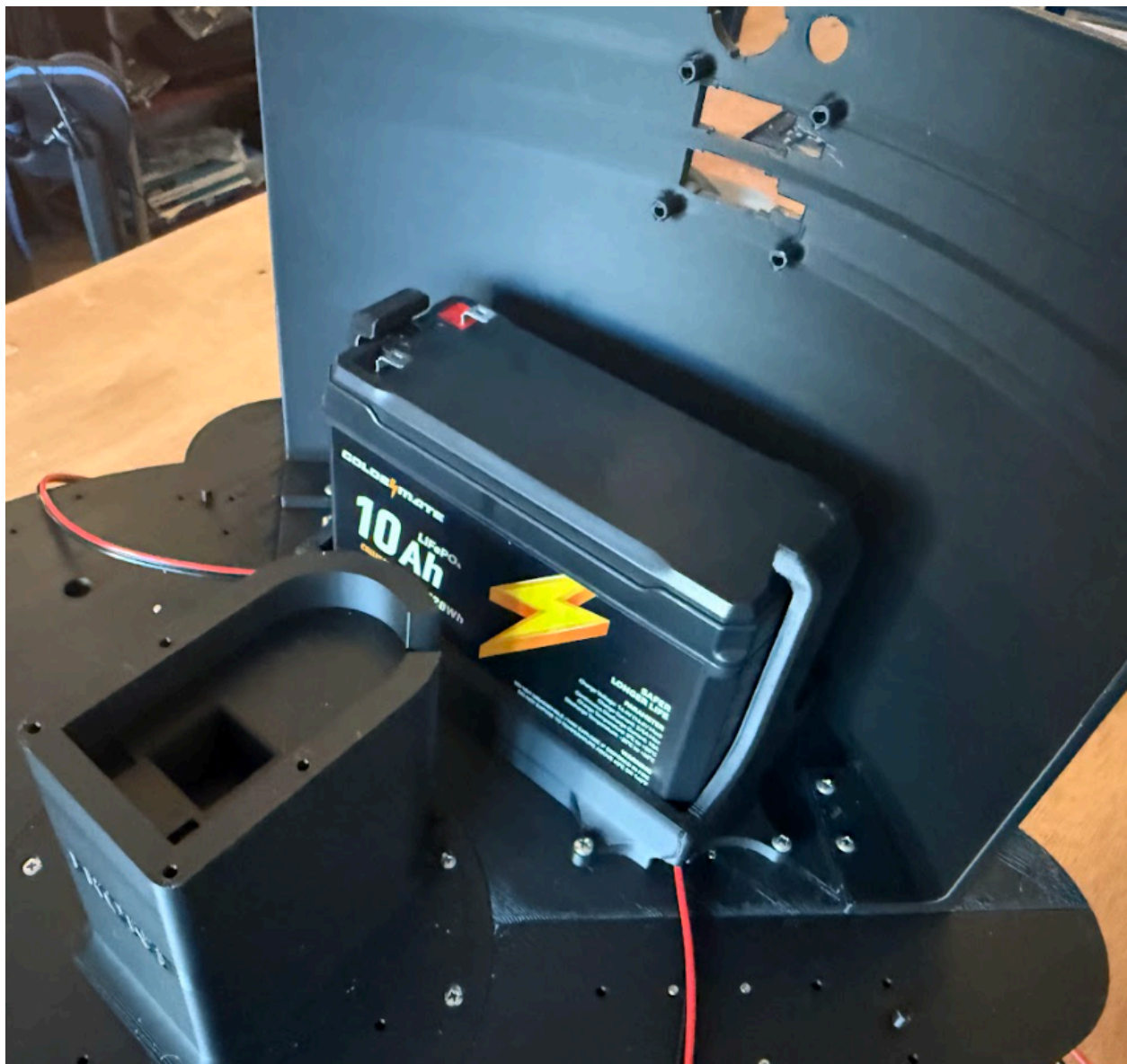


Figure 5.1 - Battery Assembly

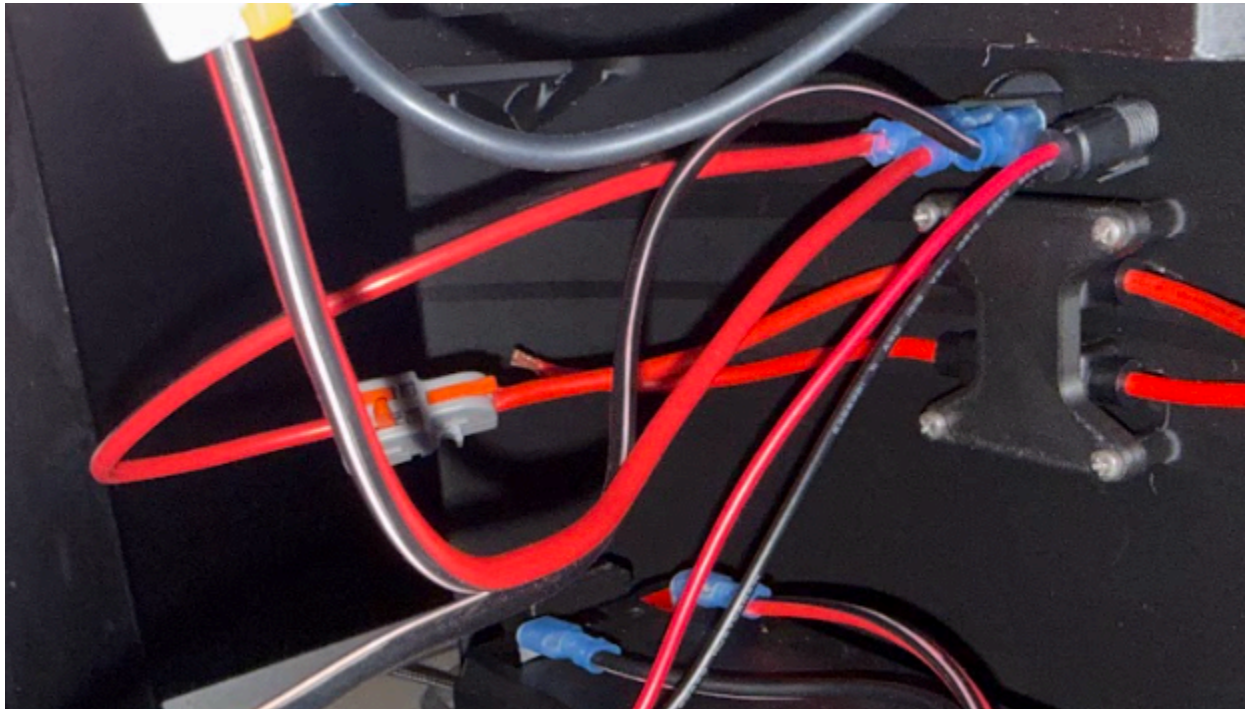


Figure 5.2 - Fuse Holder & Power Switch Assembly

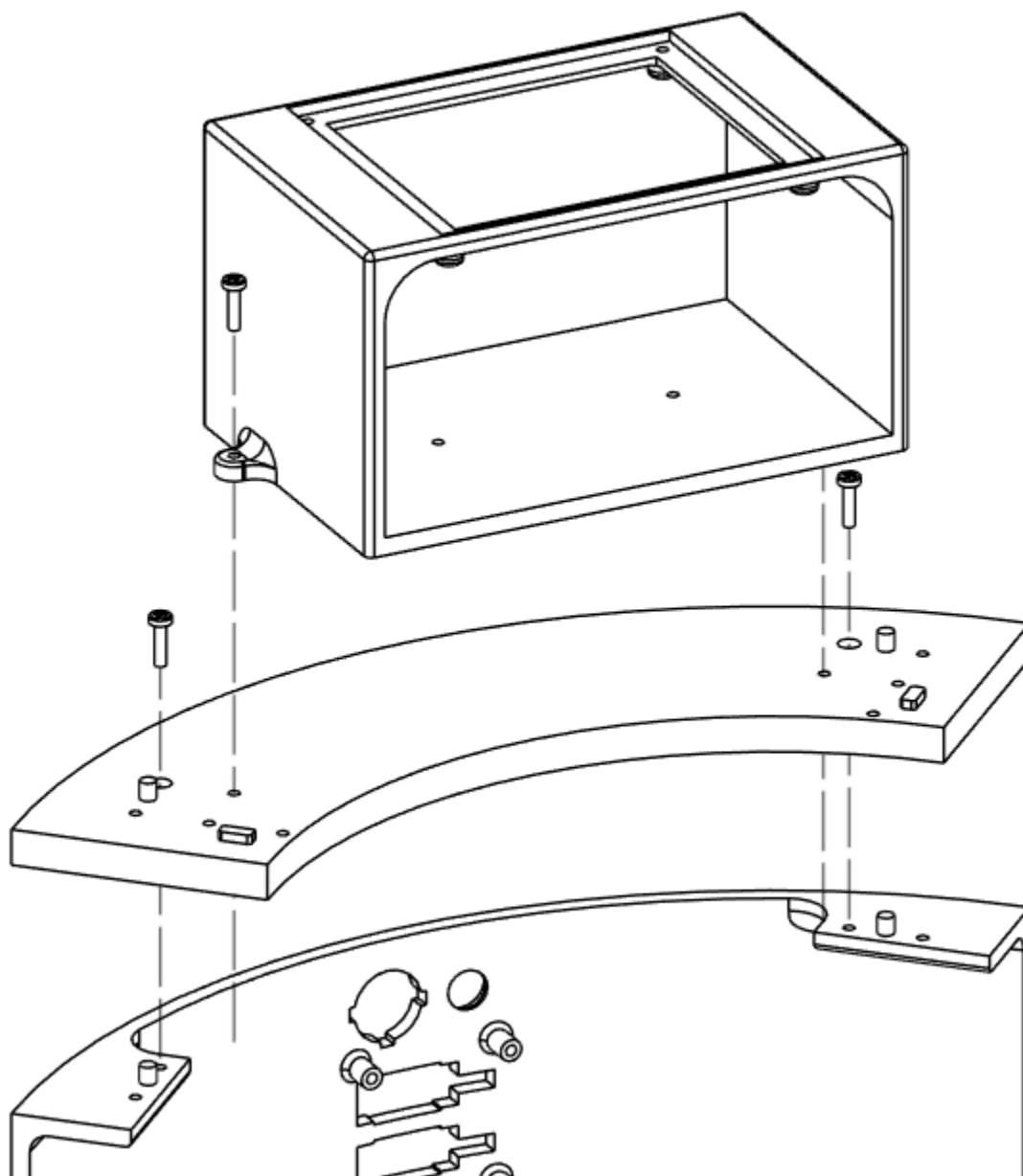


Figure 5.3 - 2nd Level Base Plate & PCB Holder Assembly

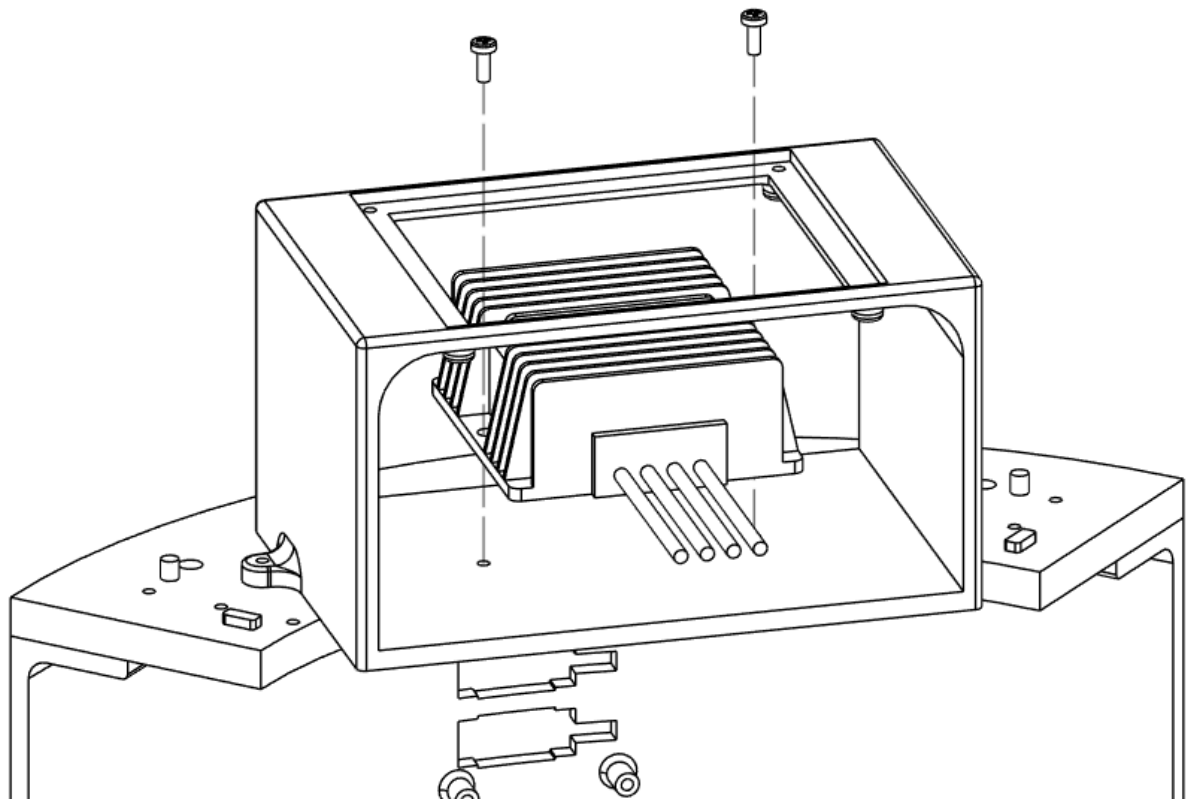


Figure 5.4 - Buck Converter Assembly

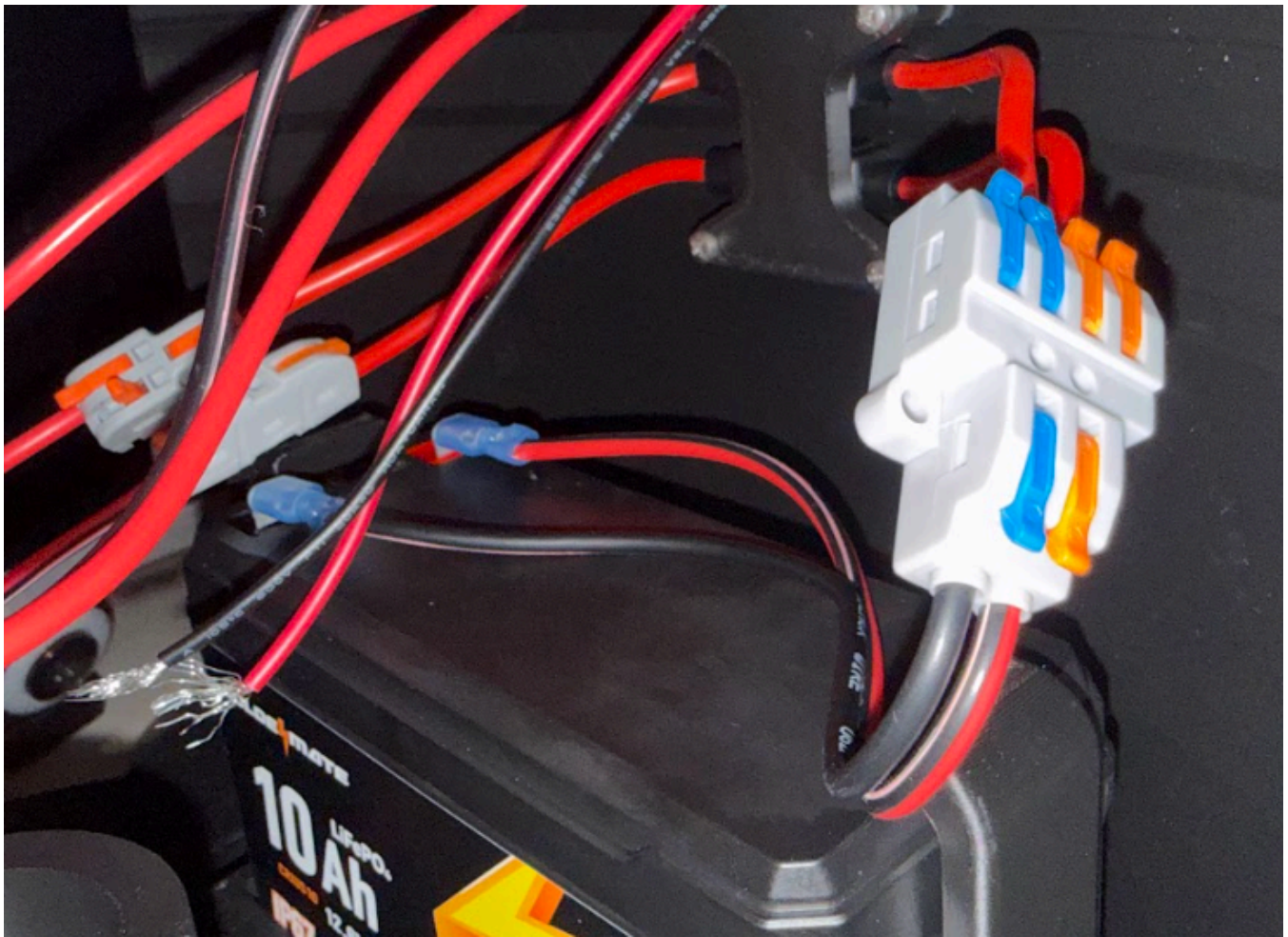


Figure 5.5 - Battery & Fuse Holder Assembly

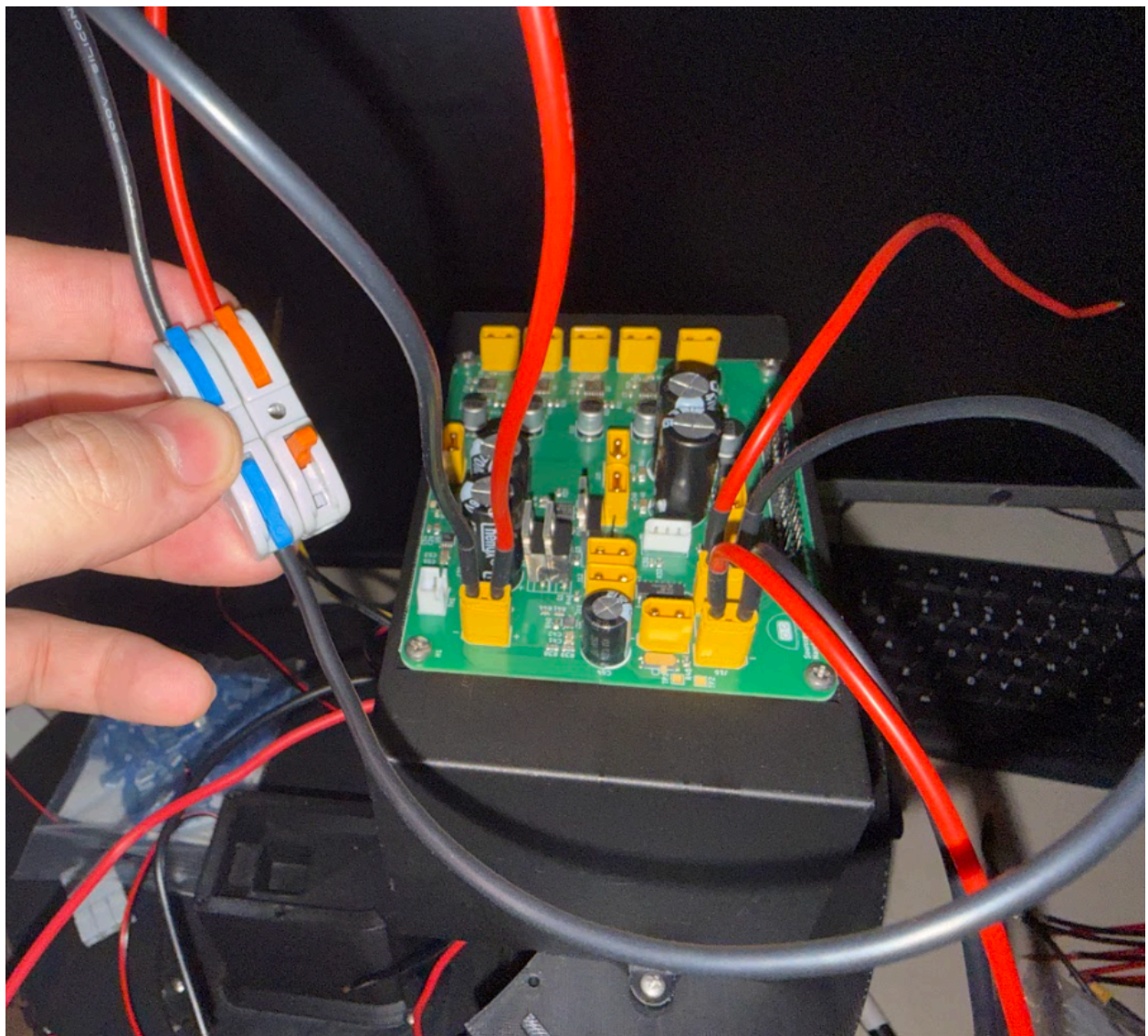


Figure 5.6 - Ground Connection Assembly

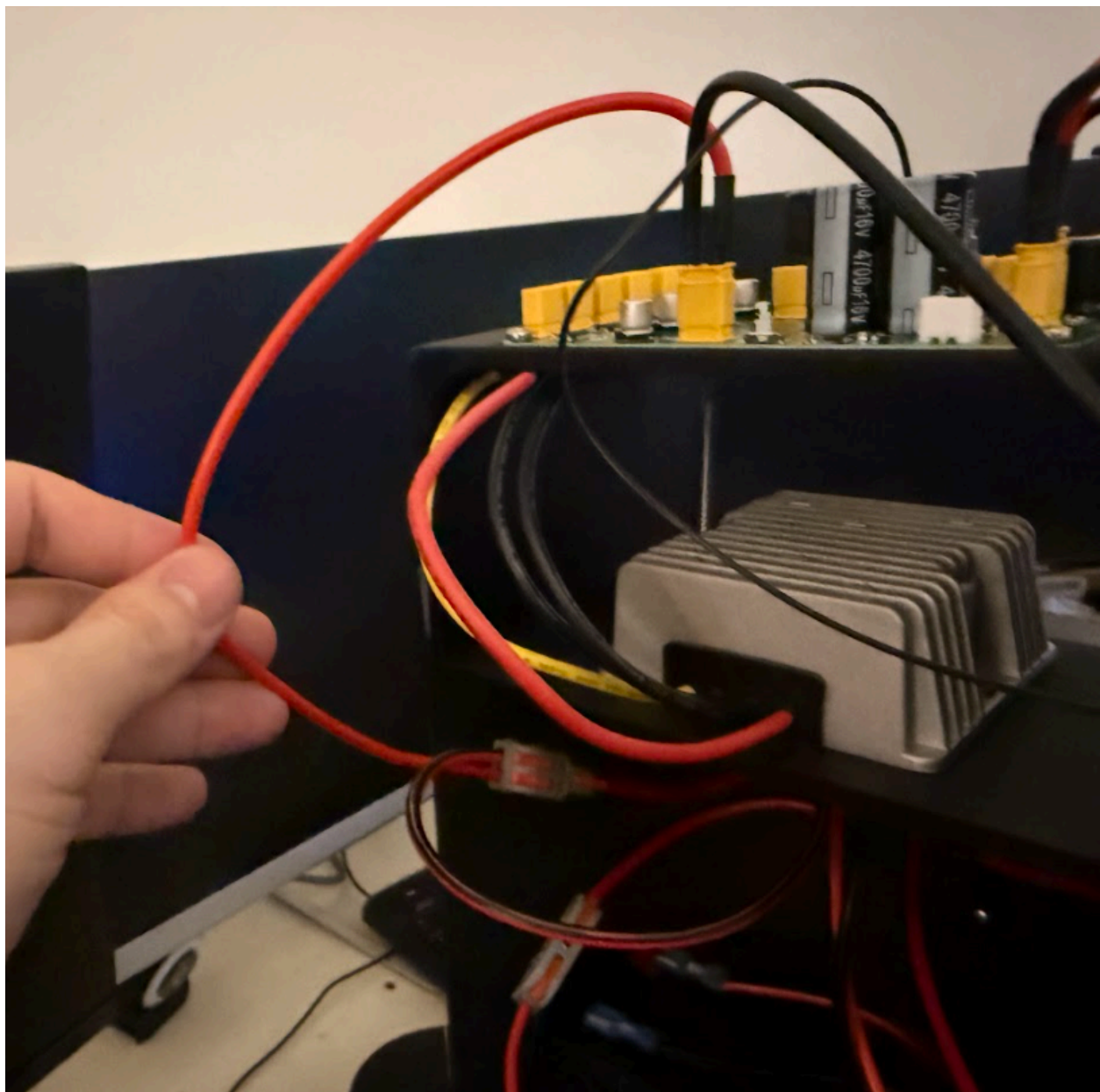


Figure 5.7 - Port Red-2 Assembly



Figure 5.8 - Thermistor Assembly

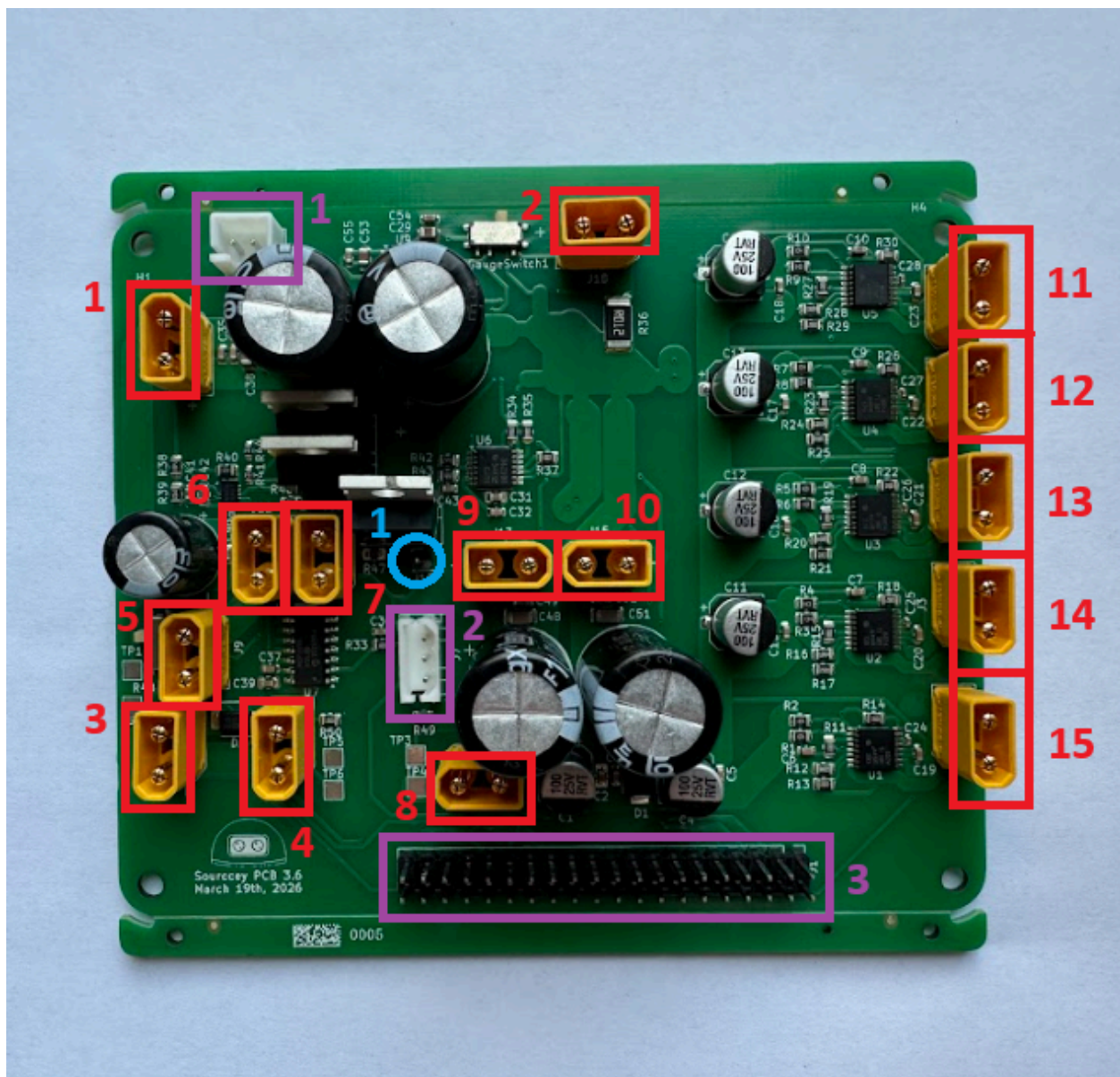


Figure 5.9 - PCB XT30 Connectors

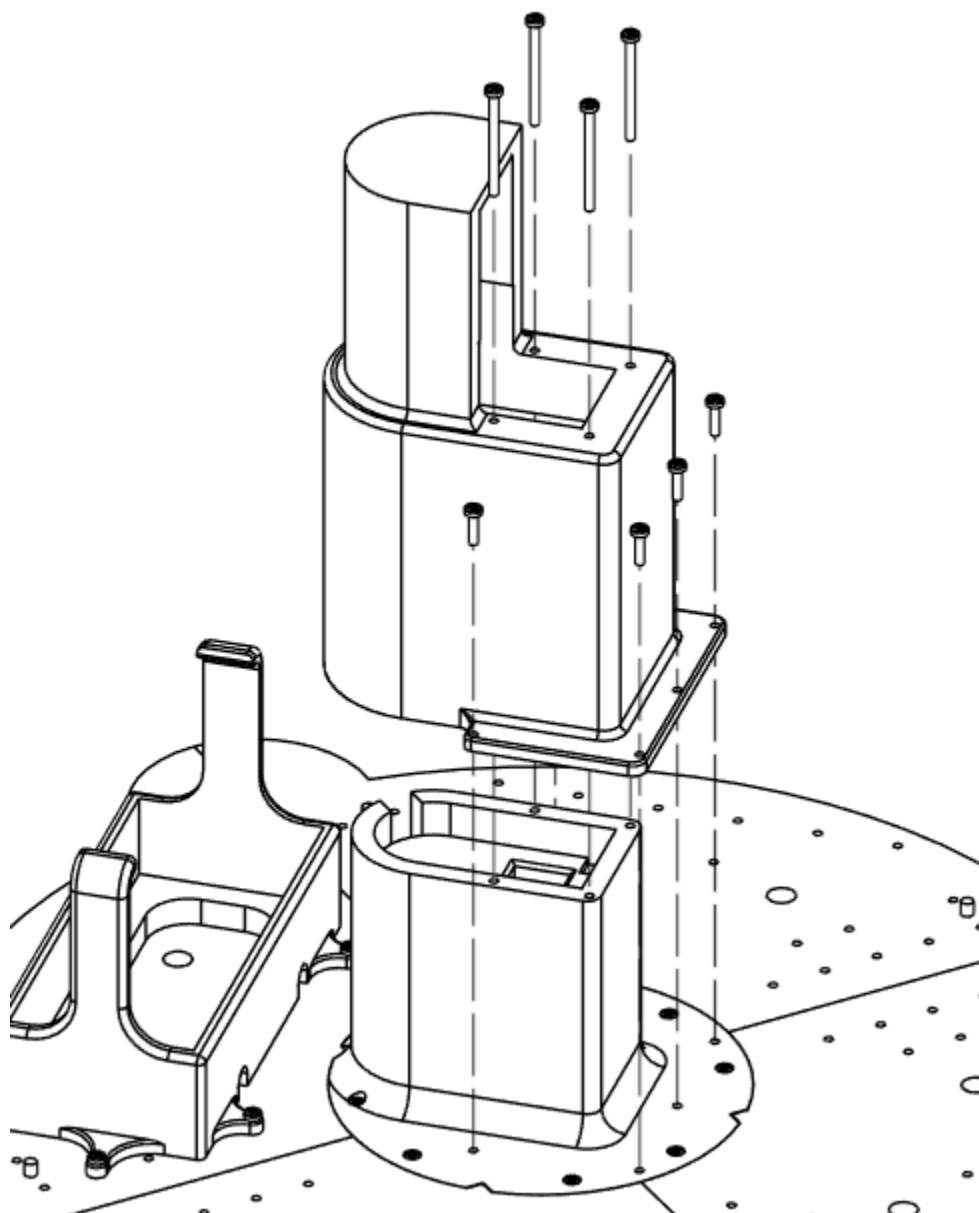


Figure 5.10 - Linear Actuator Stabilizer Assembly

Stage 6: 1st Level Body

1st Level Body

Materials

Type	Name	Count
Sub Assembly	1st Level Left Wall Assembly	1
Sub Assembly	1st Level Right Wall Assembly	1
Part	1st Level Front Wall	1
Part	2nd Level Base Plate	3
Part	Plate Connector	4
Part	Side Flares	4
Fastener	M3x12mm	22
Fastener	M3x16mm	12

Procedure

1. Fasten 1st level walls to Level 1 Base Plates as shown in Figure 6.1
 - a. (12) M3x16mm
2. Fasten (3) Level 2 Base Plates to 1st level walls as shown in Figure 6.2
 - a. (6) M3x12mm
3. Fasten (4) Plate Connectors to (4) 2nd Level Base Plates as shown in Figure 6.2
 - a. (16) M3x12mm
4. Install (4) Side Flares to 1st level walls as shown in Figure 6.3

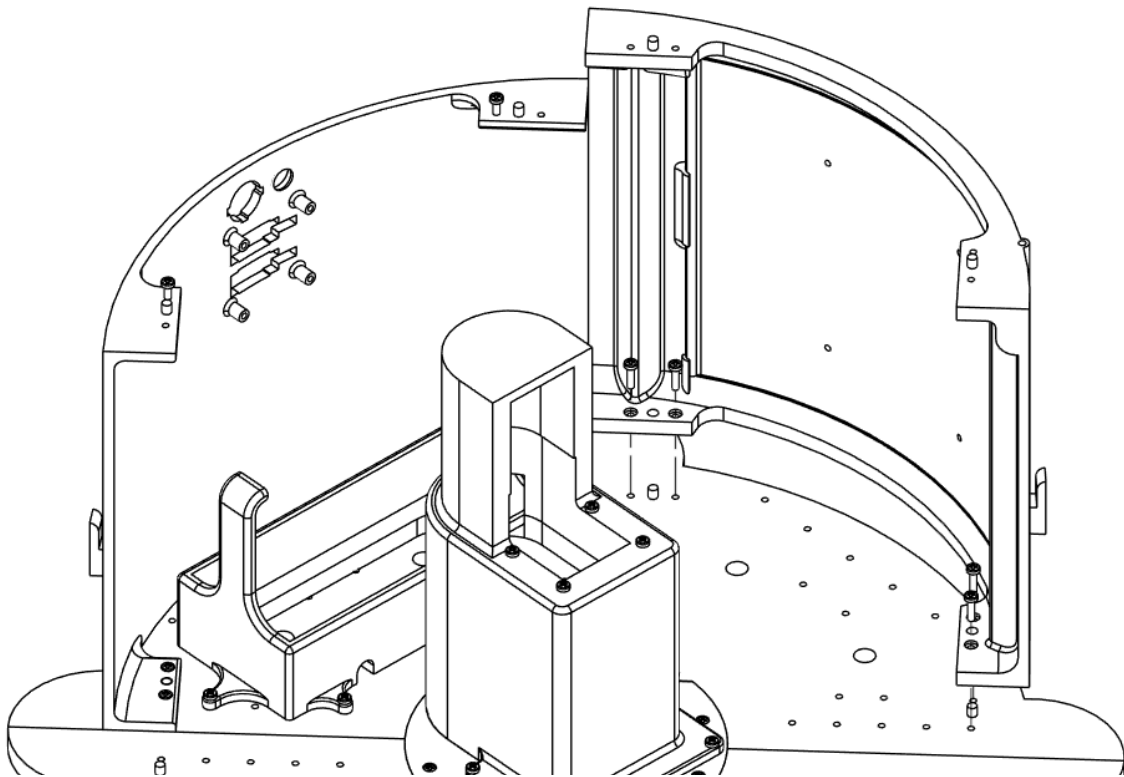


Figure 6.1 - 1st Level Wall Assembly

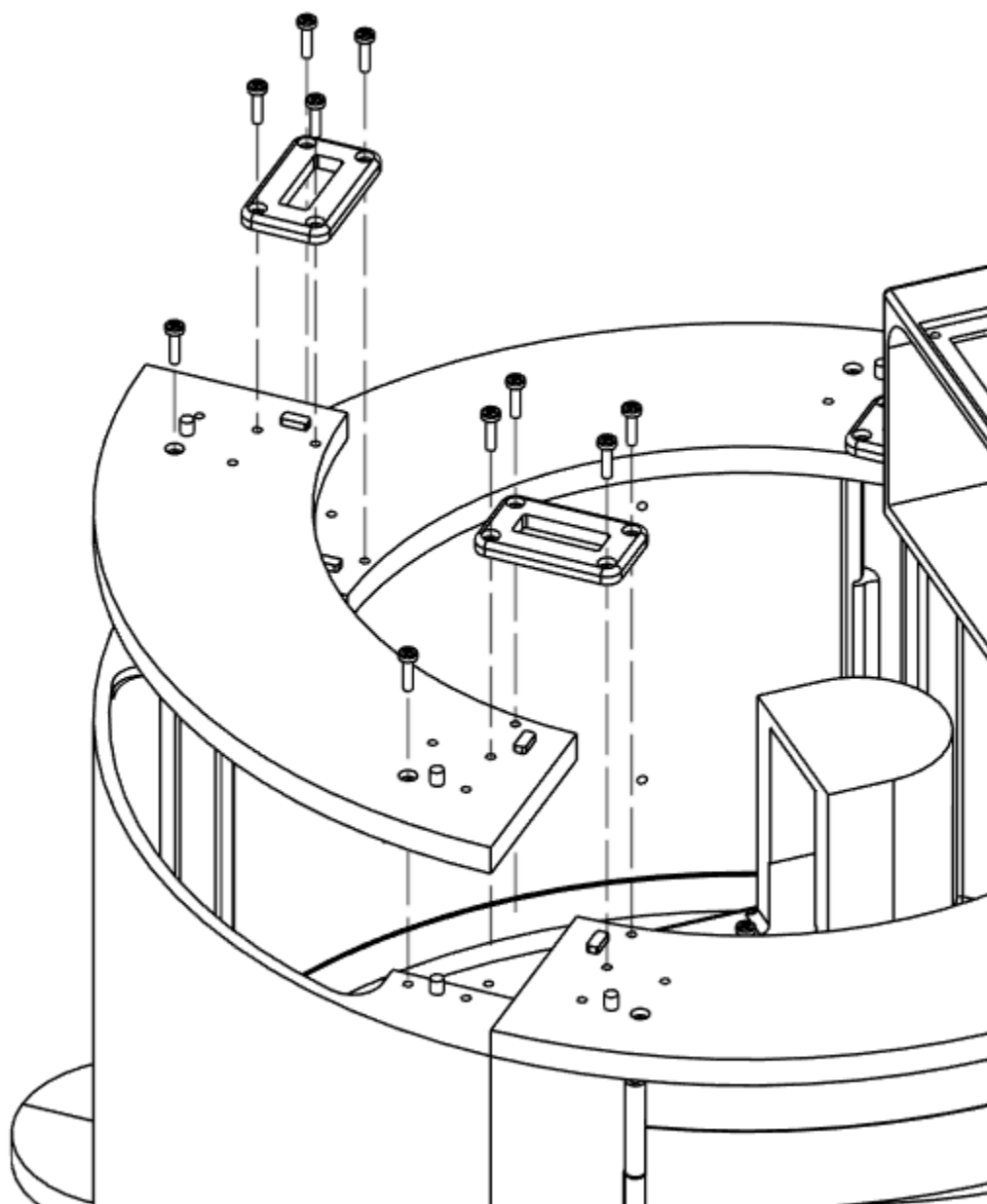


Figure 6.2 - 2nd Level Base Plates and Plate Connectors Assembly

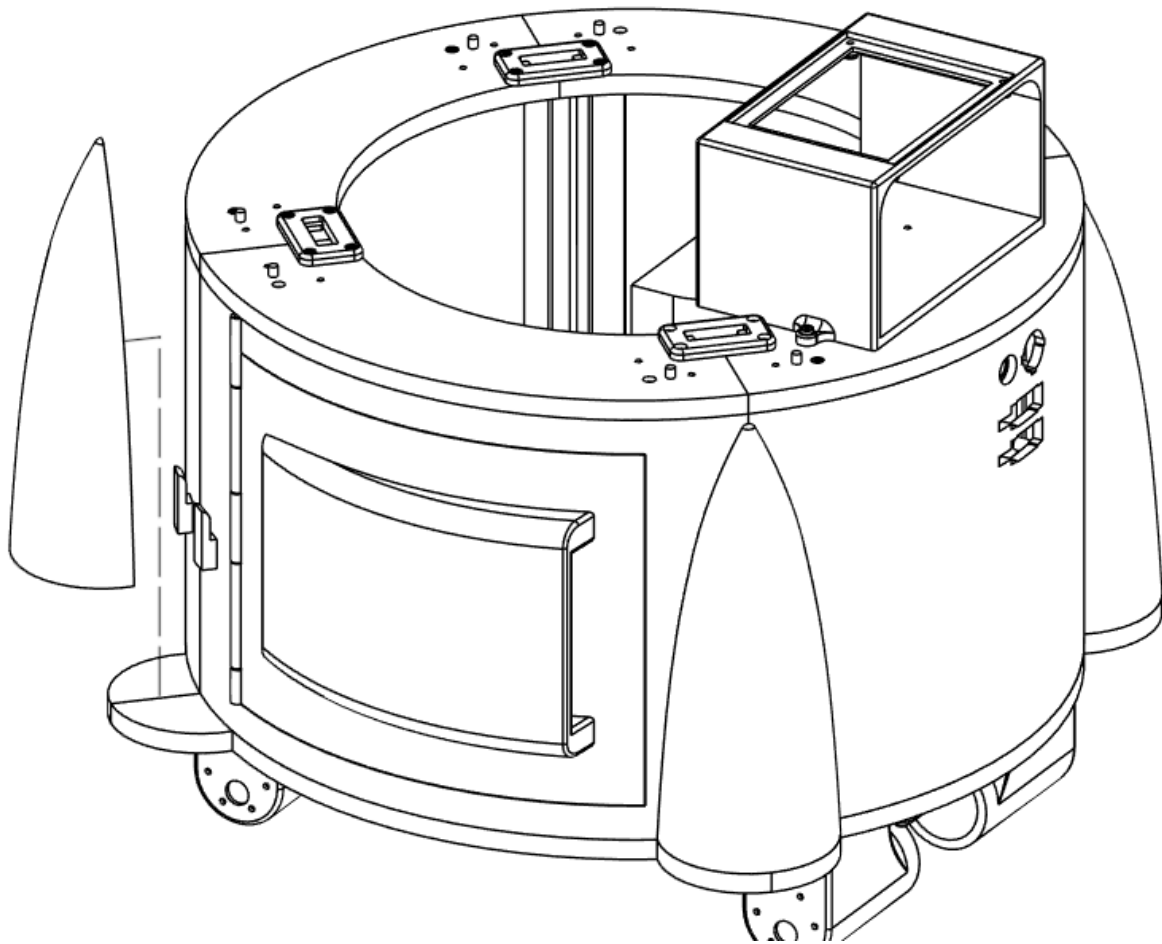


Figure 6.3 - Side Flares Assembly

Stage 7: 2nd Level Body

2nd Level Body

Materials

Type	Name	Count
Electronics	Motor Driver Board	2
Sub Assembly	2nd Level Rear Wall Assembly	1
Sub Assembly	Potentiometer Assembly	1
Sub Assembly	Raspberry Pi Assembly	1
Part	2nd Level Left Wall	1
Part	2nd Level Right Wall	1
Part	2nd Level Front Wall	1
Part	Circle Accessory Base	2
Part	3rd Level Rear Base Plate	1
Part	3rd Level Left Base Plate	1
Part	3rd Level Right Base Plate	1
Part	3rd Level Front Base Plate	1
Part	Motor Driver Holders	2
Part	Anti-Deflection Shaft	1
Fastener	M2.5x8mm	8
Fastener	M3x12mm	28
Fastener	M3x16mm	8
Fastener	M3x20mm	4
Fastener	M3x8mm Black Oxide	8

Procedure

1. Fasten 2nd Level Walls to 2nd Level Base Plates as shown in Figure 7.1
 - a. (8) M3x12mm
2. Fasten 3rd Level Base Plates to 2nd Level Walls as shown in Figure 7.2
 - a. (8) M3x12mm

- 3. Fasten (2) Circle Accessory Bases to 2nd Level Walls as shown in Figure 7.3**
 - a. (8) M3x8mm Black Oxide**
- 4. Install Motor Drivers**
 - a. Fasten (2) Motor Driver Holders to 3rd Level Base Plates as shown in Figure 7.4**
 - i. (8) M3x12mm**
 - b. Fasten (2) Motor Driver Boards to the Motor Driver Holders**
 - i. (8) M2.5x8mm**
 - c. Connect power cables (barrel jacks) to Motor Driver Boards**
- 5. Fasten Potentiometer Assembly to 3rd Level Front Base Plate as shown in Figure 7.4**
 - a. (4) M3x20mm**
 - b. Plug potentiometer wire into port on PCB, threading in between the 2 arm braces**
- 6. Fasten Anti-Deflection Shaft to Level 3 Rear Plate as shown in Figure 7.5**
 - a. (8) M3x16mm**
- 7. Fasten Raspberry Pi Assembly to 3rd Level Rear Base Plate as shown in Figure 7.6**
 - a. (4) M3x12mm**
 - b. Connect GPIO Ribbon Cable from Raspberry Pi Assembly to the PCB**
- 8. Connect all USB Cables to Raspberry Pi as shown in Figure 7.7**
 - a. Port 1: Speakers**
 - b. Port 2: Right Arm Motor Driver**
 - c. Port 3: Left Arm Motor Driver**
 - d. Port 4: Right Eye Camera**
 - e. Port 5: Left Eye Camera**
 - f. Port 6: Right Arm Camera**
 - g. Port 7: Left Arm Camera**

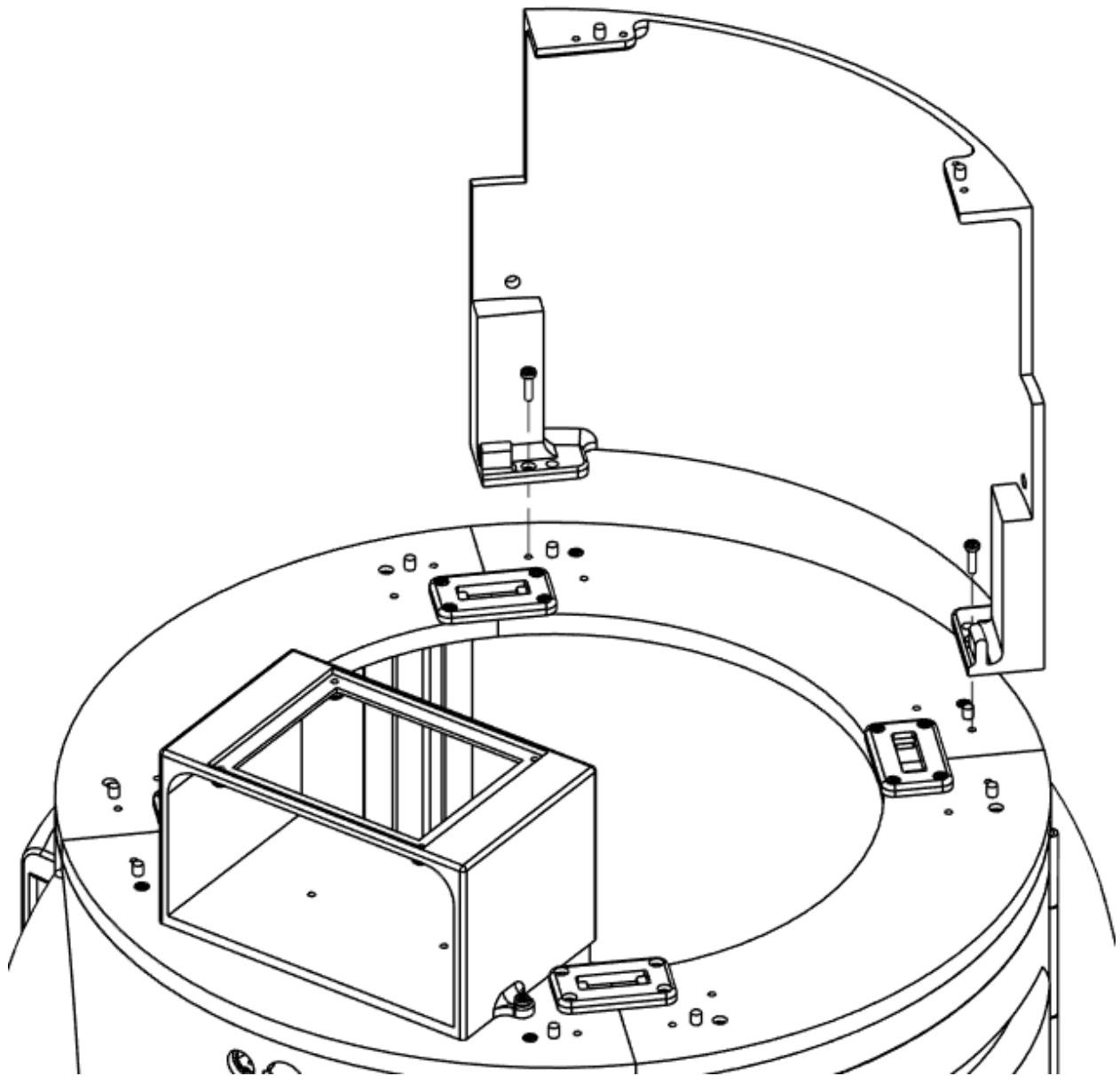


Figure 7.1 - 2nd Level Walls Assembly

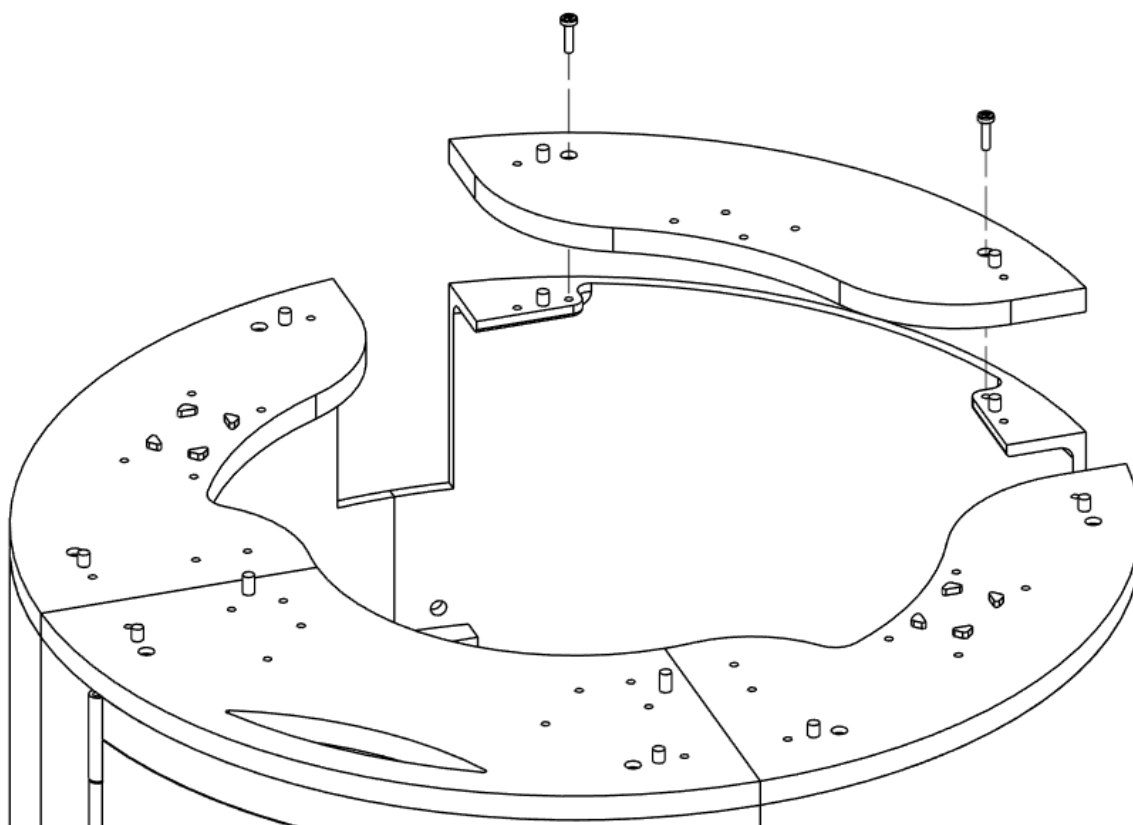


Figure 7.2 - 3rd Level Base Plate Assembly

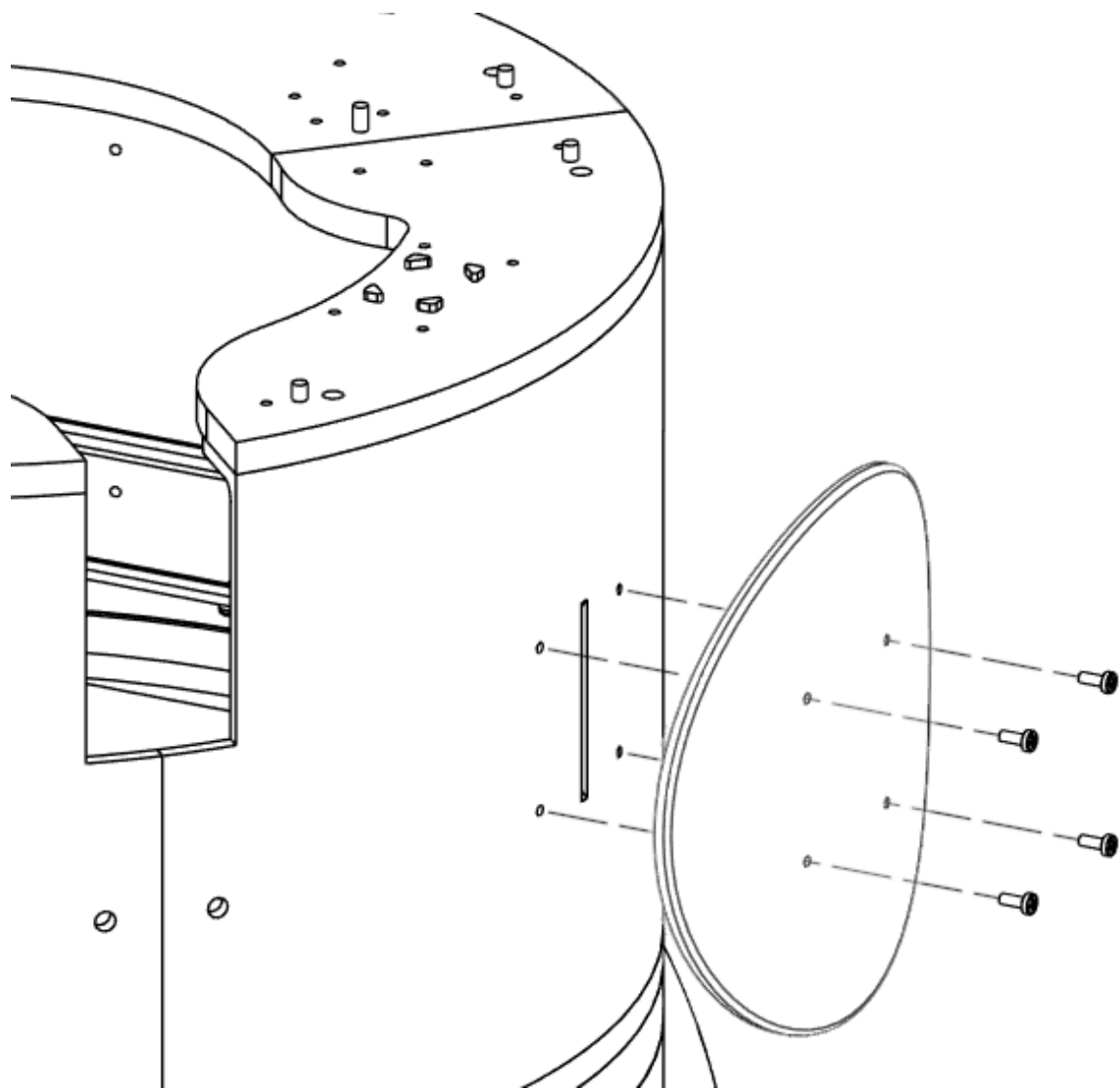


Figure 7.3 - 2nd Level Circle Accessory Base Assembly

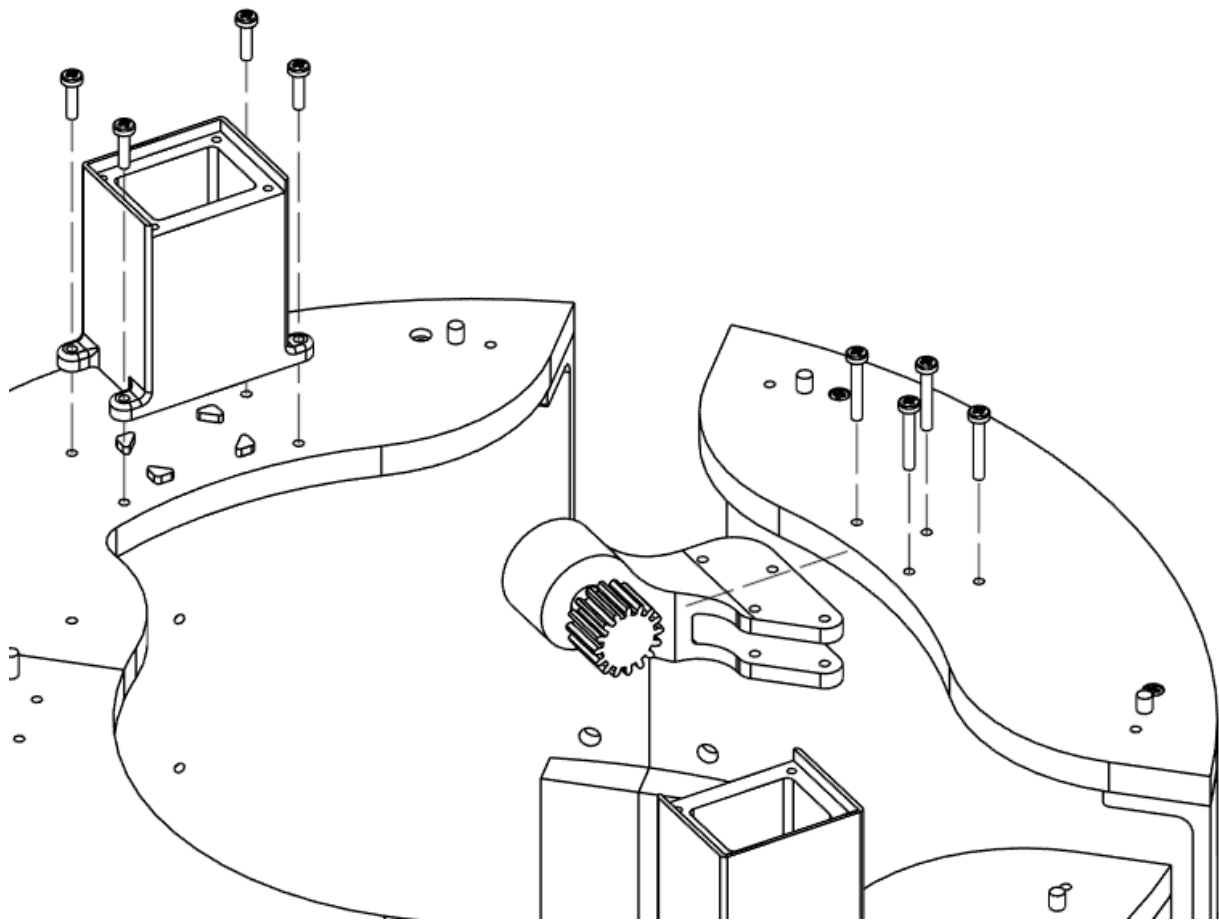


Figure 7.4 - Motor Driver & Potentiometer Assembly

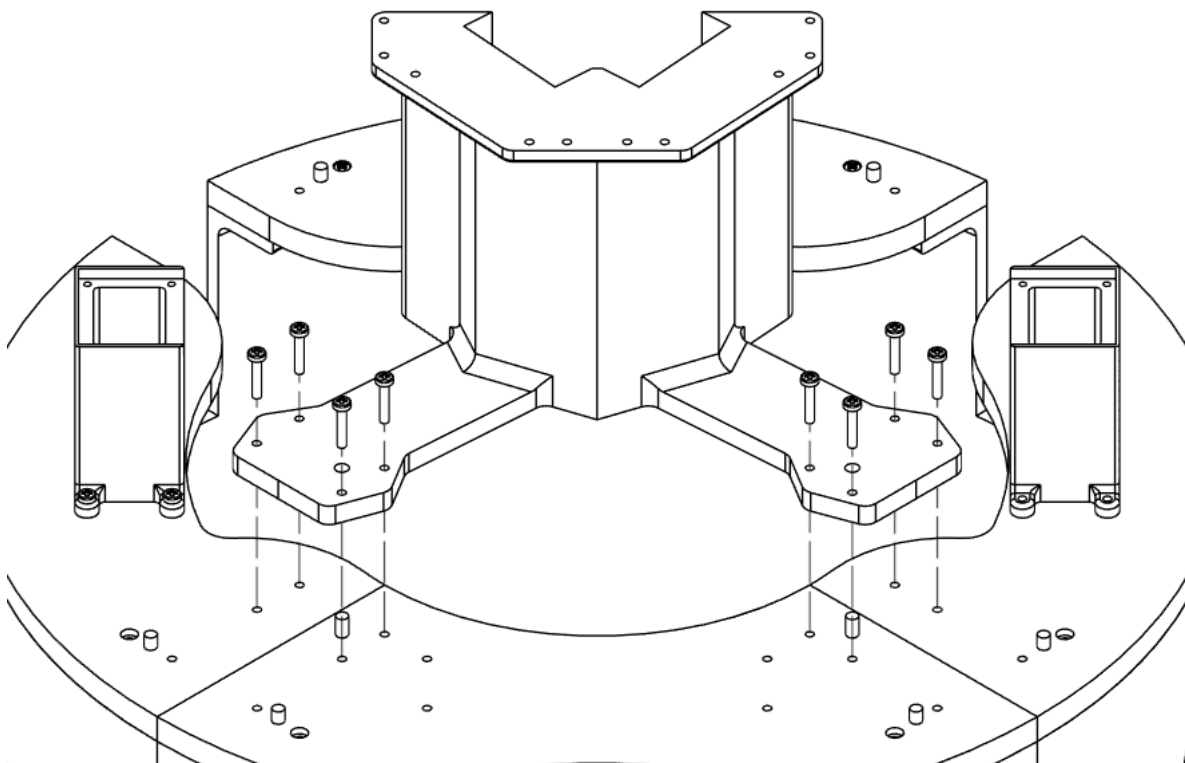


Figure 7.5 - Lower Anti-Deflection Shaft Assembly

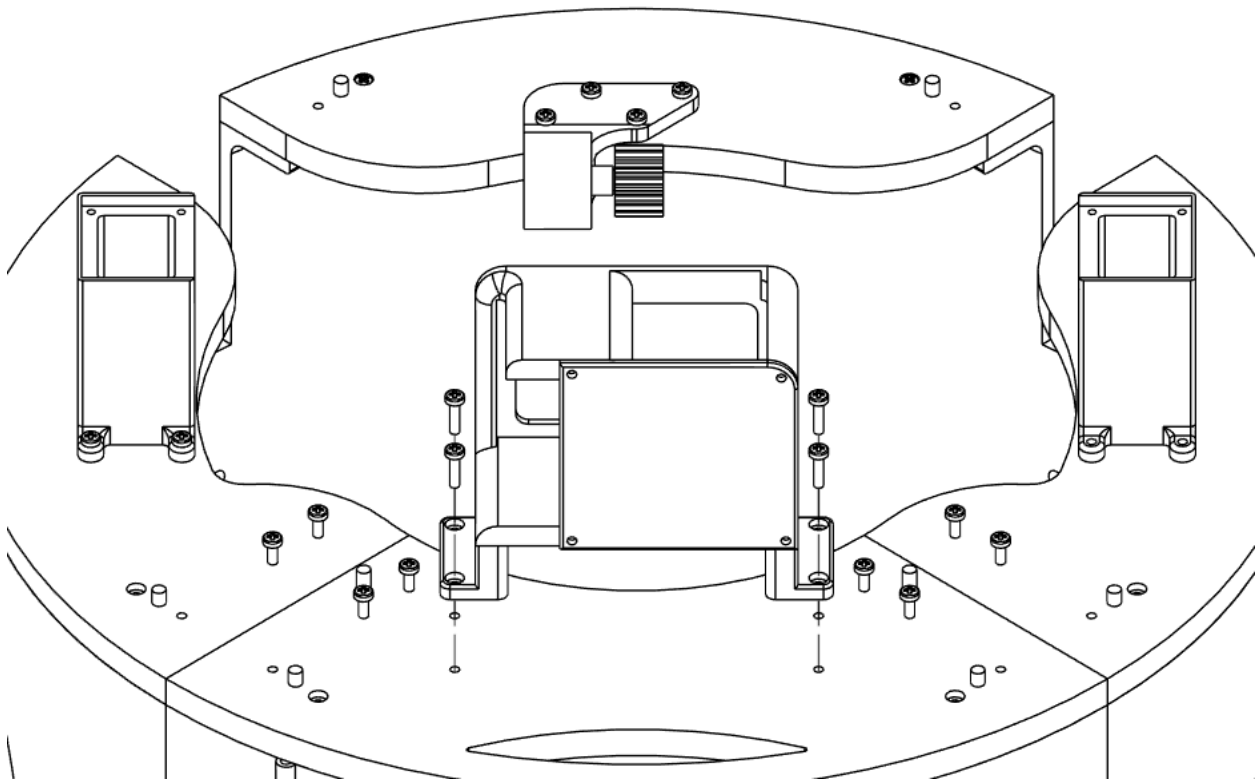


Figure 7.6 - Raspberry Pi & Level 3 Base Plate Assembly

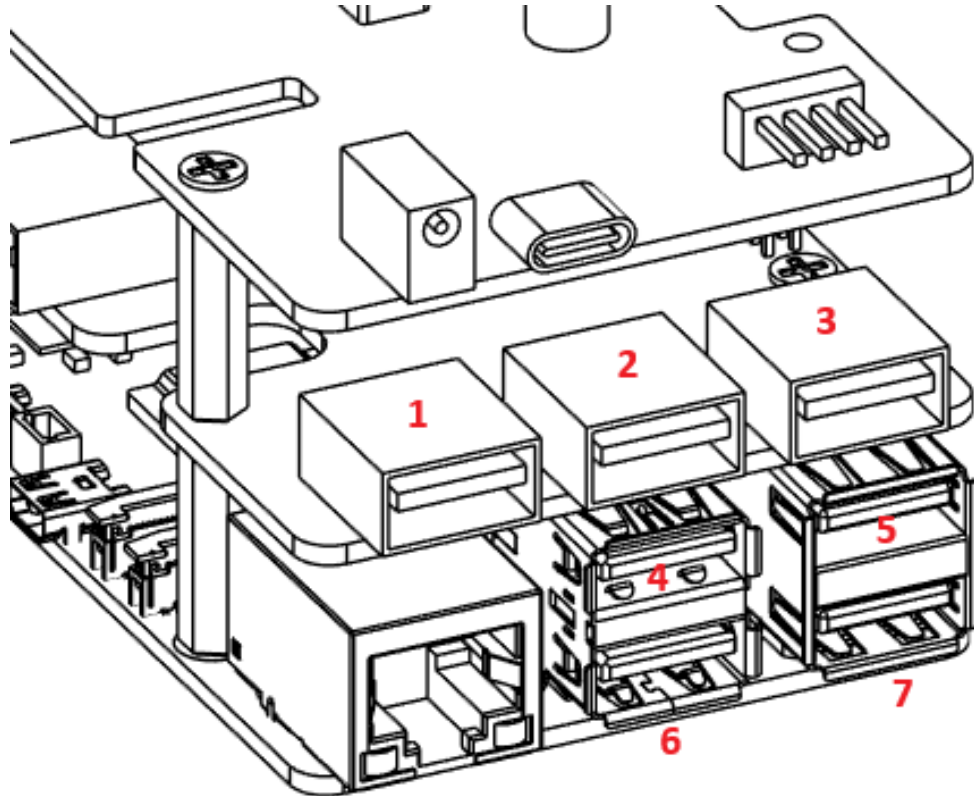


Figure 7.7 - Raspberry Pi USB Ports

Stage 8: 3rd Level Body

3rd Level Body

Materials

Type	Name	Count
Electronics	Speaker Set	1
Sub Assembly	3rd Level Rear Wall Assembly	1
Part	3rd Level Left Wall	1
Part	3rd Level Right Wall	1
Part	3rd Level Front Wall	1
Part	Rectangle Design	3
Part	Breastplate	1
Part	Carry Handle	2
Part	Circle Accessory Base	2
Part	Dome Level Rear Plate	1
Part	Dome Level Left Plate	1
Part	Dome Level Right Plate	1
Part	Anti-Deflection Shaft	1
Fastener	M3x8mm	8
Fastener	M3x10mm	10
Fastener	M3x12mm	26
Fastener	M3x16mm	8
Fastener	M3x16mm Black Oxide	8
Fastener	M3 Washers	8

Procedure

1. Prepare 3rd Level Front Wall

- a. Fasten (3) Rectangle Designs and Breastplate to 3rd Level Front Wall as shown in Figure 8.1
 - i. (8) M3x8mm

2. Install speakers on 3rd Level Walls as shown in Figure 8.2

- a. (8) M3x12mm
- b. (8) M3 Washers
- 3. Fasten 3rd Level Walls to 3rd Level Base Plates as shown in Figure 8.3
 - a. (8) M3x12mm
 - b. Plug Speakers into Raspberry Pi Assembly as shown in Figure 7.7
- 4. Fasten Carry Handle and Circle Accessory Base on 3rd Level Left Wall as shown in Figure 8.4
 - a. (4) M3x16mm Black Oxide
 - b. Repeat for opposite side of robot
- 5. Fasten Dome Level Base Plates on 3rd Level Walls as shown in Figure 8.5
 - a. (10) M3x12mm
- 6. Fasten Anti-Deflection Shaft to lower Anti-Deflection Shaft and Dome Level Rear Base Plate as shown in Figure 8.6
 - a. (10) M3x10mm
 - b. (8) M3x16mm

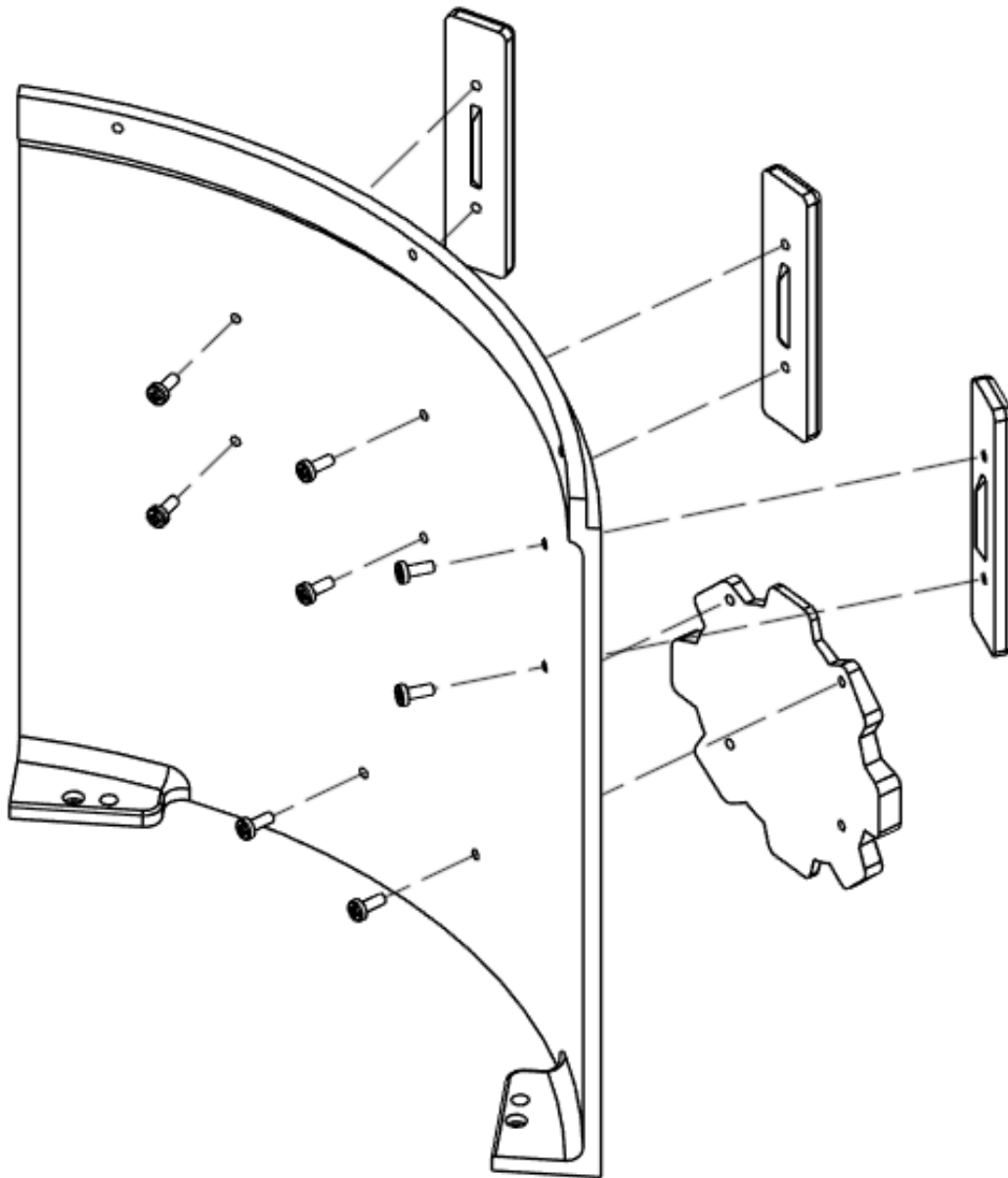


Figure 8.1 - 3rd Level Front Wall Assembly

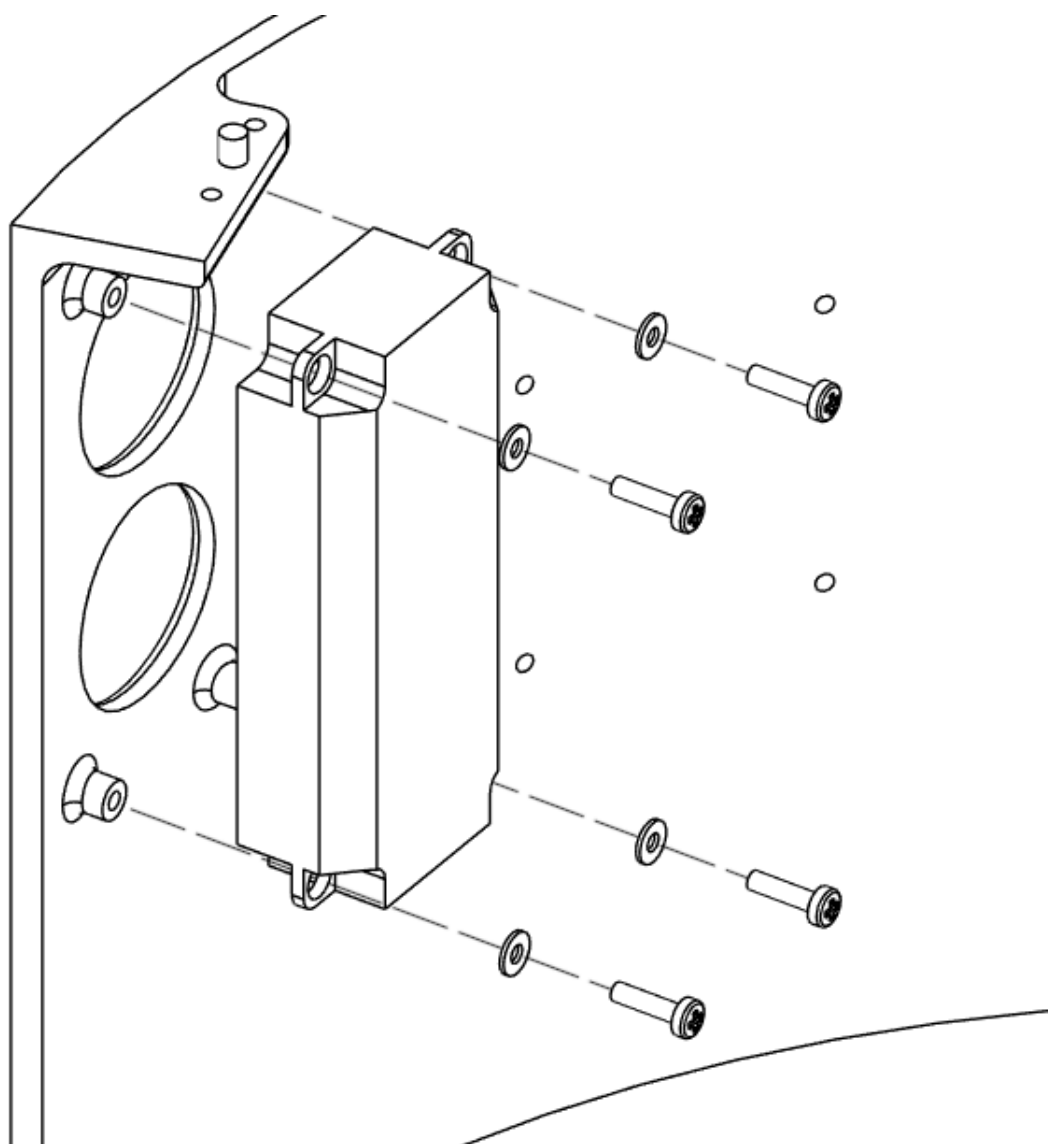


Figure 8.2 - Speaker Assembly

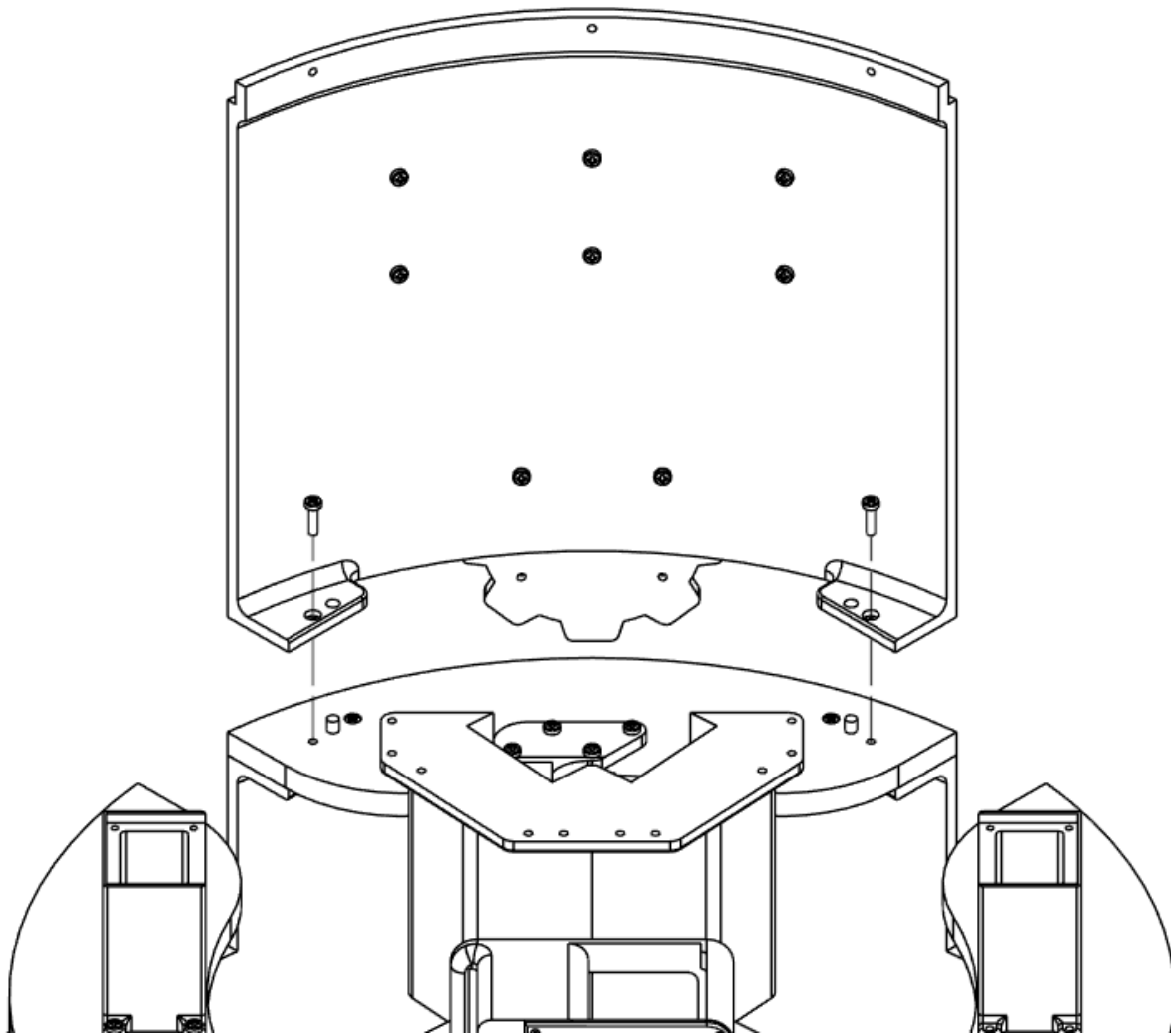


Figure 8.3 - 3rd Level Walls Assembly

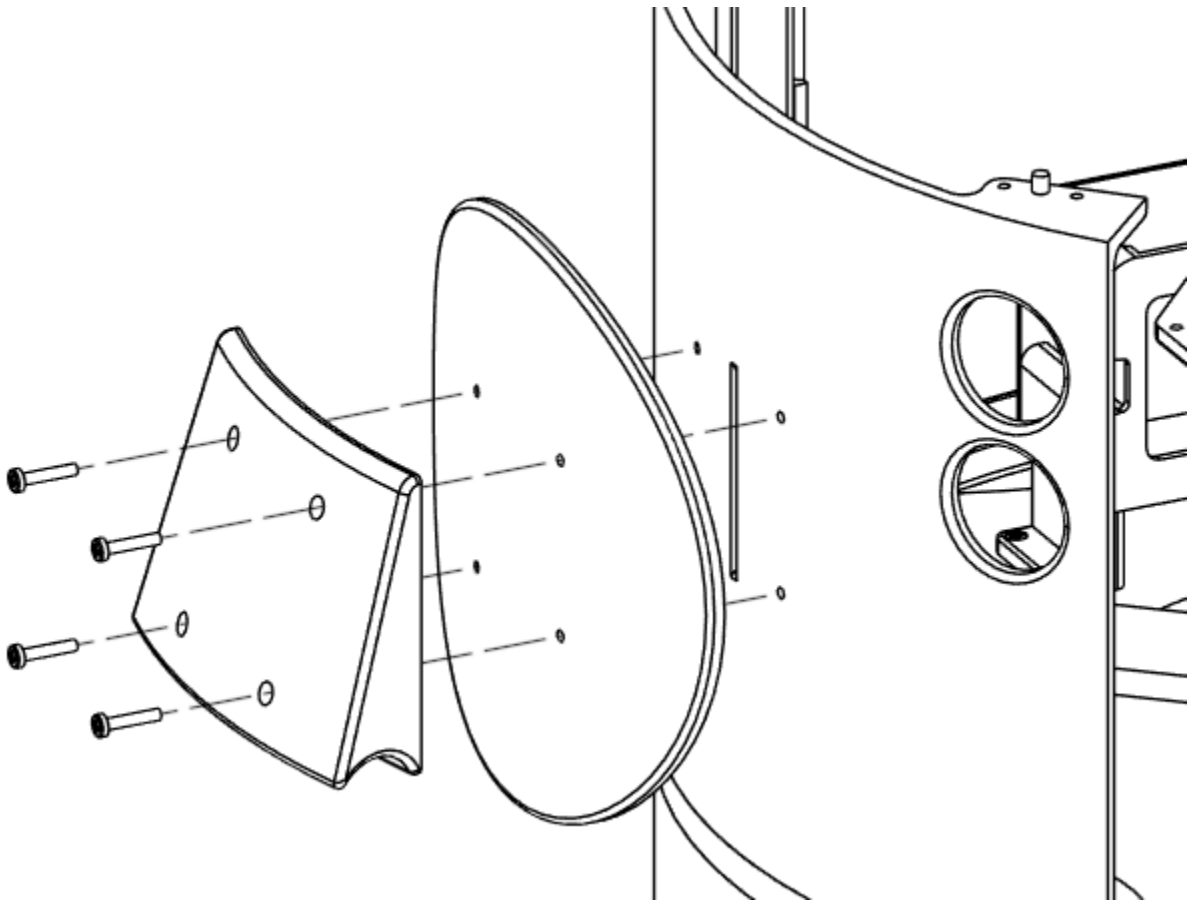


Figure 8.4 - Carry Handle Assembly

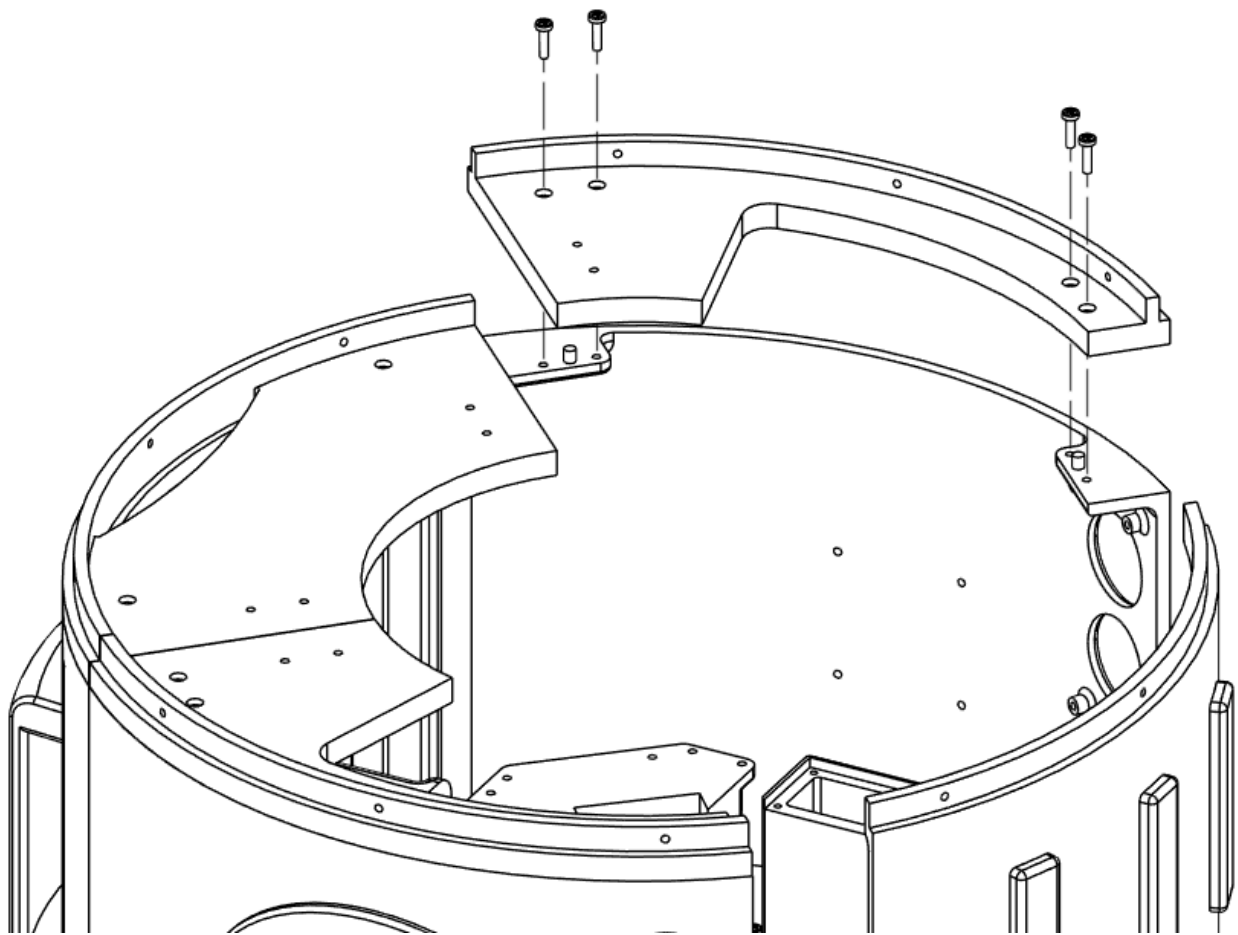


Figure 8.5 - Dome Level Base Plates Assembly

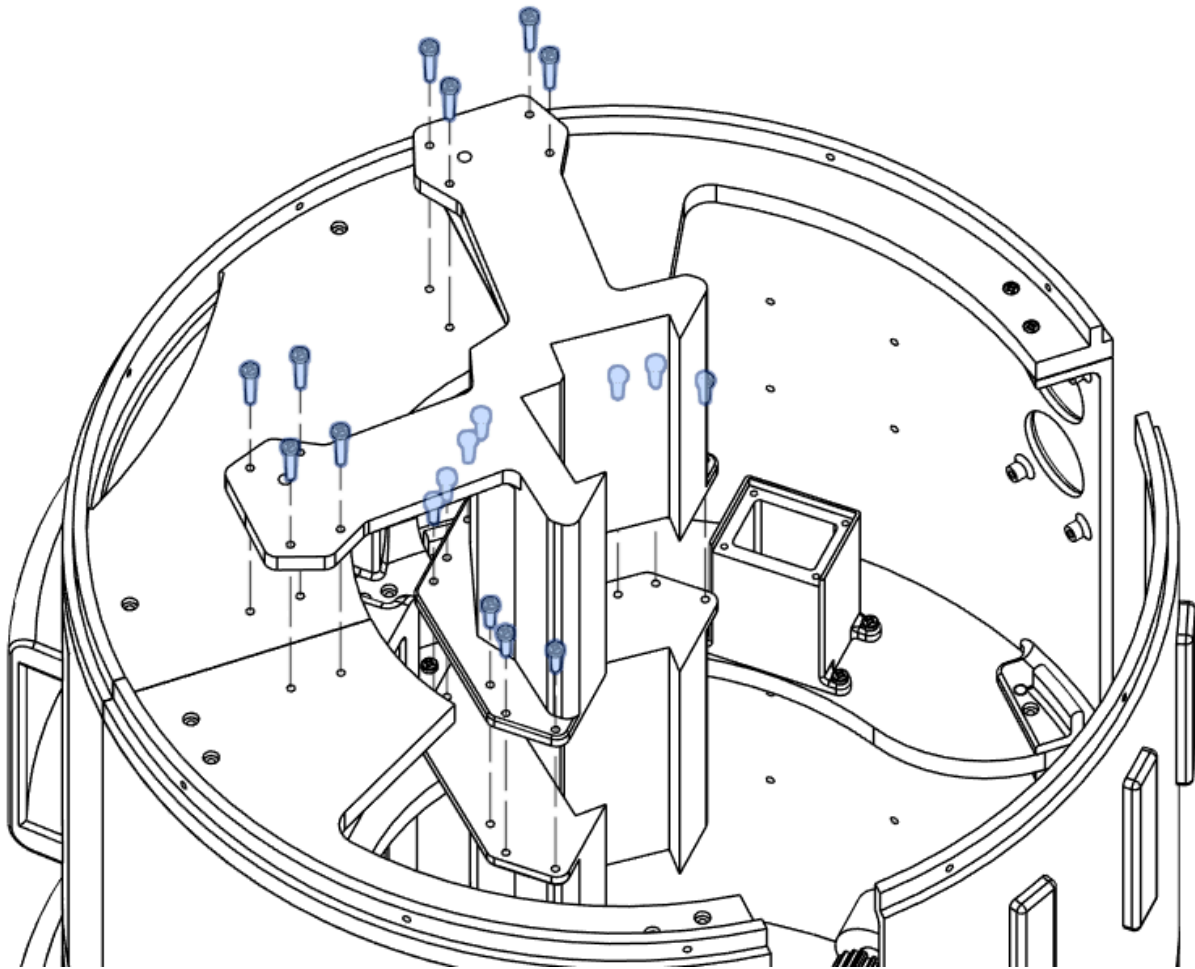


Figure 8.6- Upper Anti-Deflection Shaft Assembly

Stage 9: Linear Actuator

Linear Actuator

Materials

Type	Name	Count
Electronics	Arm Motor Cable - 400mm	2
Electronics	Camera Cable	2
Part	Linear Actuator Core - Upper	1
Part	Linear Actuator Core - Lower	1
Part	Linear Actuator Connector	4
Part	Rack - Upper	1
Part	Rack - Lower	1
Part	Linear Actuator Top Brace	2
Part	Linear Actuator Hypotenuse Brace	2
Part	Grommet Strip	2
Fastener	Linear Actuator Pin (included with Linear Actuator)	1
Fastener	Grommet Strip Screw (included with Grommet Strip kit)	4
Fastener	Grommet Strip Nut (included with Grommet Strip kit)	4
Fastener	M3x8mm	32
Fastener	M3x10mm	12
Fastener	M3x20mm	10

Procedure

1. Fasten Core Upper, Core Lower, and Connector as shown in Figure 9.1
 - a. (32) M3x8mm
2. Fasten Rack Upper to Rack Lower as shown in Figure 9.2
 - a. (2) M3x20mm
3. Fasten Rack to Linear Actuator Core as shown in Figure 9.3
 - a. (4) M3x20mm
4. Fasten Top Brace, Hypotenuse Brace, and Grommet Strip as shown in Figure 9.4

- a. (2) M3x20mm
 - b. Repeat this step for other Linear Actuator Arm
 - c. **NOTE:** Ensure that Grommet Strips have the flat side oriented outwards
- 5. Fasten Linear Actuator Arm to Core as shown in Figure 9.5
 - a. (6) M3x10mm
 - b. Repeat this step for other Linear Actuator Arm
- 6. Insert Linear Actuator Assembly into Anti-Deflection Shaft as shown in Figure 9.6
 - a. **NOTE:** Ensure that Linear Actuator is fully extended, and fits into the hole at the top of the Core
- 7. Fasten Core to Linear Actuator as shown in Figure 9.7
 - a. (1) Linear Actuator Pin
 - b. **NOTE:** This is a tight fit, a small mallet is recommended
- 8. Fasten Grommet Strip to 2nd Level Walls as shown in Figure 9.8
 - a. (2) Grommet Strip screw
 - b. (2) Grommet Strip nut
 - c. Repeat this step for other Grommet Strip
- 9. Feed Arm Motor Cable and Arm Camera Cable through channel in Top Brace
 - a. Repeat this step for other Linear Actuator Arm

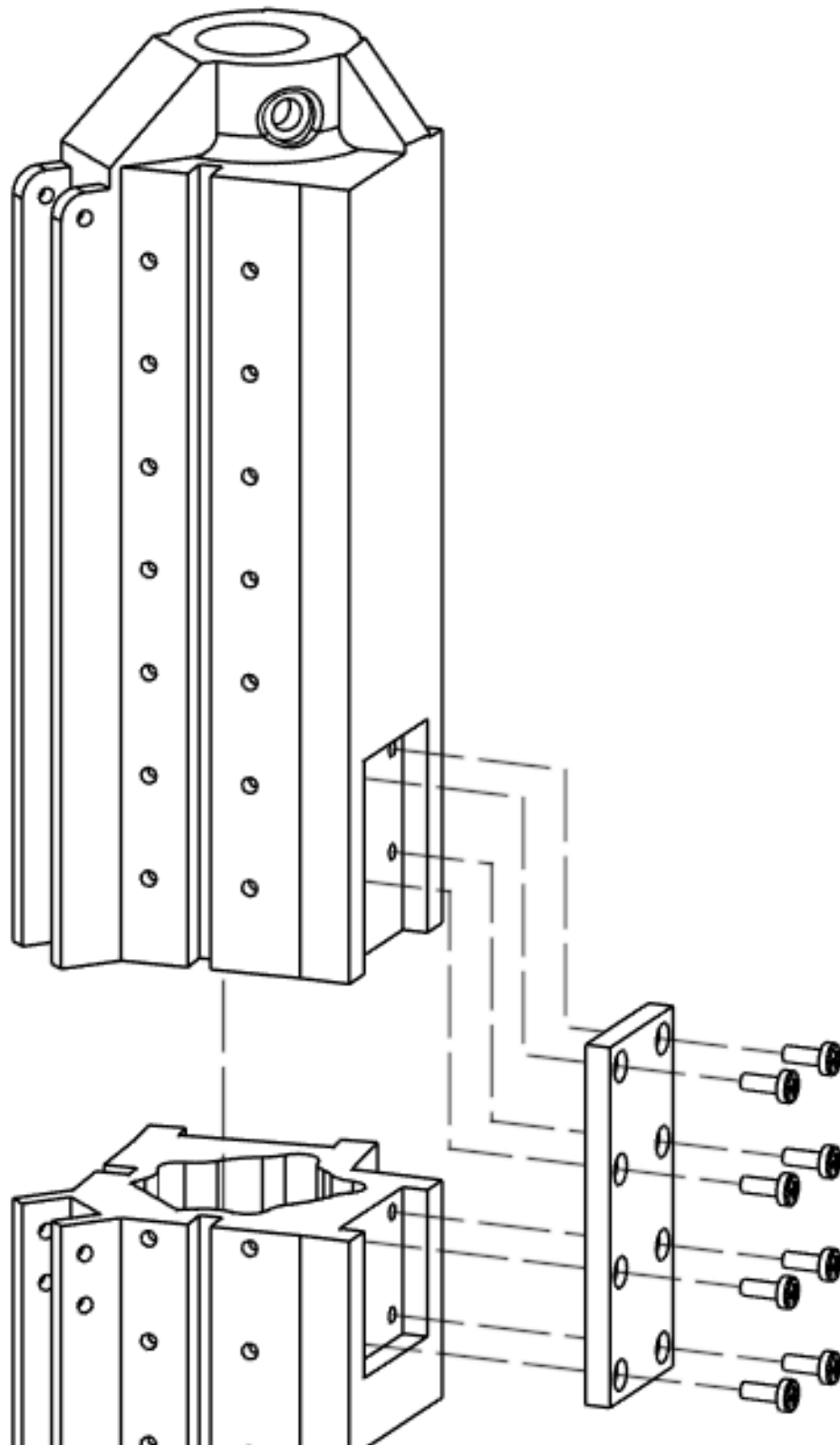


Figure 9.1 - Linear Actuator Core Assembly

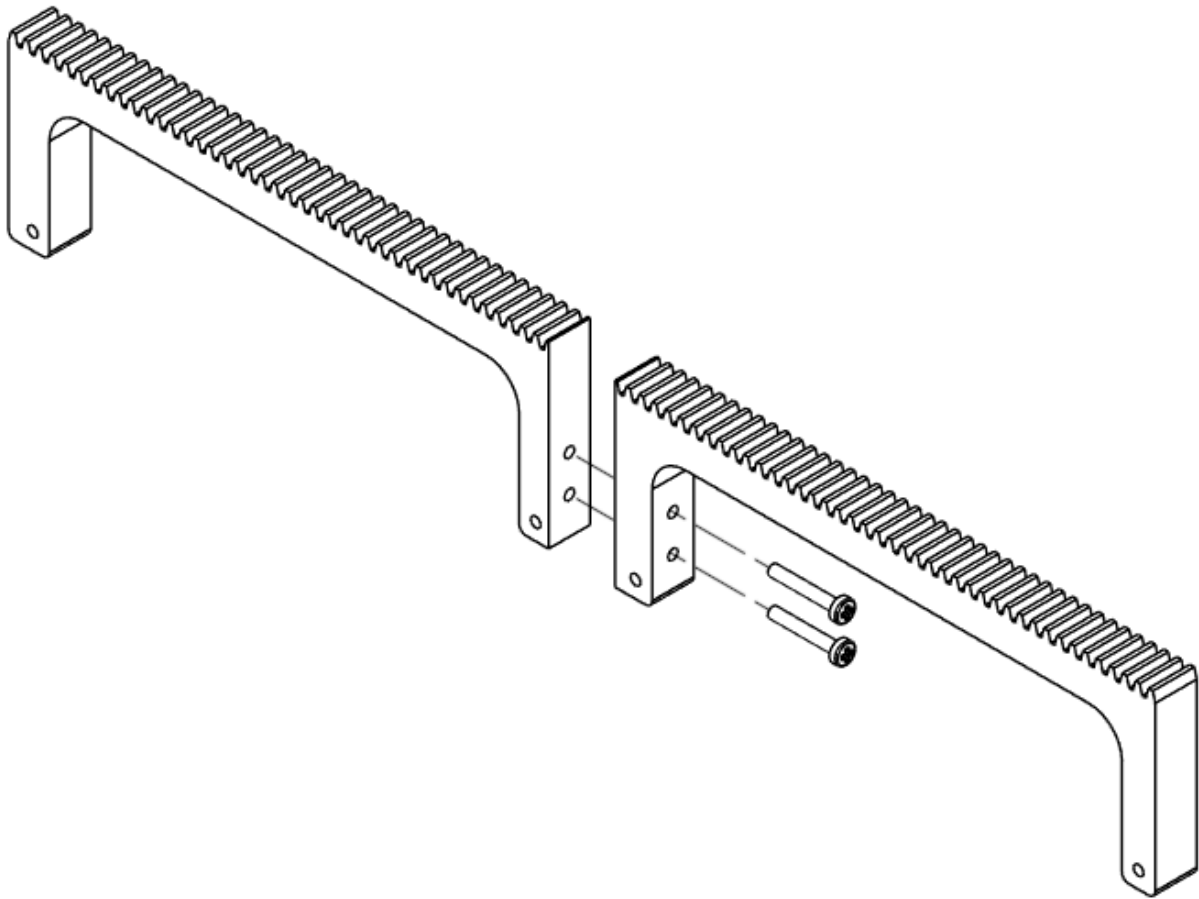


Figure 9.2 - Rack Assembly

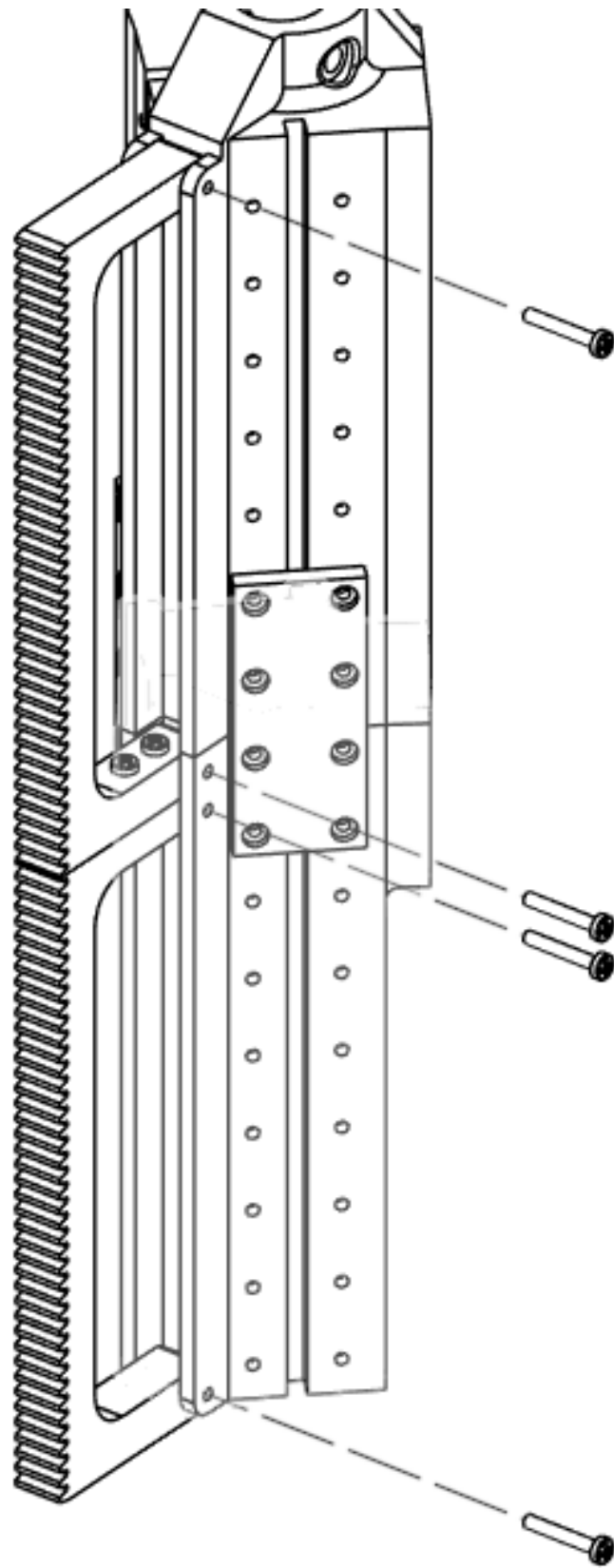


Figure 9.3 - Rack & Linear Actuator Assembly

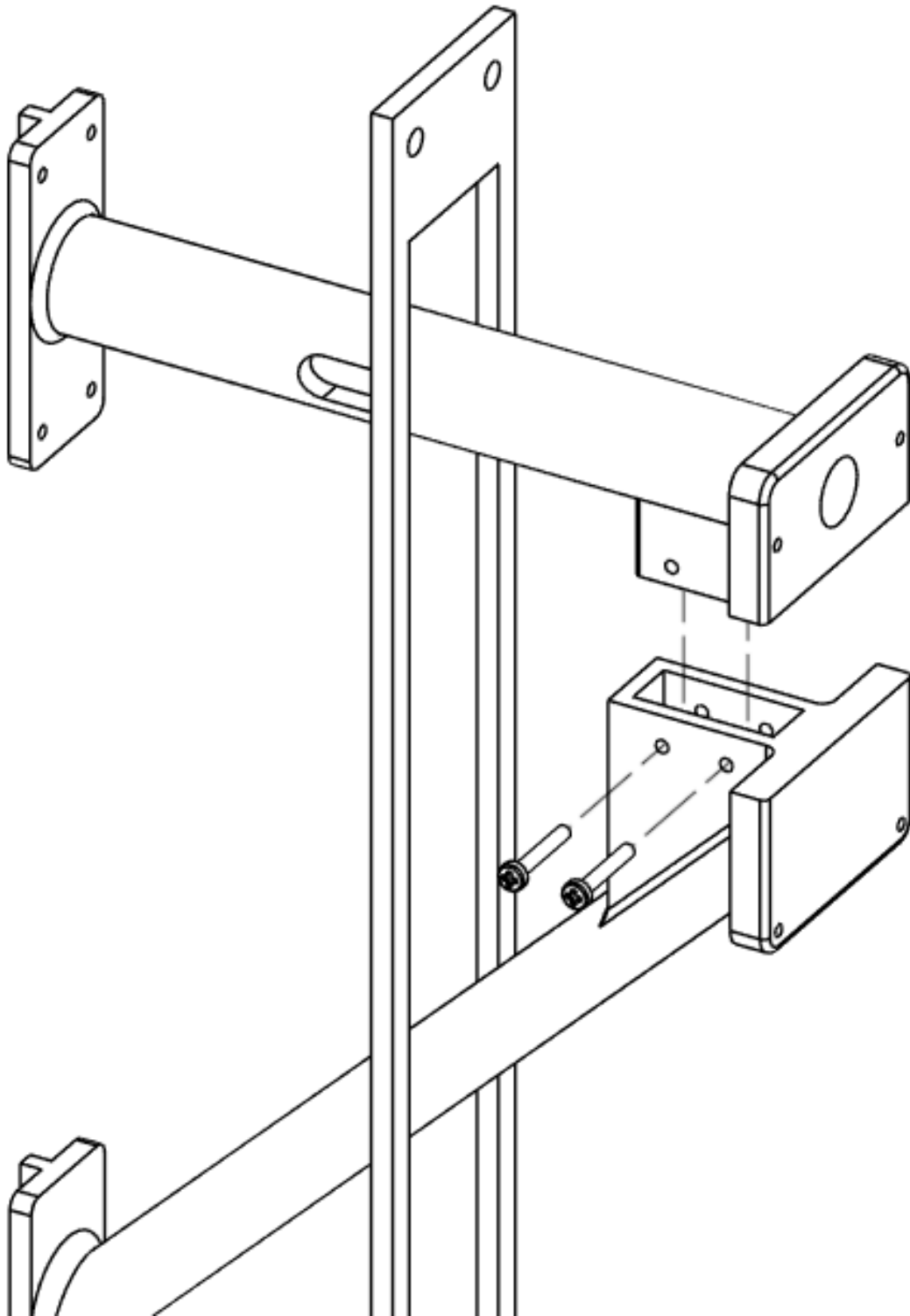


Figure 9.4 - Linear Actuator Arm Assembly

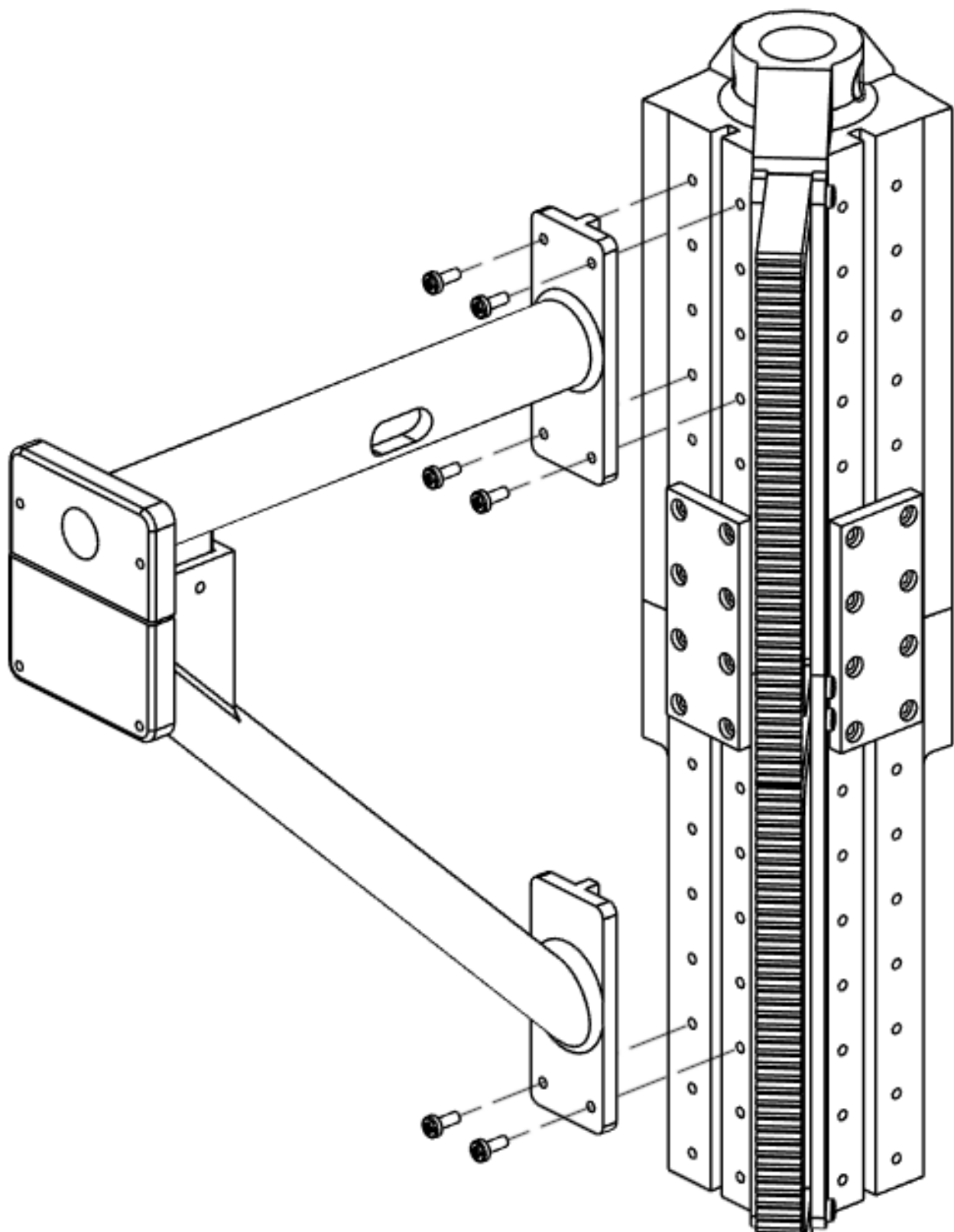


Figure 9.5 - Linear Actuator Arm & Core Assembly

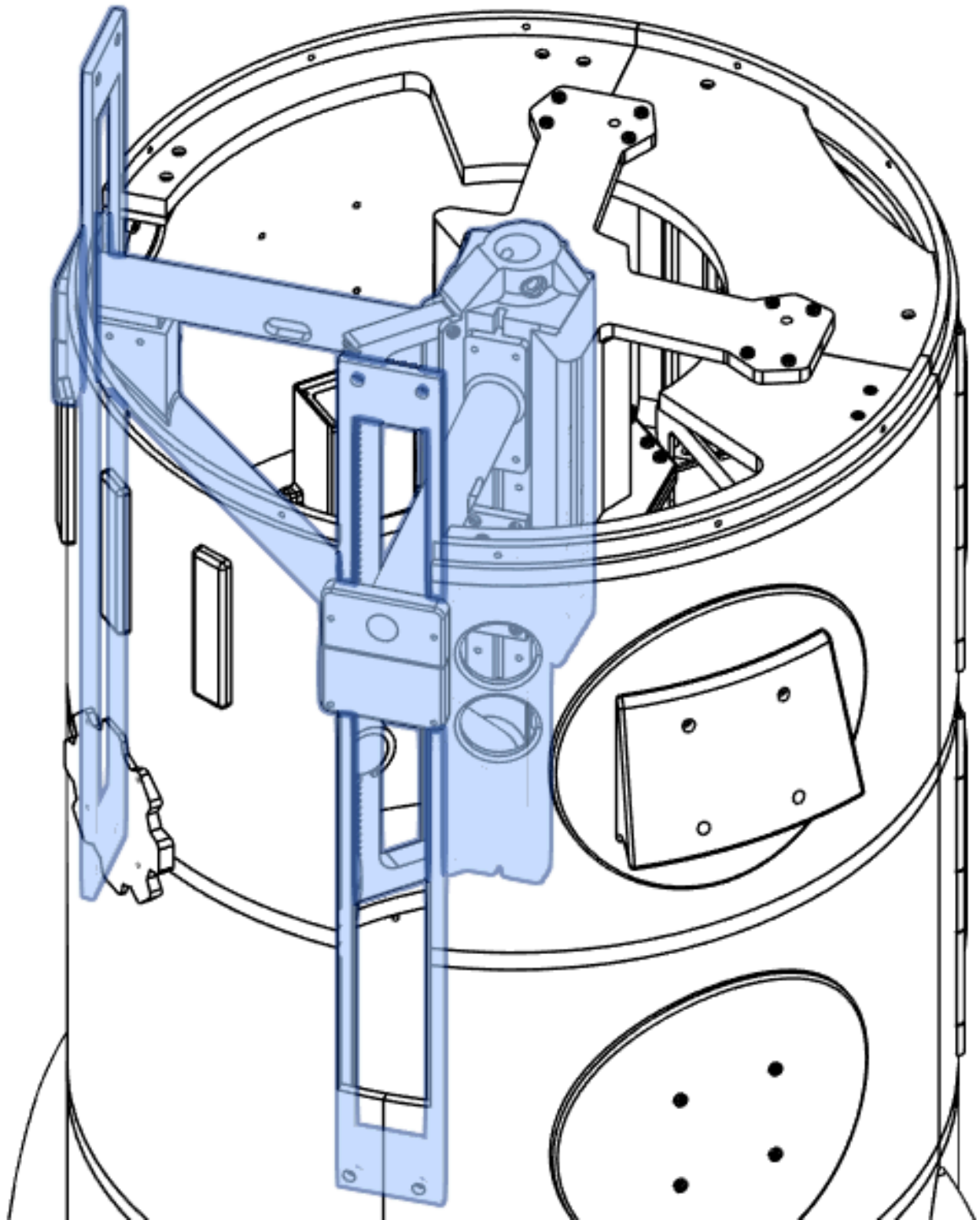


Figure 9.6 - Linear Actuator & Body Assembly

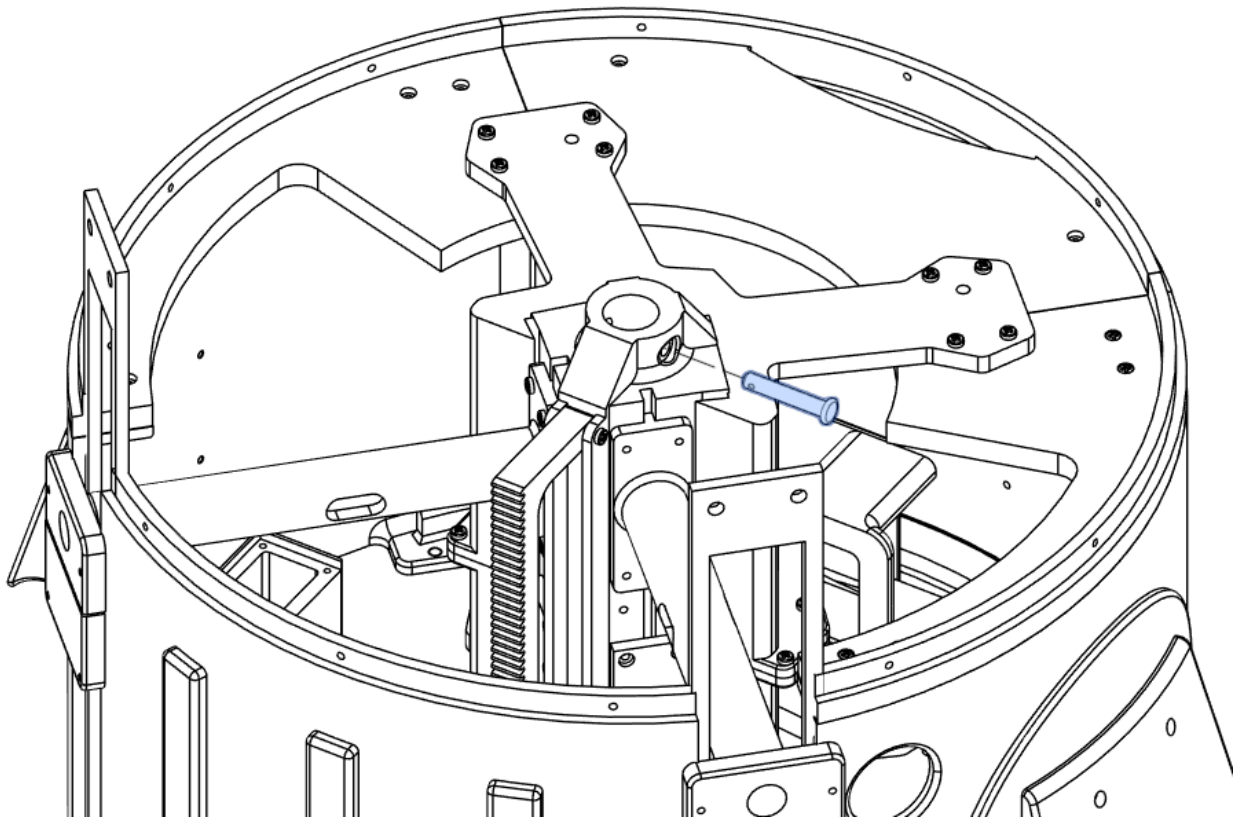


Figure 9.7 - Linear Actuator Pin Assembly

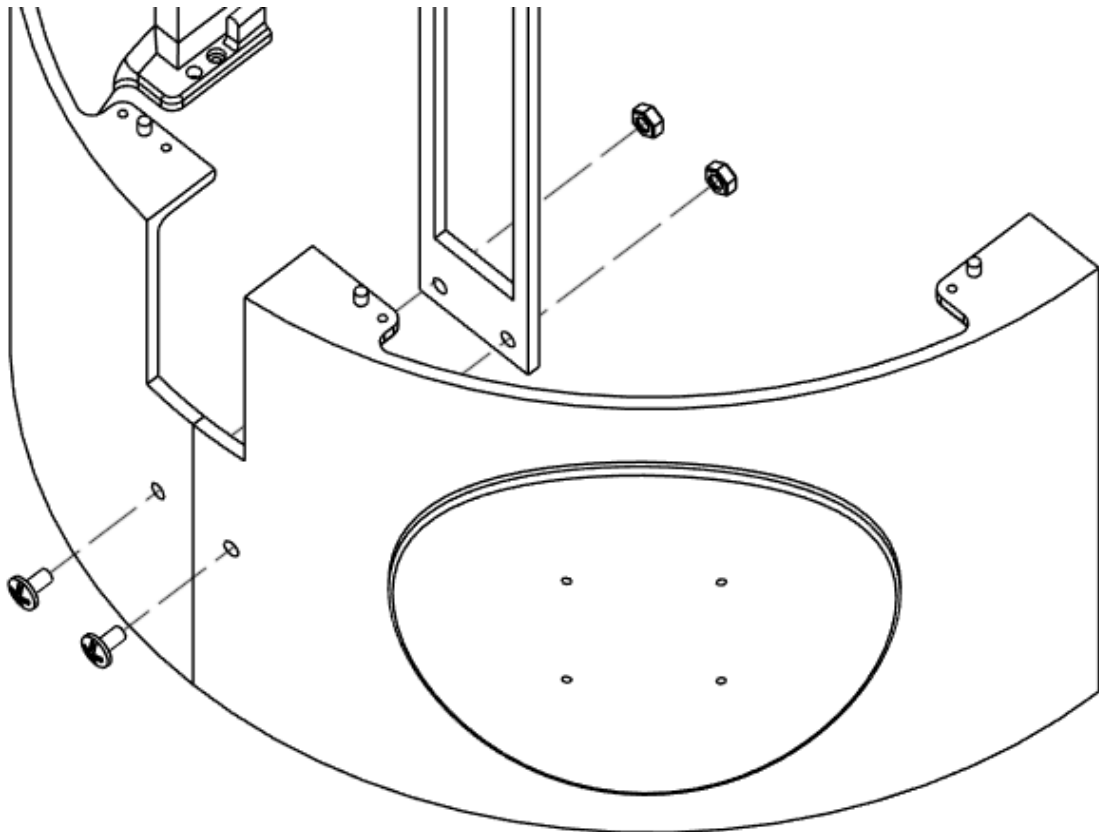


Figure 9.8 - Lower Grommet Strip Assembly

Stage 10: Robot Arms

Robot Arms

Materials

Type	Name	Count
Actuator	Feetech Servo Motor STS3215	8
Actuator	Feetech Servo Motor STS3250	4
Electronics	Arm Motor Wire - 100mm	2
Electronics	Arm Motor Wire - 120mm	2
Electronics	Arm Motor Wire - 170mm	2
Electronics	Arm Motor Wire - 200mm	4
Part	Arm Base	2
Part	Planet Gear	8
Part	Planet Bearing	8
Part	Shoulder	2
Part	Motor 1 Support	2
Part	Motor 2 Support	2
Part	Sun Gear	2
Part	Bicep	2
Part	Bicep Ring Gear	2
Part	Motor 3 Support	2
Part	Forearm	2
Part	Motor 4 Support	2
Part	Wrist	2
Part	Motor 5 Support	2
Part	Motor 5 Wire Clip	2
Part	Gripper Base Left	1

Type	Name	Count
Part	Gripper Base Right	1
Part	Gripper Finger	2
Part	Motor 6 Support Left	1
Part	Motor 6 Support Right	1
Part	Motor 6 Wire Clip Left	1
Part	Motor 6 Wire Clip Right	1
Part	Motor 4 Wire Clip	2
Part	Motor 3 Wire Clip	2
Part	Motor 2 Wire Clip	2
Part	Motor 1 Wire Clip	2
Part	Arm Camera Cover	2
Fastener	M2x5mm	18
Fastener	M2.5x8mm	6
Fastener	M3x5mm	10
Fastener	M3x8mm	34
Fastener	M3x10mm	4
Fastener	M3x12mm	16
Fastener	M3x16mm	68
Fastener	M3x20mm	10
Fastener	M3x25mm	12

Procedure

NOTE: Pay careful attention to the orientation of the parts and motors in this stage. Some parts may seem symmetrical at first, but they are not. The arm illustrated in this section's figures is the left arm

NOTE: Some parts are left/right specific. Ensure the correct part is used

NOTE: Each Servo Motor comes with 2 horns that attach to either side of the motor. The one with the teeth on the inside is referred to as the "Driven Motor Horn". The one without teeth is referred to as the "Idle Motor Horn". Both of these horns are to be placed on the motors prior to installation unless otherwise specified

1. Insert Motor 1 into Arm Base as shown in Figure 10.1
2. Fasten Planets to Shoulder as shown in Figure 10.2
 - a. (4) M3x20mm
 - b. (4) Planet Bearings
3. Fasten Shoulder to Motor 1 as shown in Figure 10.3
 - a. (8) M3x8mm
4. Fasten Motor 1 Support to Arm Base as shown in Figure 10.4
 - a. (4) M3x12mm
5. Insert Motor 2 into Shoulder as shown in Figure 10.5
 - a. **NOTE:** Do NOT attach the Driven Motor Horn to Motor 2
6. Fasten Motor 2 Support to Shoulder as shown in Figure 10.6
 - a. (2) M3x16mm
7. Fasten Driven Motor Horn to Sun Gear as shown in Figure 10.7
 - a. (4) M3x16mm
8. Fasten Sun Gear Assembly to Motor 2 as shown in Figure 10.8
 - a. (1) M3x8mm
9. Fasten Bicep to Motor 2 as shown in Figure 10.9
 - a. (4) M3x16mm
10. Fasten Bicep Ring Gear to Bicep and Shoulder as shown in Figure 10.10
 - a. (6) M3x25mm
11. Fasten Motor 3 and Motor 3 Support to Bicep as shown in Figure 10.11
 - a. (2) M3x10mm
12. Fasten Forearm to Motor 3 as shown in Figure 10.12
 - a. (8) M3x16mm
13. Fasten Motor 4 and Motor 4 Support to Forearm as shown in Figure 10.13
 - a. (2) M3x5mm
14. Fasten Motor 5 and Motor 5 Support to Wrist as shown in Figure 10.14
 - a. (3) M3x5mm
 - b. **NOTE:** Do NOT attach the Driven Motor Horn to Motor 5
15. Connect a 100mm Arm Wire and 170mm Arm Wire to Motor 5
16. Fasten Motor 5 Wire Clip to Wrist as shown in Figure 10.15
 - a. (1) M2x5mm
 - b. **NOTE:** Arm Wires are not shown in Figure 10.15
17. Fasten Wrist to Motor 4 as shown in Figure 10.16
 - a. (8) M3x8mm
18. Fasten Driven Motor Horn to Gripper Base as shown in Figure 10.17
 - a. (4) M3x16mm
 - b. **NOTE:** Gripper bases are left/right specific. Ensure the correct one is used
19. Fasten Gripper Base Assembly to Motor 5 as shown in Figure 10.18
 - a. (1) M3x20mm
20. Insert Motor 6 into Gripper Base as shown in Figure 10.19
21. Fasten Gripper Finger to Motor 6 as shown in Figure 10.20
 - a. (8) M3x16mm
22. Fasten Motor 6 Support to Gripper Base as shown in Figure 10.21
 - a. (4) M3x12mm

- b. NOTE:** Motor 6 Supports are left/right specific. Ensure the correct one is used
- 23.** Connect the longer of the 2 Arm Motor Wires (170mm) from Motor 5 to Motor 6
- 24.** Connect the shorter of the 2 Arm Motor Wires (100mm) from Motor 5 to Motor 4
- 25.** Fasten Motor 6 Wire Clip to Gripper Base as shown in Figure 10.22
 - a.** (2) M2x5mm
 - b. NOTE:** Motor 6 Wire Clips are left/right specific. Ensure the correct one is used
 - c. NOTE:** Arm Wires are not shown in Figure 10.22
- 26.** Connect a 200mm Arm Motor Wire from Motor 4 to Motor 3
- 27.** Fasten Motor 4 Wire Clip to Forearm as shown in Figure 10.23
 - a.** (1) M2x5mm
 - b. NOTE:** Arm Wires are not shown in Figure 10.23
- 28.** Connect a 200mm Arm Motor Wire from Motor 3 to Motor 2
- 29.** Fasten Motor 3 Wire Clip to Bicep as shown in Figure 10.24
 - a.** (1) M2x5mm
 - b. NOTE:** Arm Wires are not shown in Figure 10.24
- 30.** Connect a 120mm Arm Motor Wire from Motor 2 to Motor 1
- 31.** Fasten Motor 2 Wire Clip to Motor 2 Support as shown in Figure 10.25
 - a.** (2) M2x5mm
 - b. NOTE:** Arm Wires are not shown in Figure 10.25
- 32.** Fasten Arm Base to the end of Linear Actuator Arm as shown in Figure 10.26
 - a.** (4) M3x16mm
 - b. NOTE:** Take care to not overtighten screws
- 33.** Connect the 400mm Arm Motor Wire to Motor 1
- 34.** Fasten Motor 1 Wire Clip to Motor 1 Support as shown in Figure 10.27
 - a.** (2) M2x5mm
 - b. NOTE:** Arm Wires are not shown in Figure 10.27
- 35.** Thread Camera Cable through arm channels
- 36.** Fasten Camera and Arm Camera Cover to Gripper Base as shown in Figure 10.28
 - a.** (3) M2.5x8mm
- 37.** Repeat Steps 1-36 for the Right Arm
 - a. NOTE:** Motor IDs will be different for the right arm. It will start with Motor 7 in place of Motor 1, and continue to increment down the arm

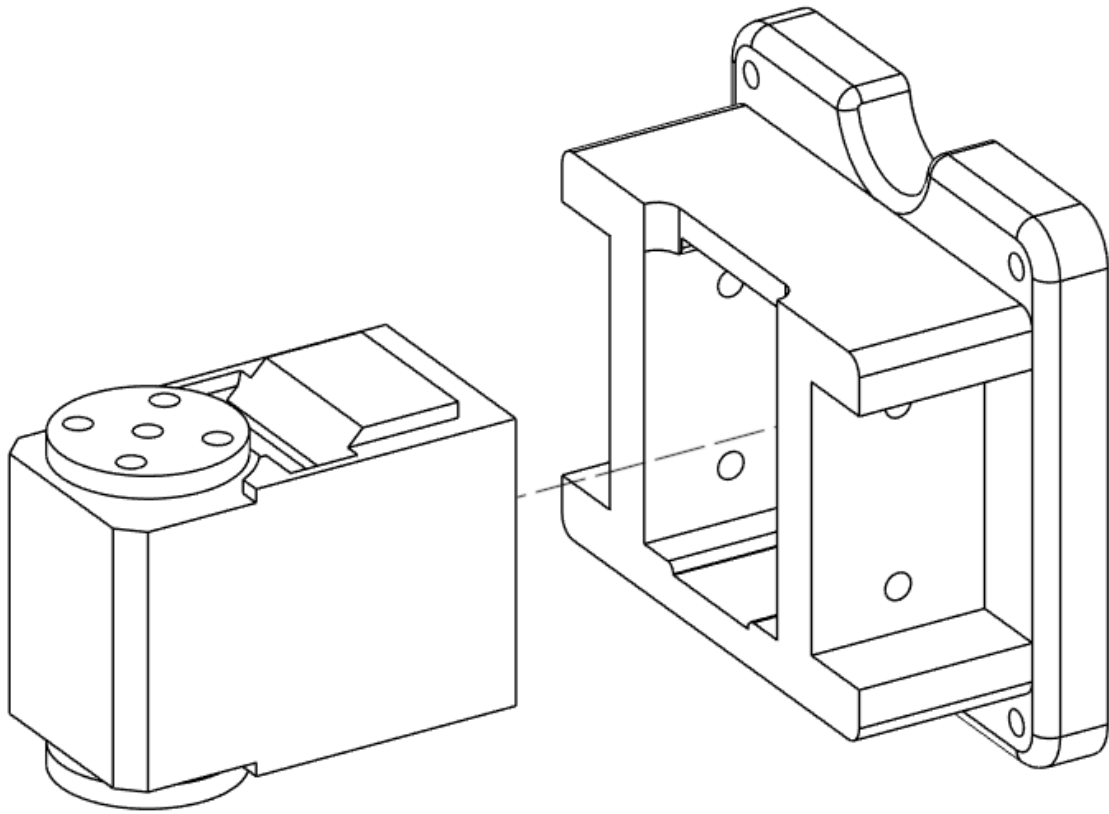


Figure 10.1 - Motor 1 Assembly

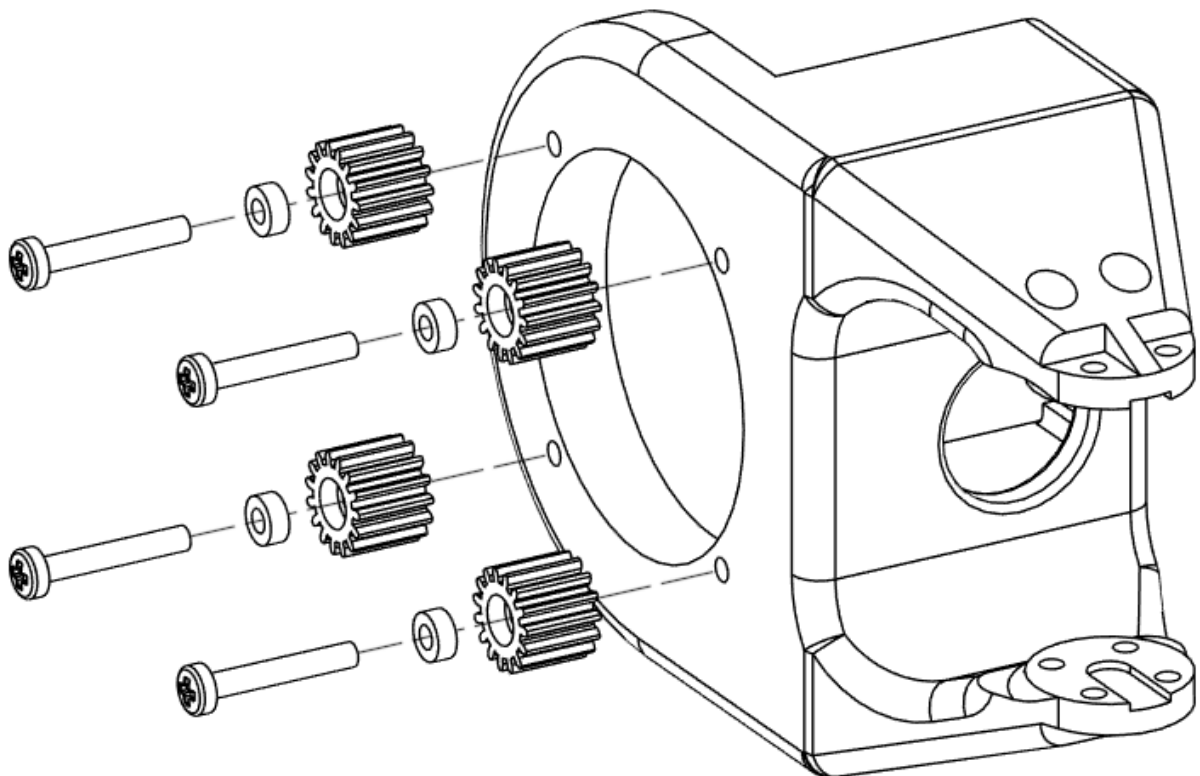


Figure 10.2 - Planets Assembly

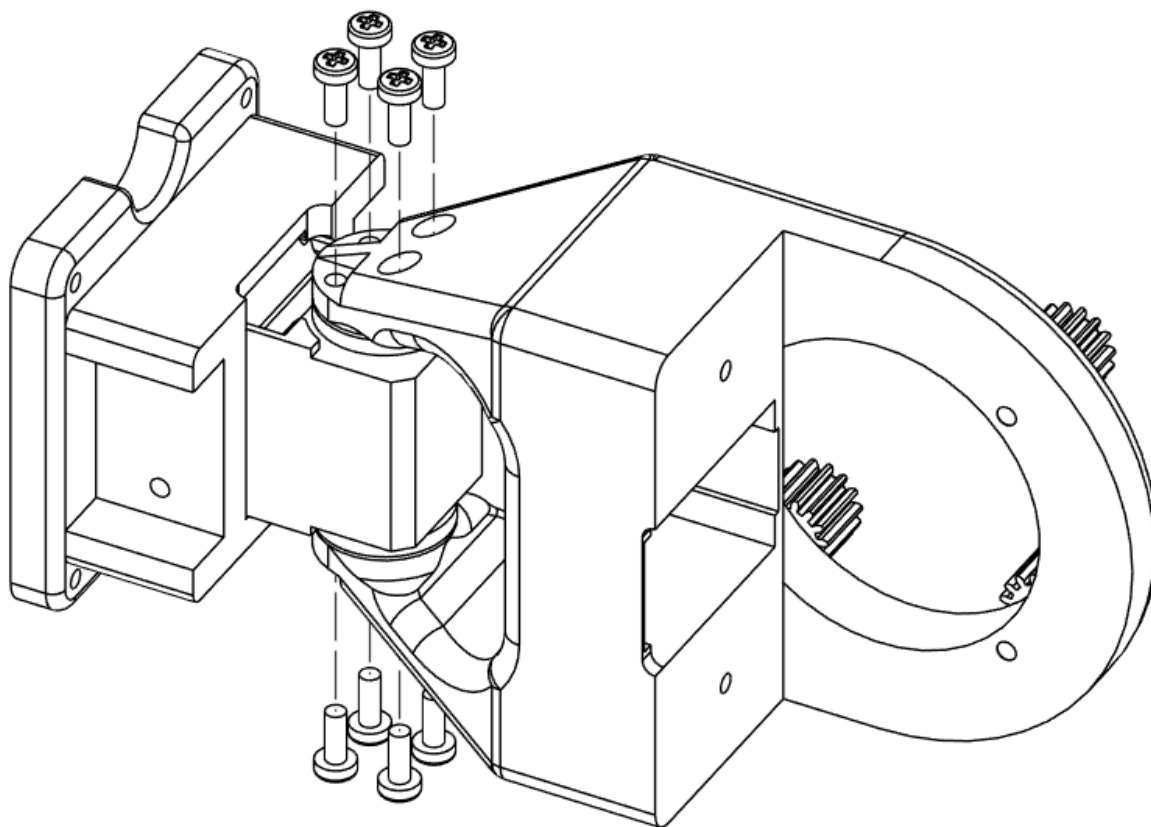


Figure 10.3 - Shoulder & Motor 1 Assembly

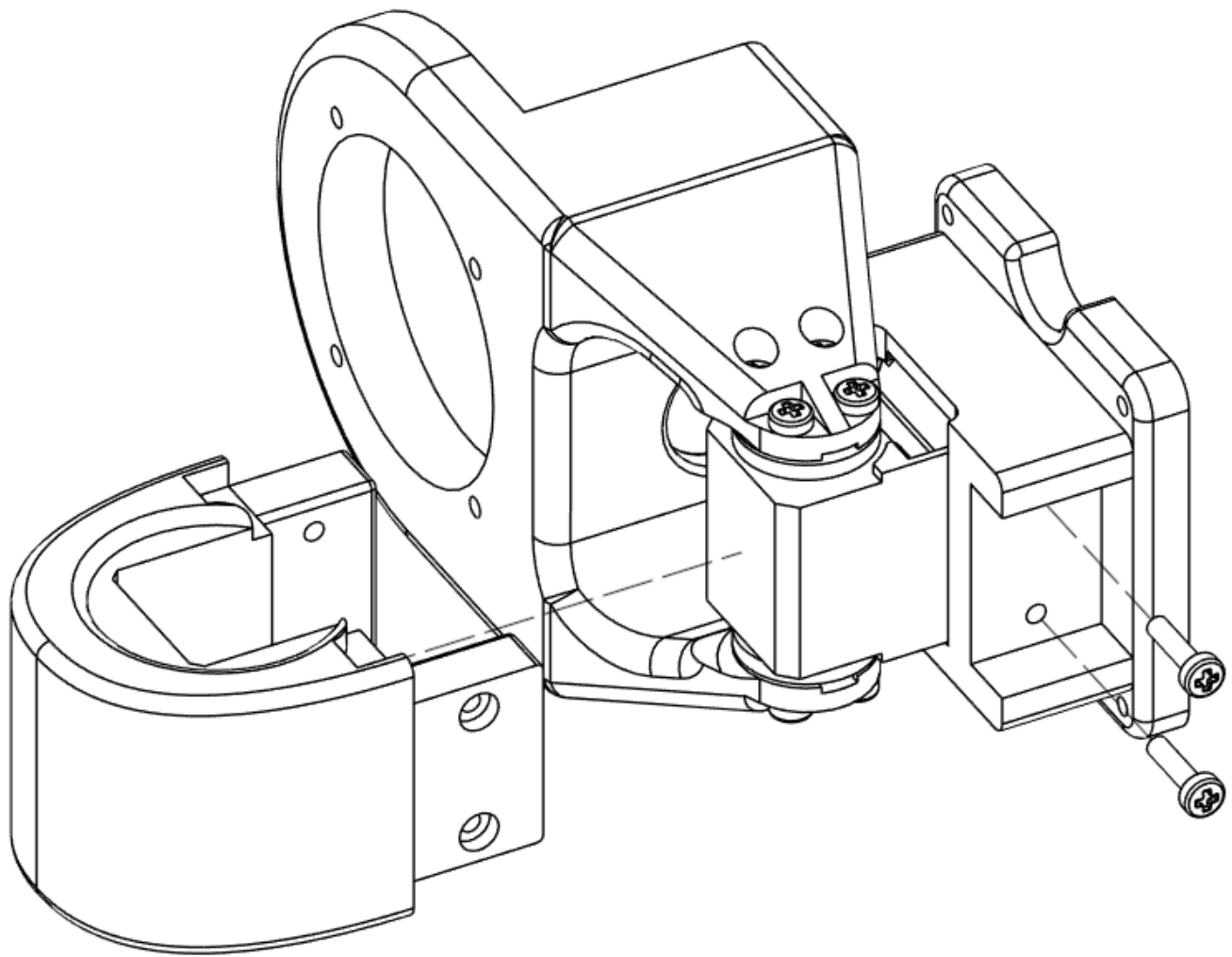


Figure 10.4 - Motor 1 Support Assembly

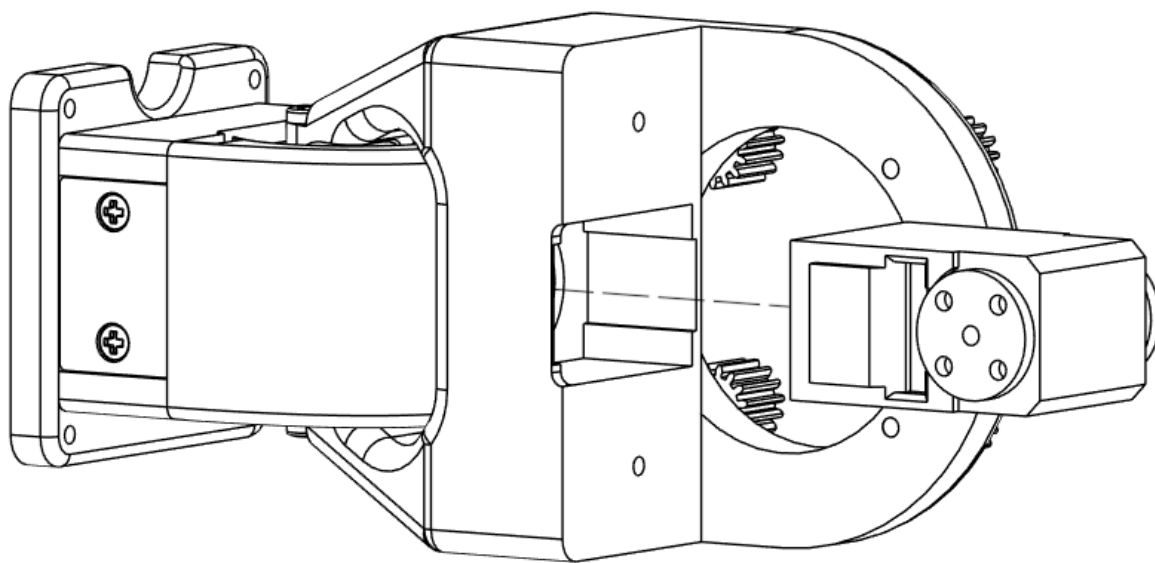


Figure 10.5 - Motor 2 Assembly

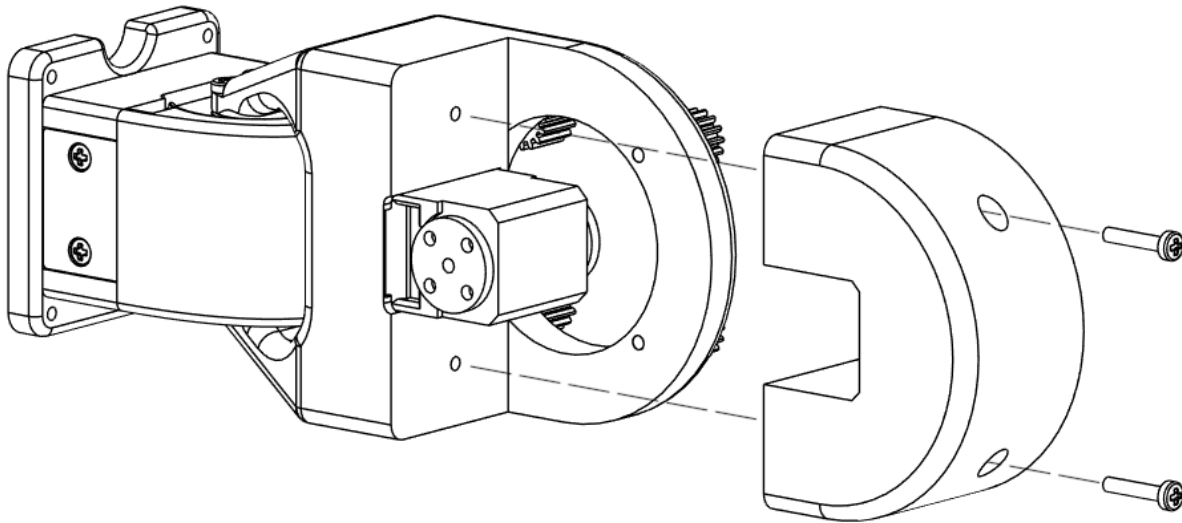


Figure 10.6 - Motor 2 Support Assembly

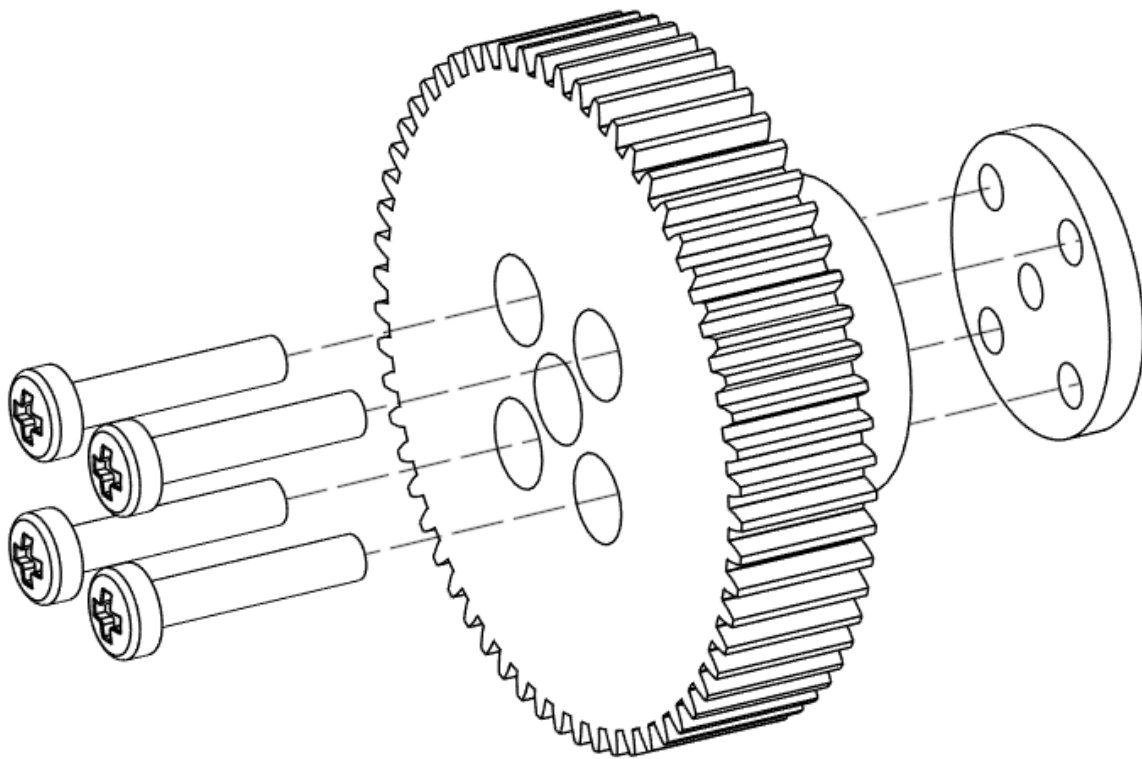


Figure 10.7 - Sun Gear & Driven Motor Horn Assembly

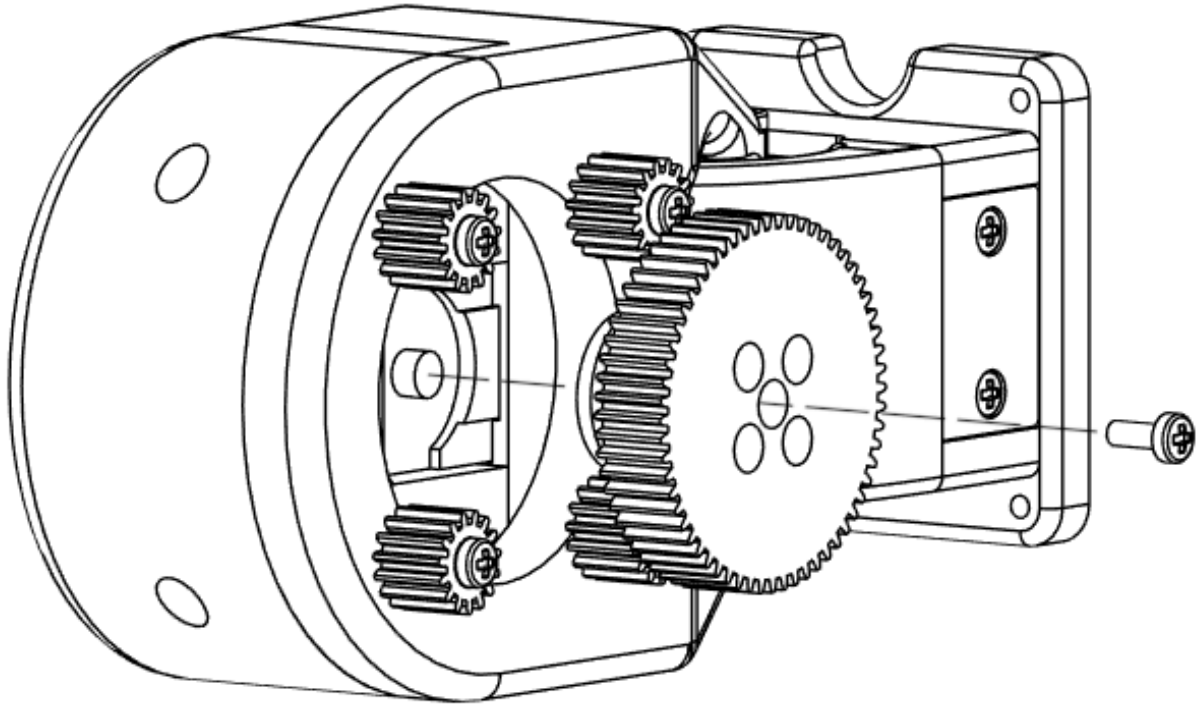


Figure 10.8 - Sun Gear Assembly

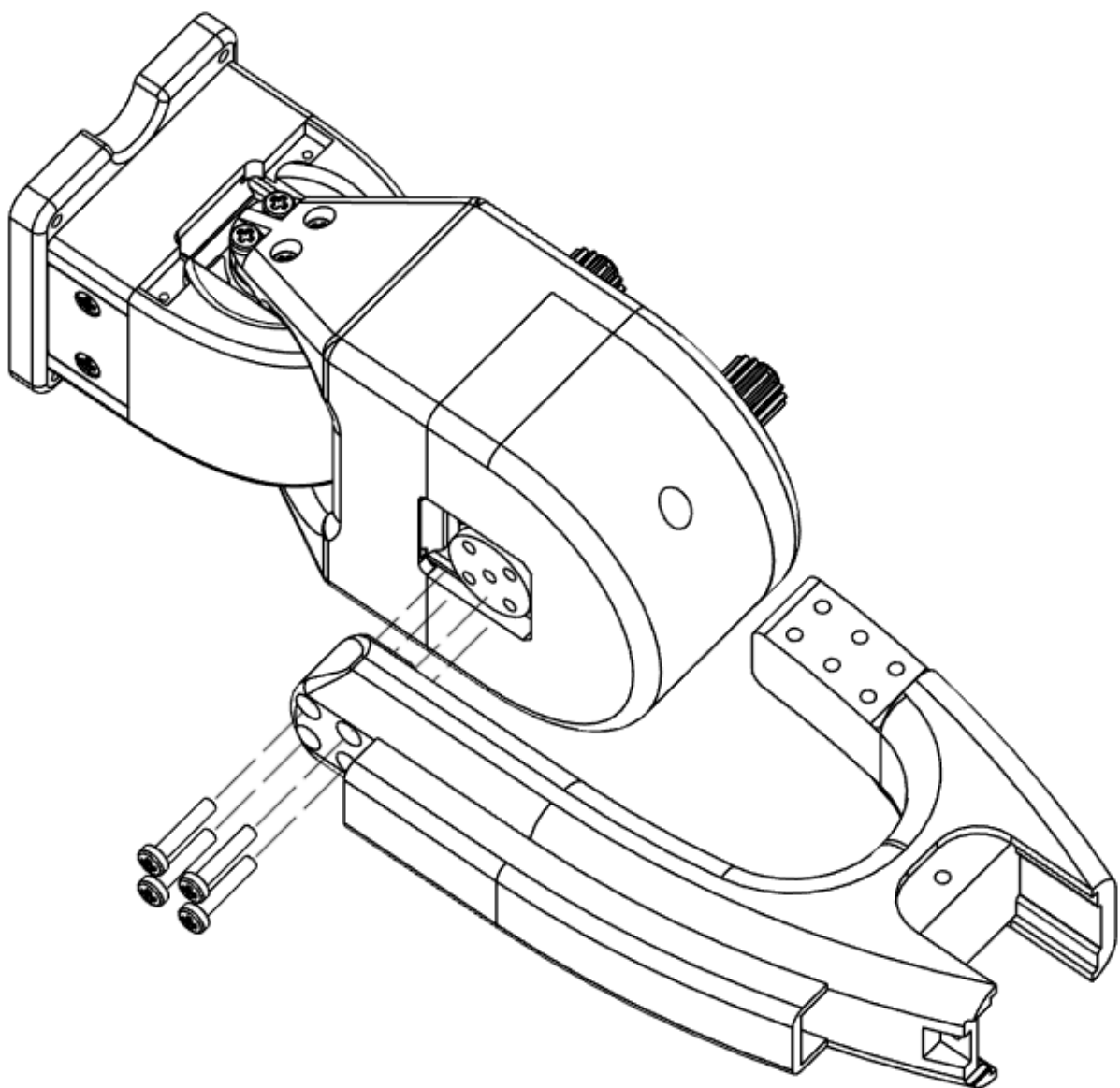


Figure 10.9 - Bicep Assembly

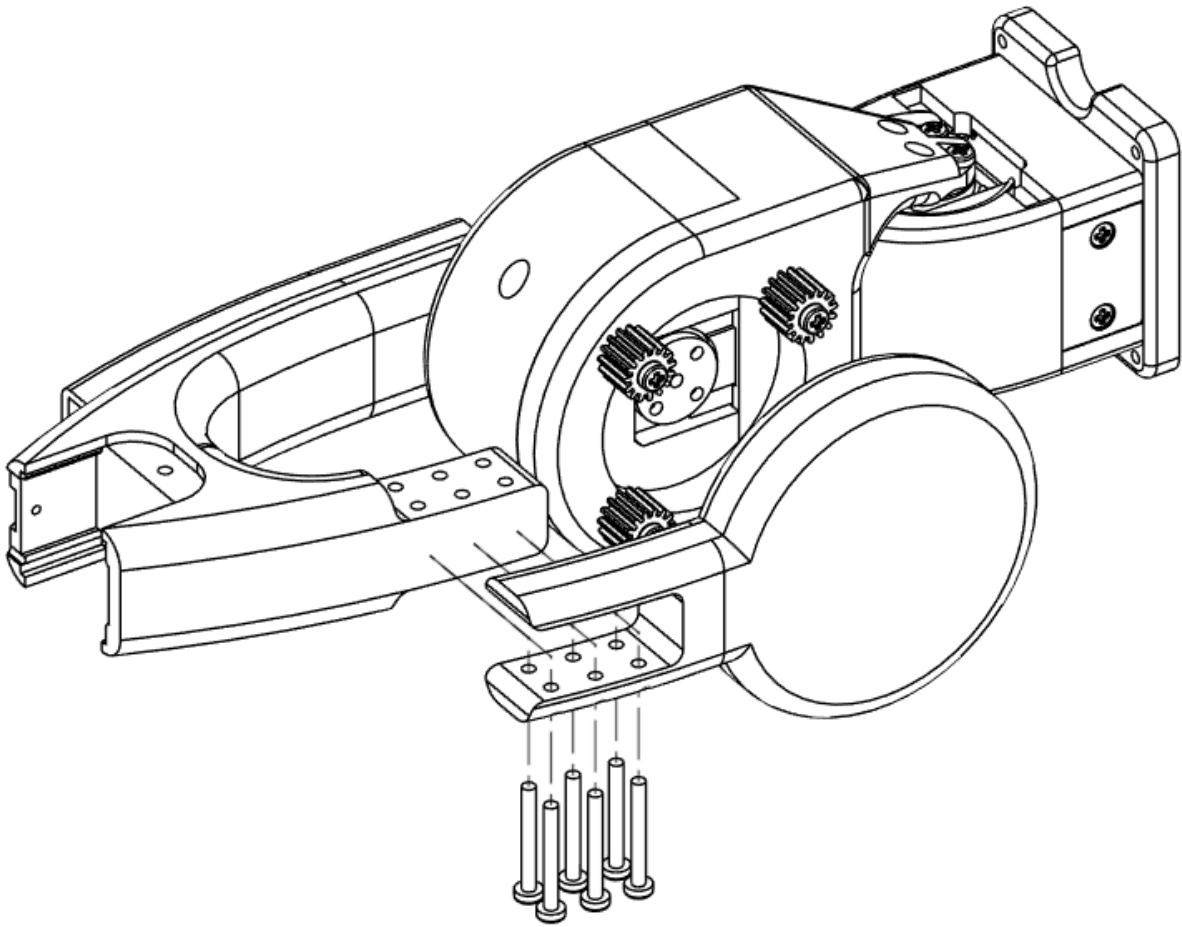


Figure 10.10 - Bicep Ring Gear Assembly

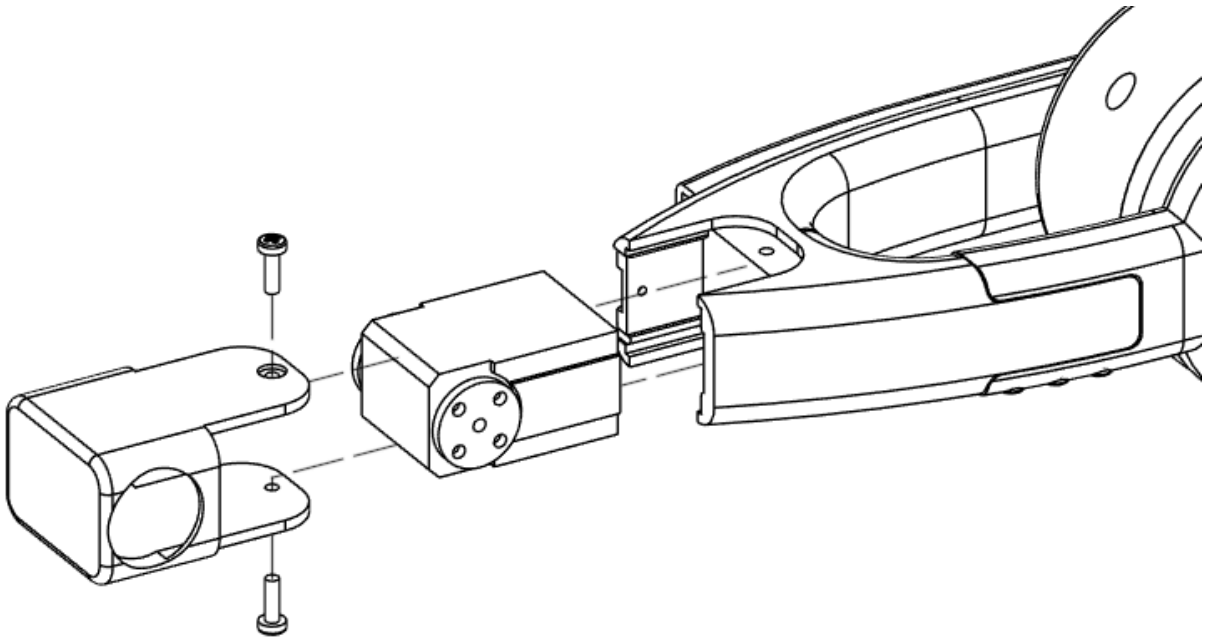


Figure 10.11 - Motor 3 Assembly

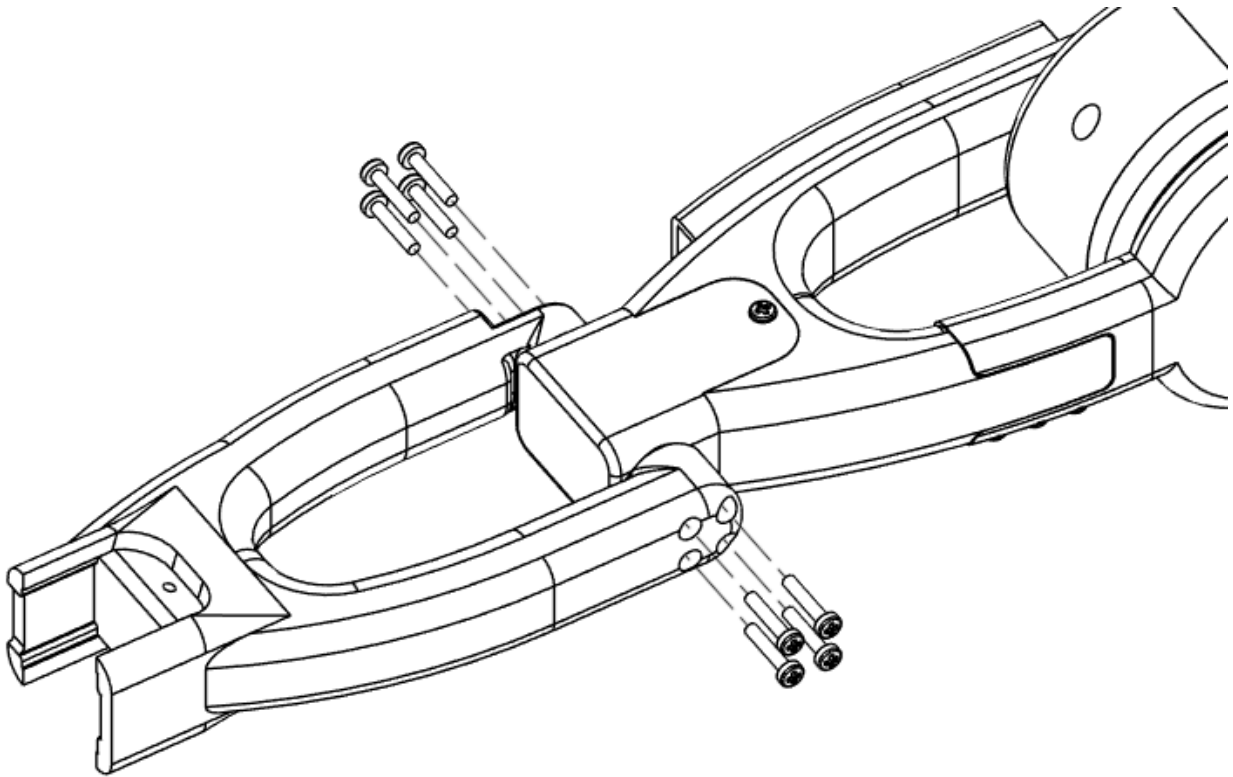


Figure 10.12 - Forearm Assembly

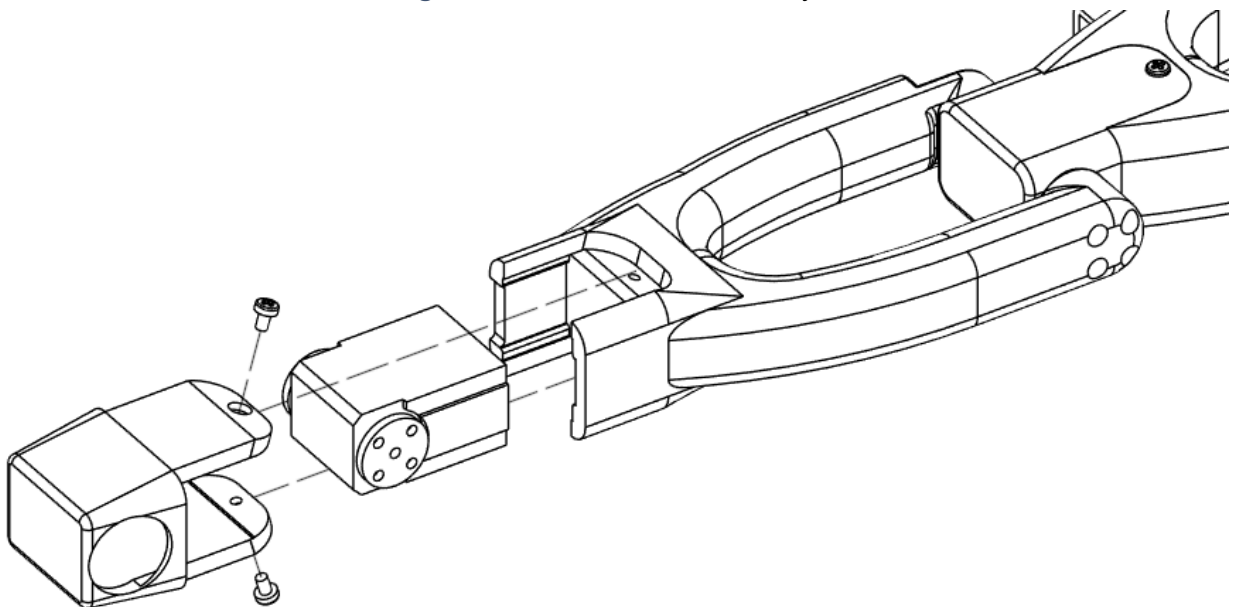


Figure 10.13 - Motor 4 Assembly

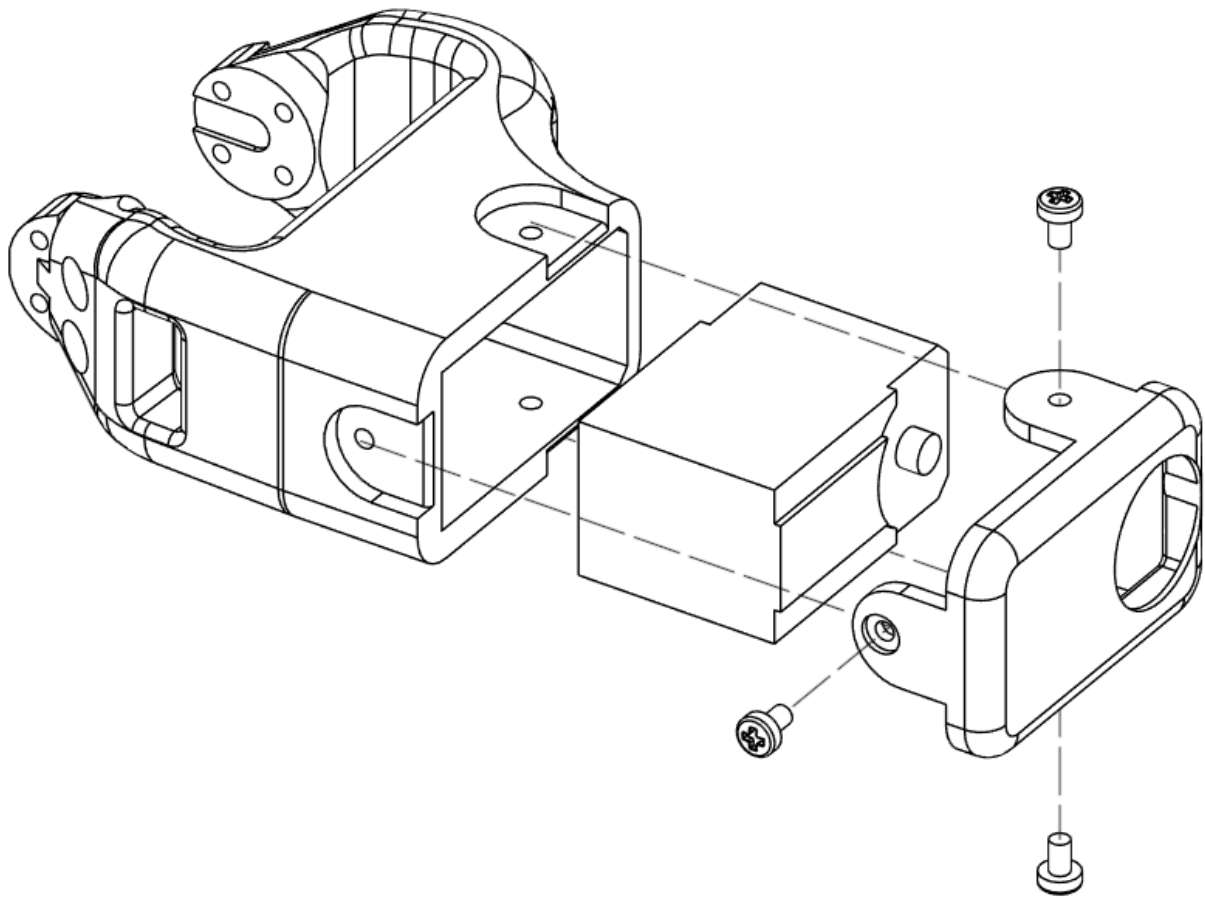


Figure 10.14 - Motor 5 Assembly

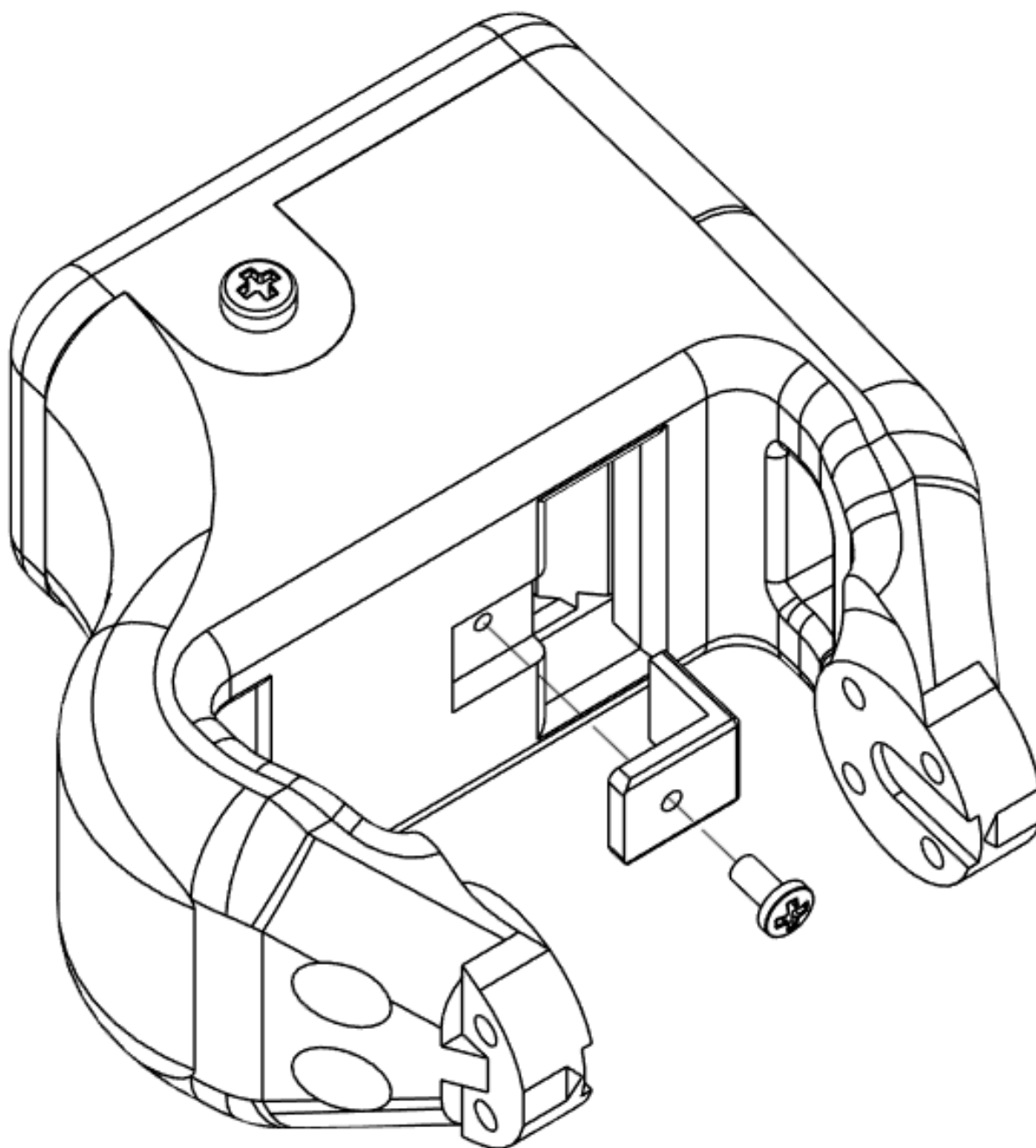


Figure 10.15 - Motor 5 Wire Clip Assembly

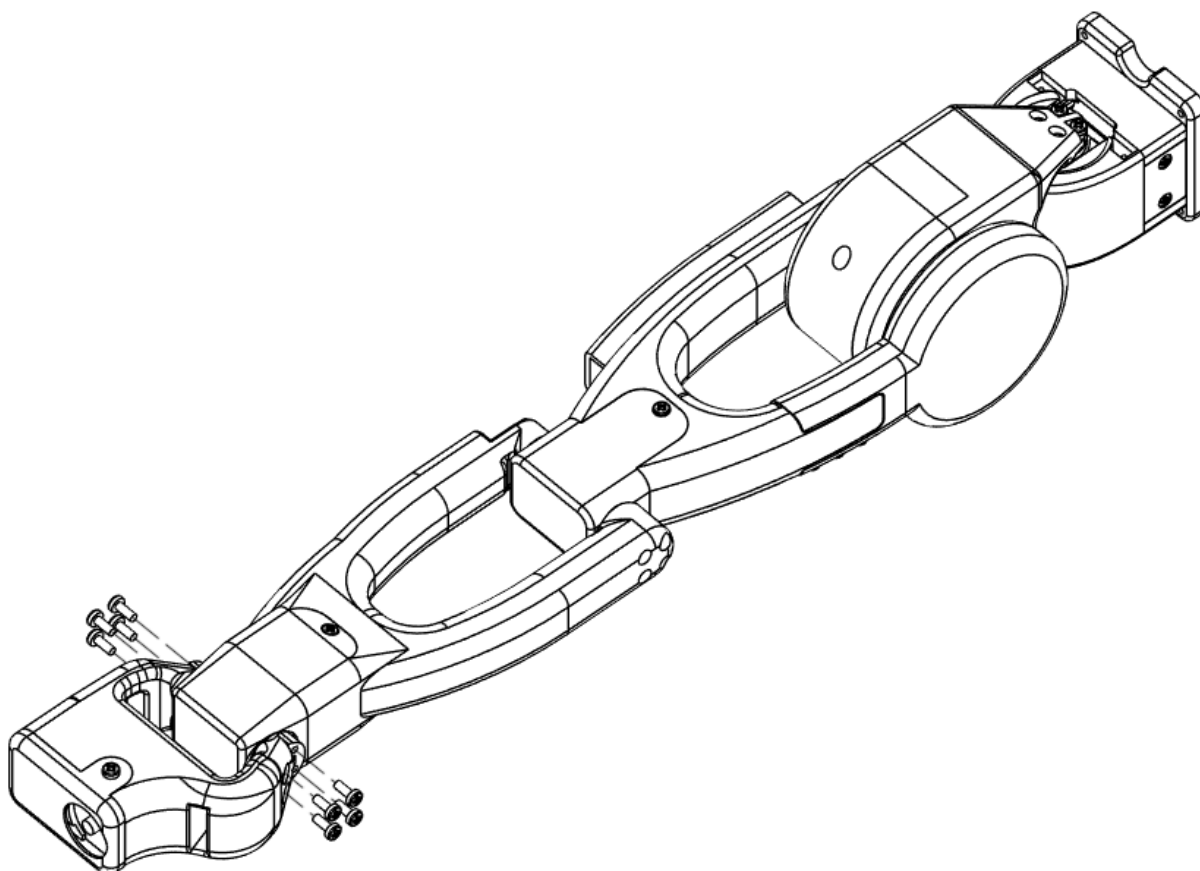


Figure 10.16 - Wrist Assembly

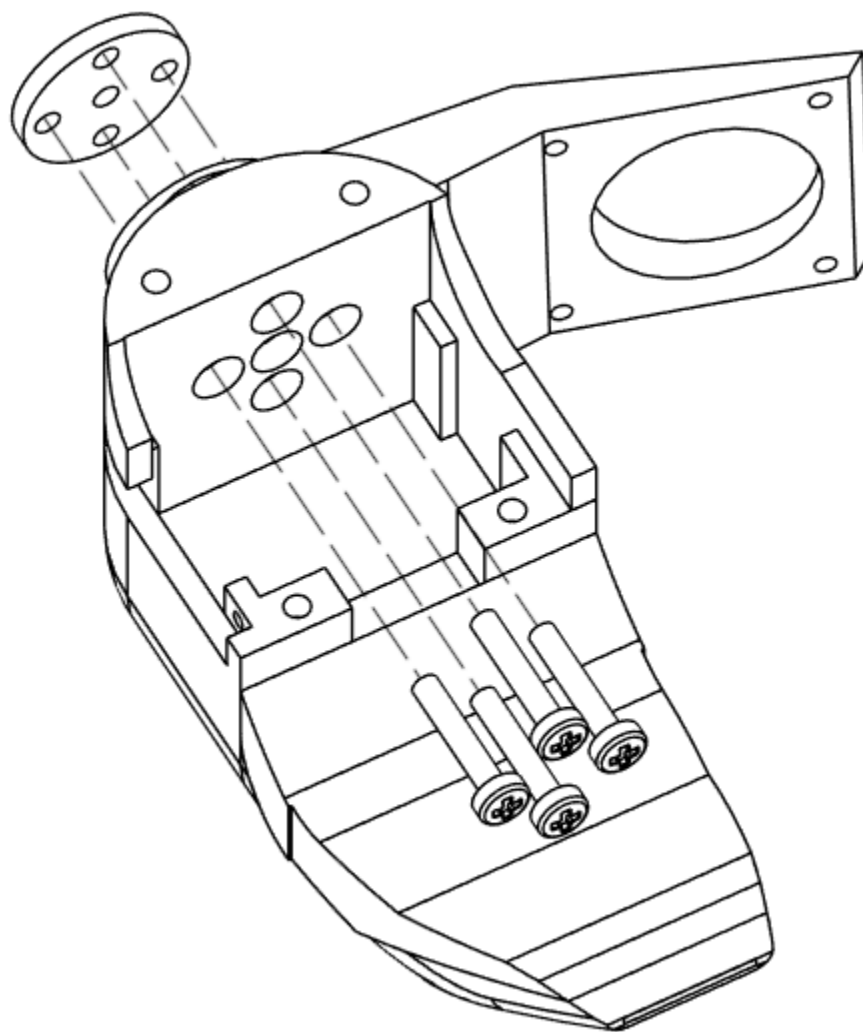


Figure 10.17 - Gripper Base & Motor Horn Assembly

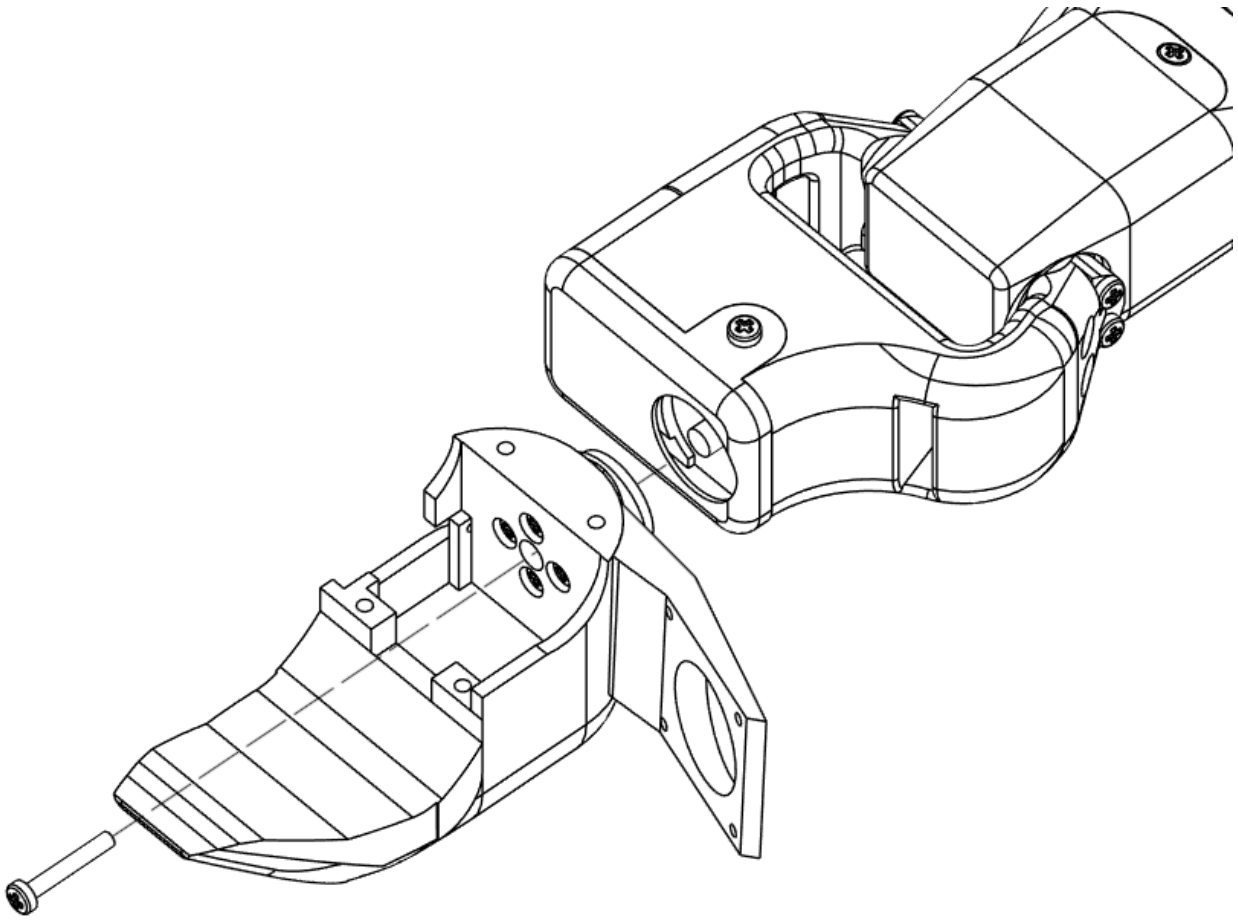


Figure 10.18 - Gripper Assembly

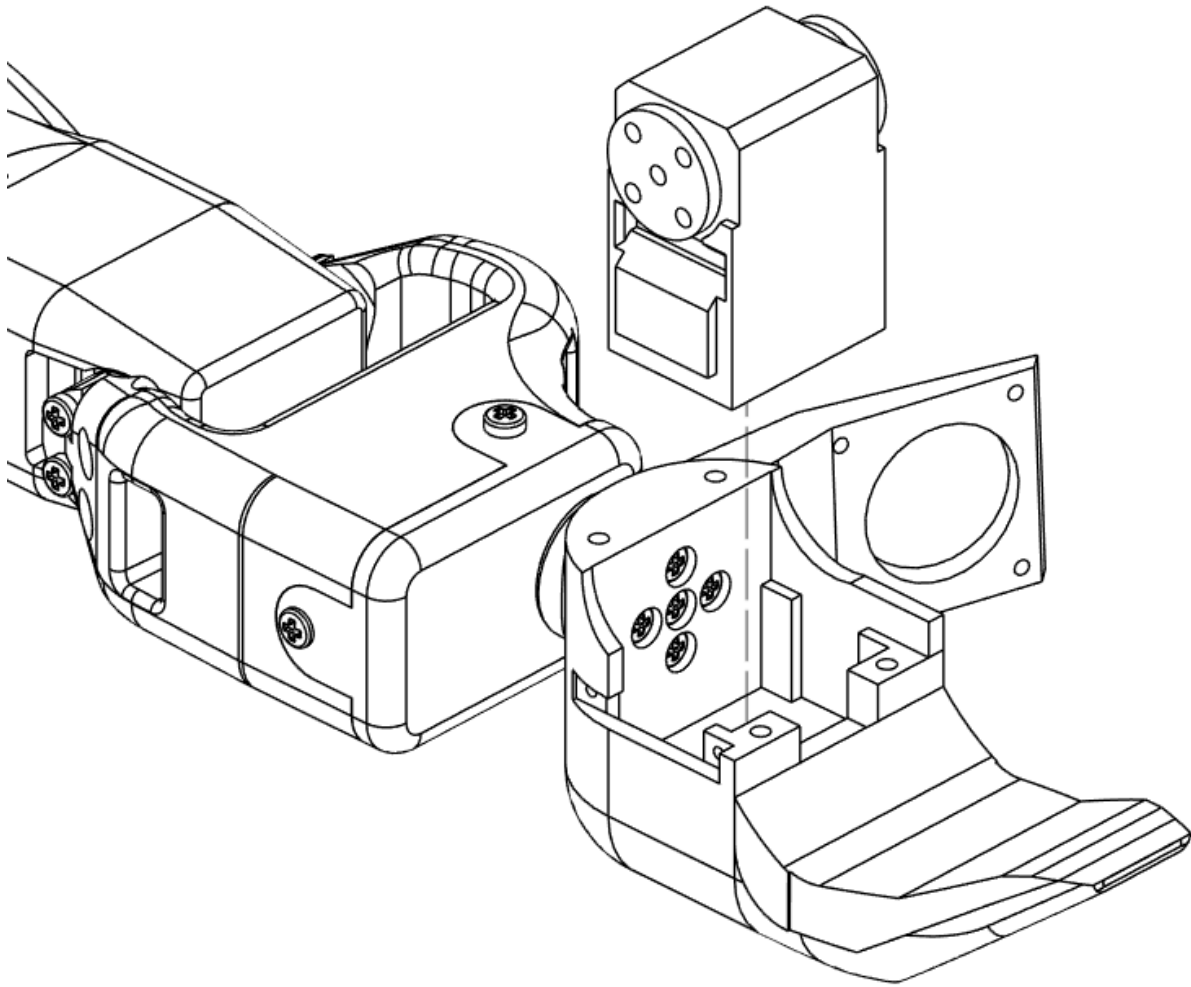


Figure 10.19 - Motor 6 Assembly

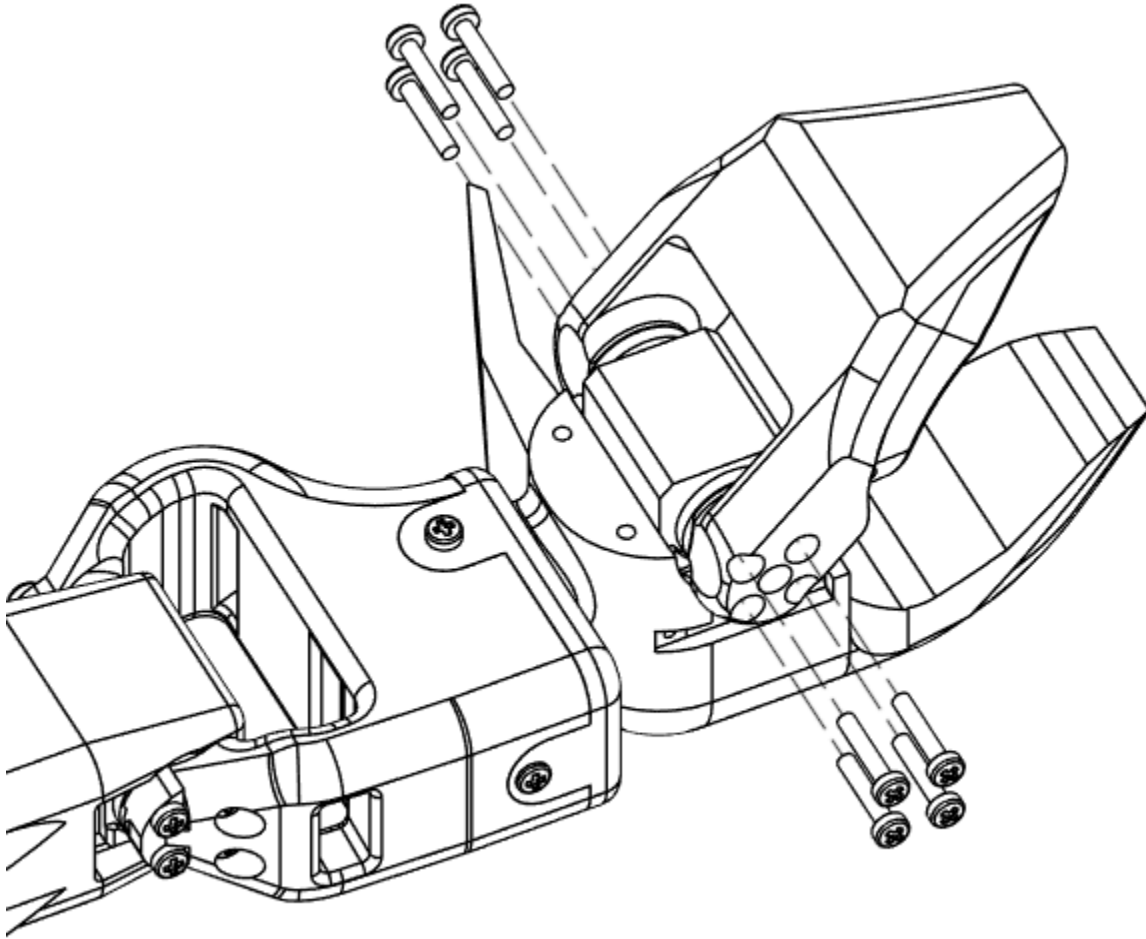


Figure 10.20 - Gripper Finger Assembly

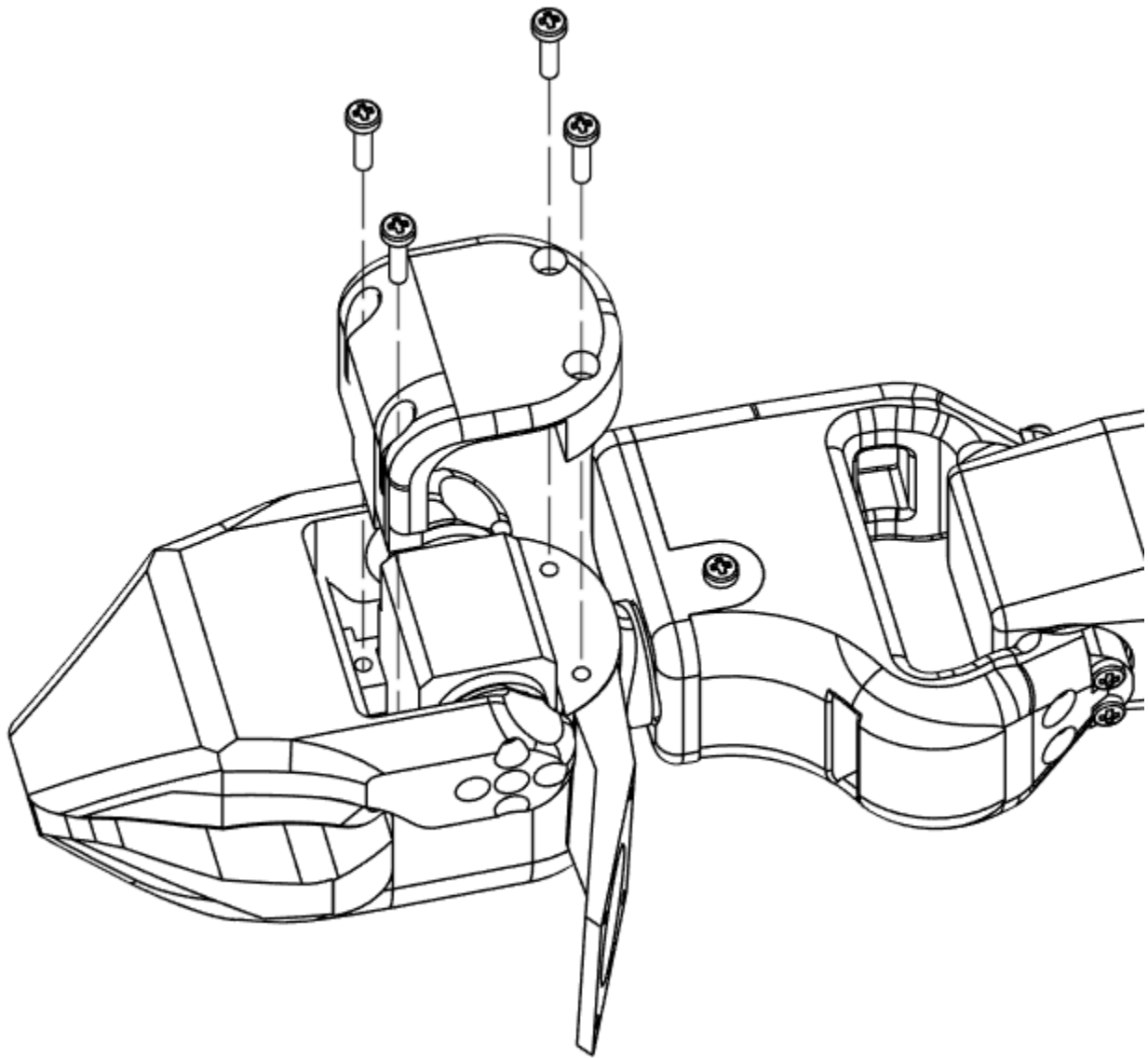


Figure 10.21 - Motor 6 Support Assembly

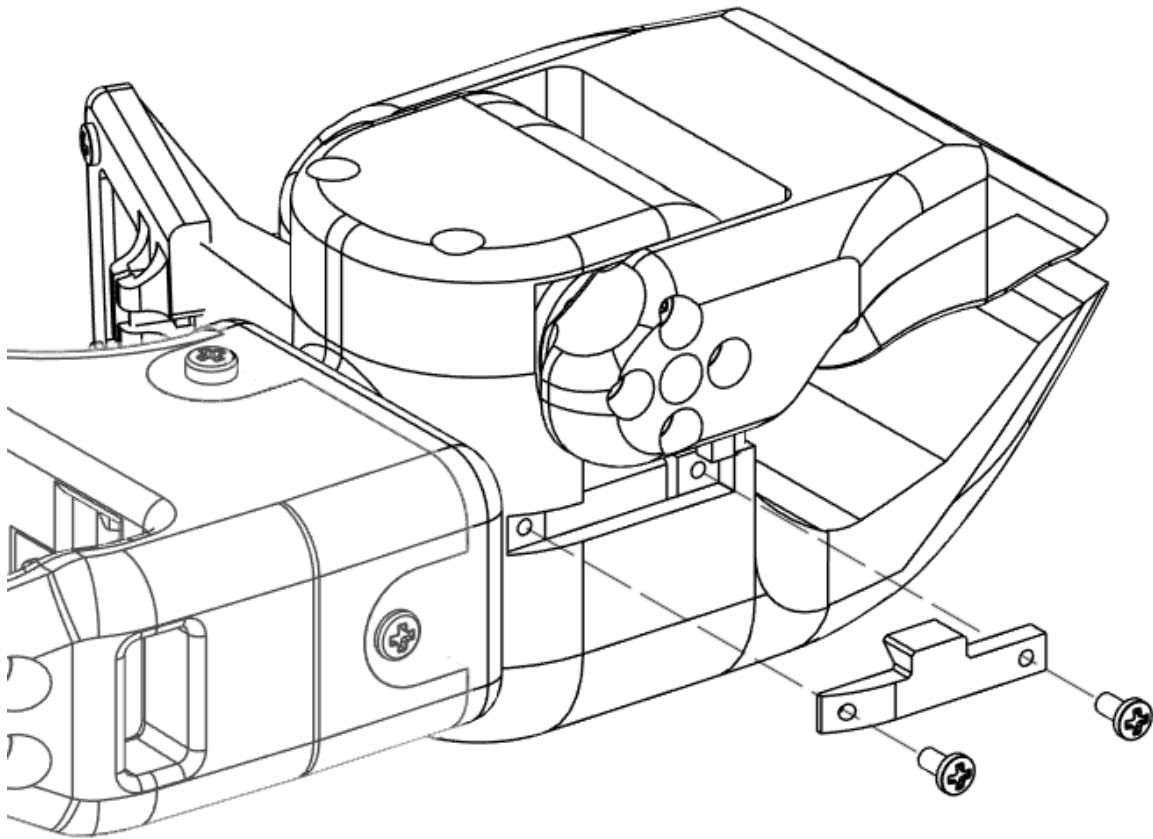


Figure 10.22 - Motor 6 Wire Clip Assembly

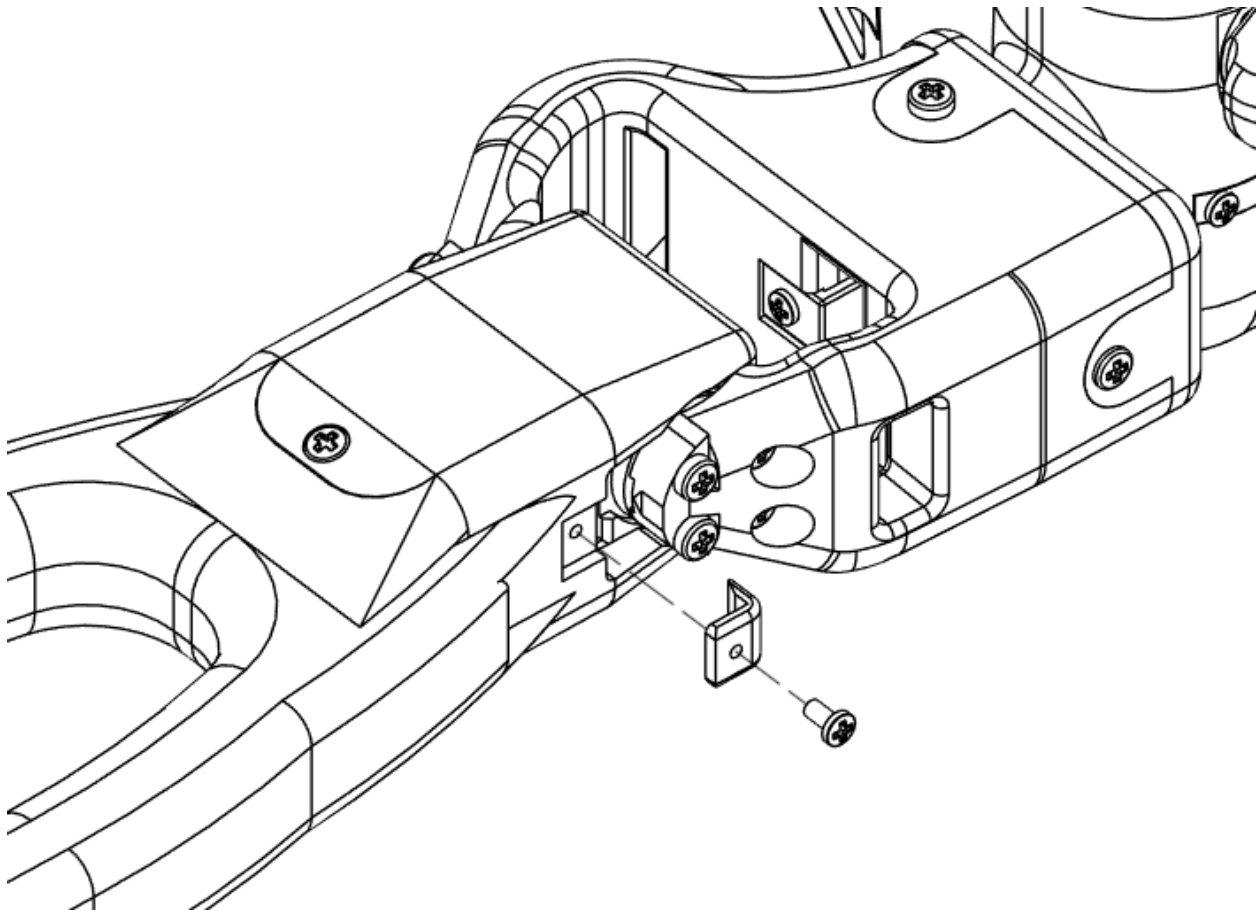


Figure 10.23 - Motor 4 Wire Clip Assembly

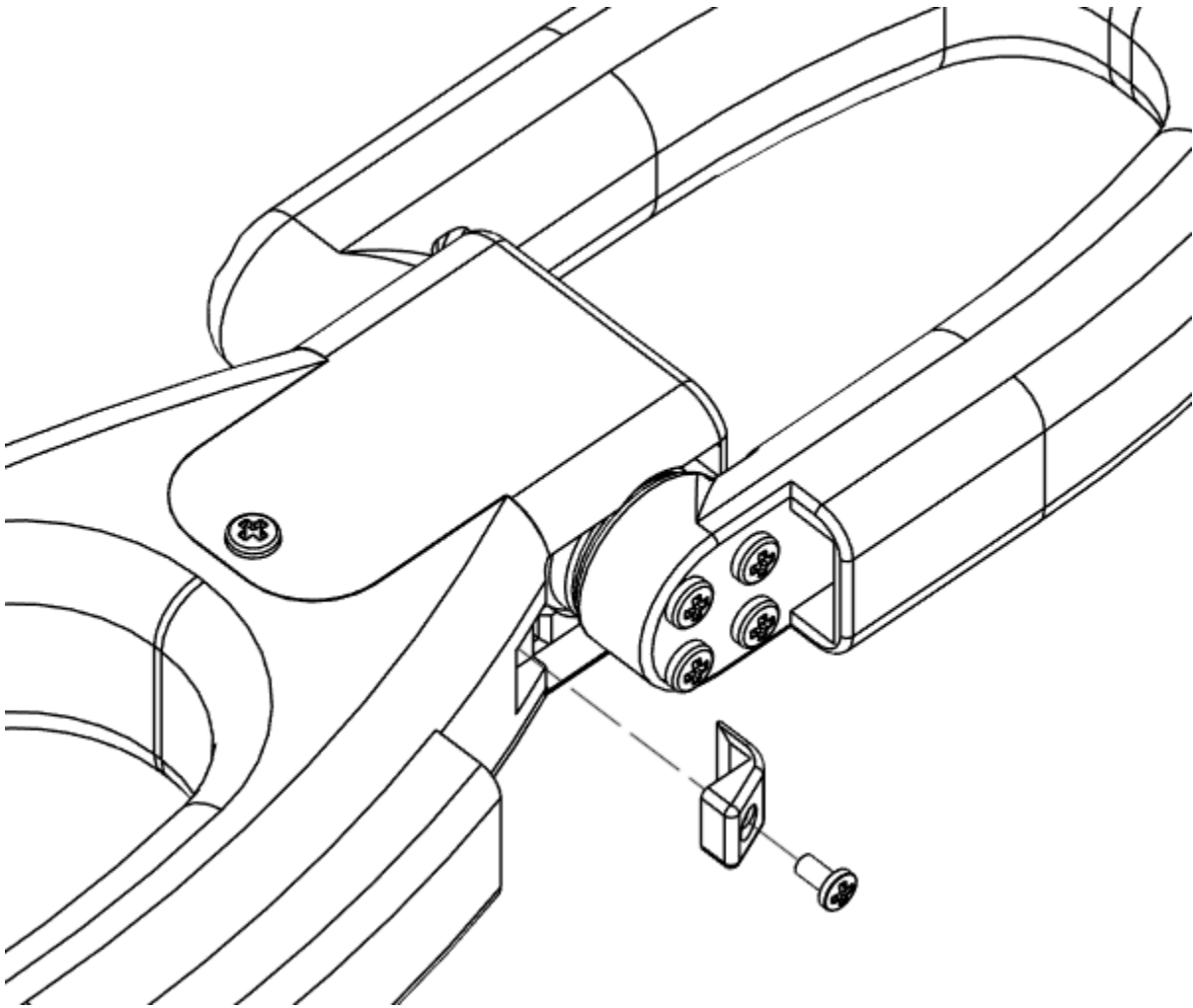


Figure 10.24 - Motor 3 Wire Clip Assembly

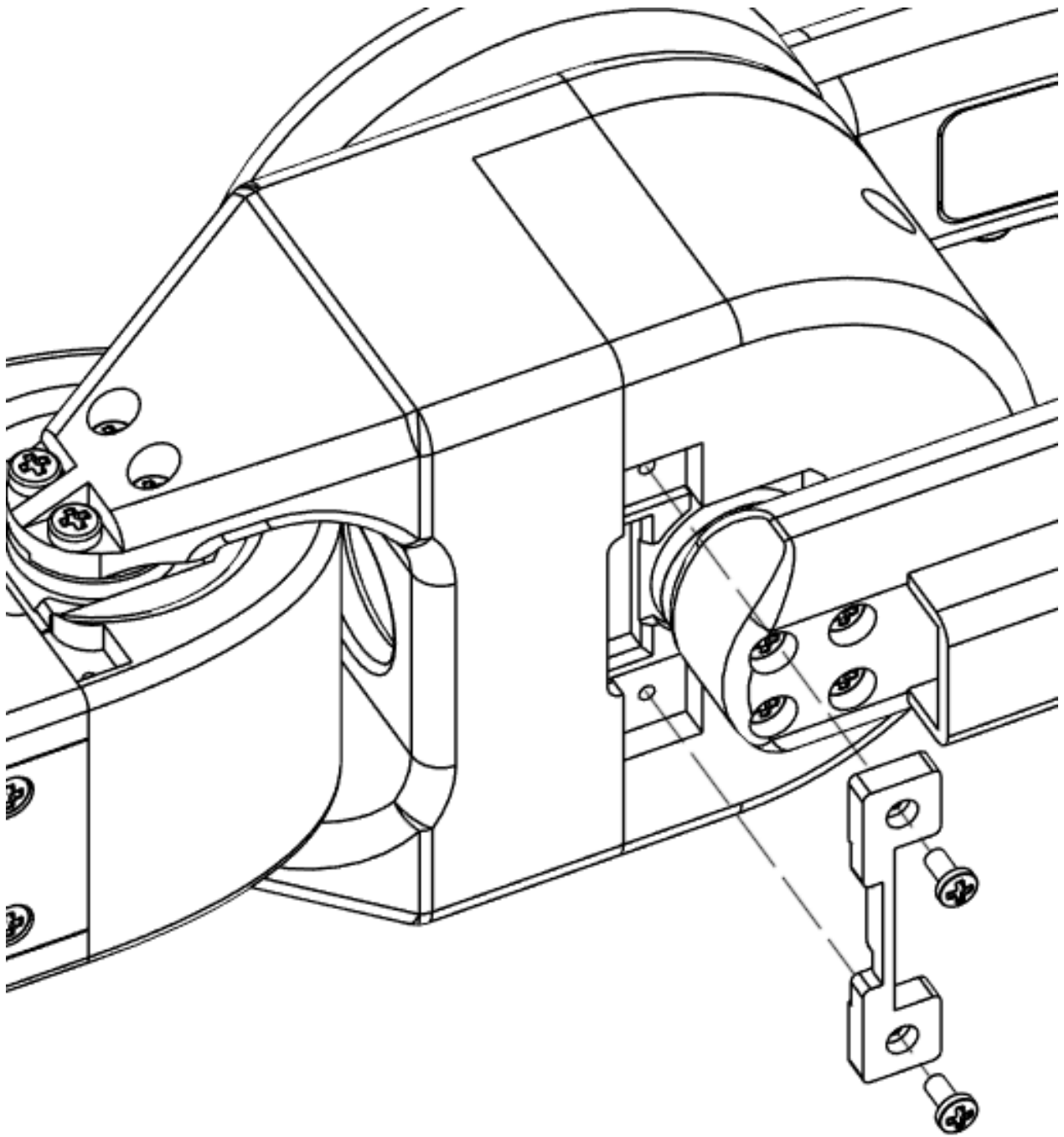


Figure 10.25 - Motor 2 Wire Clip Assembly

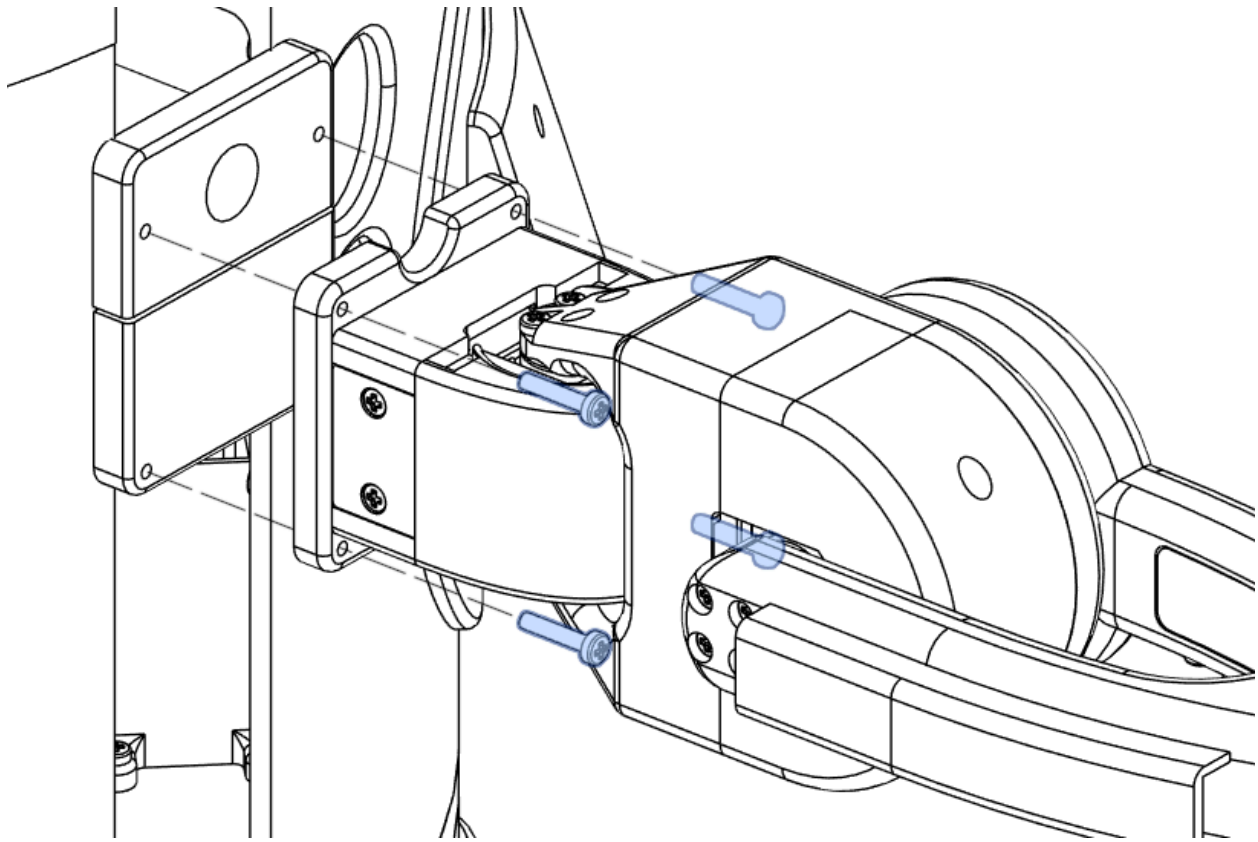


Figure 10.26 - Arm & Body Assembly

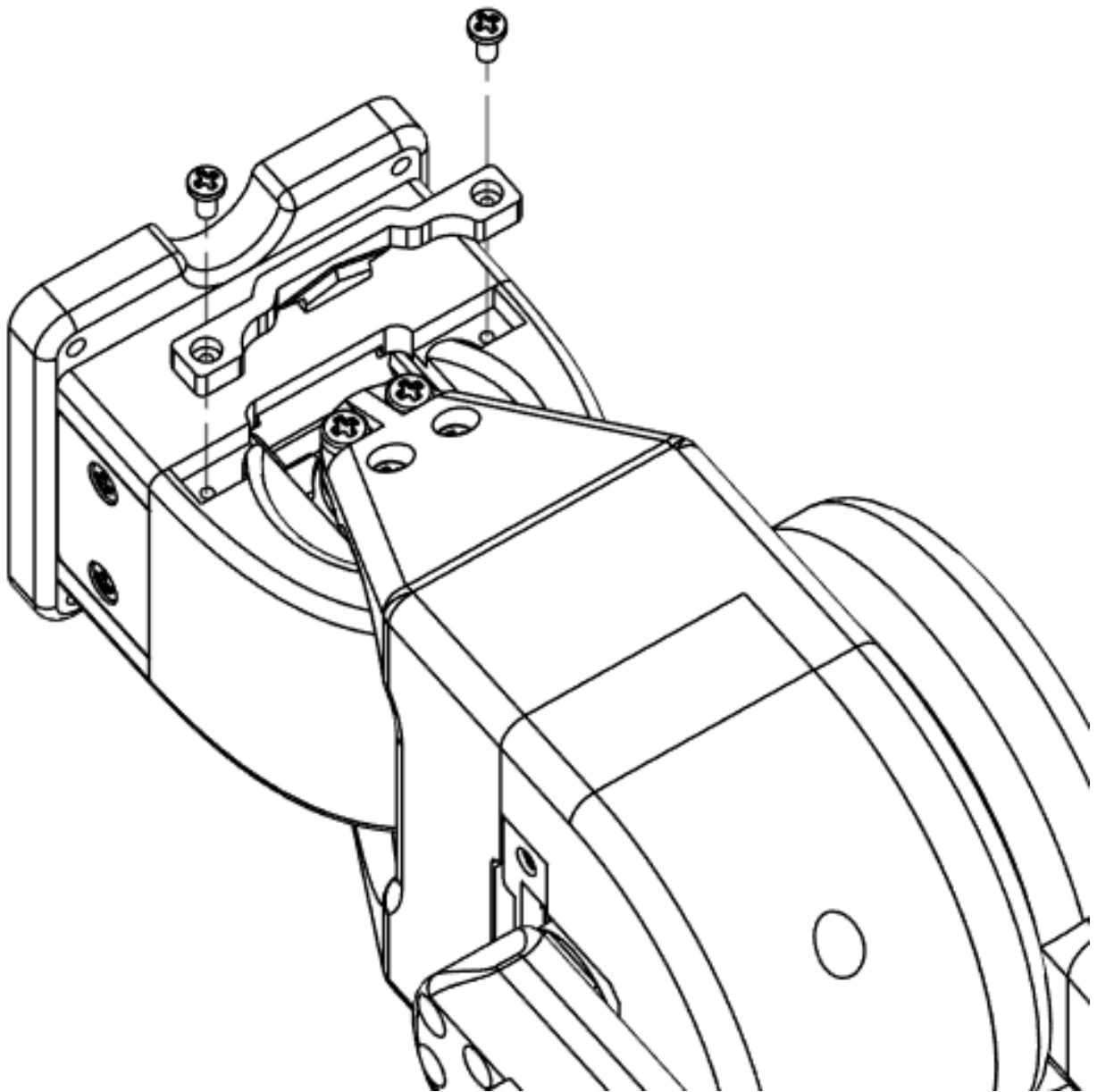


Figure 10.27 - Motor 1 Wire Clip Assembly

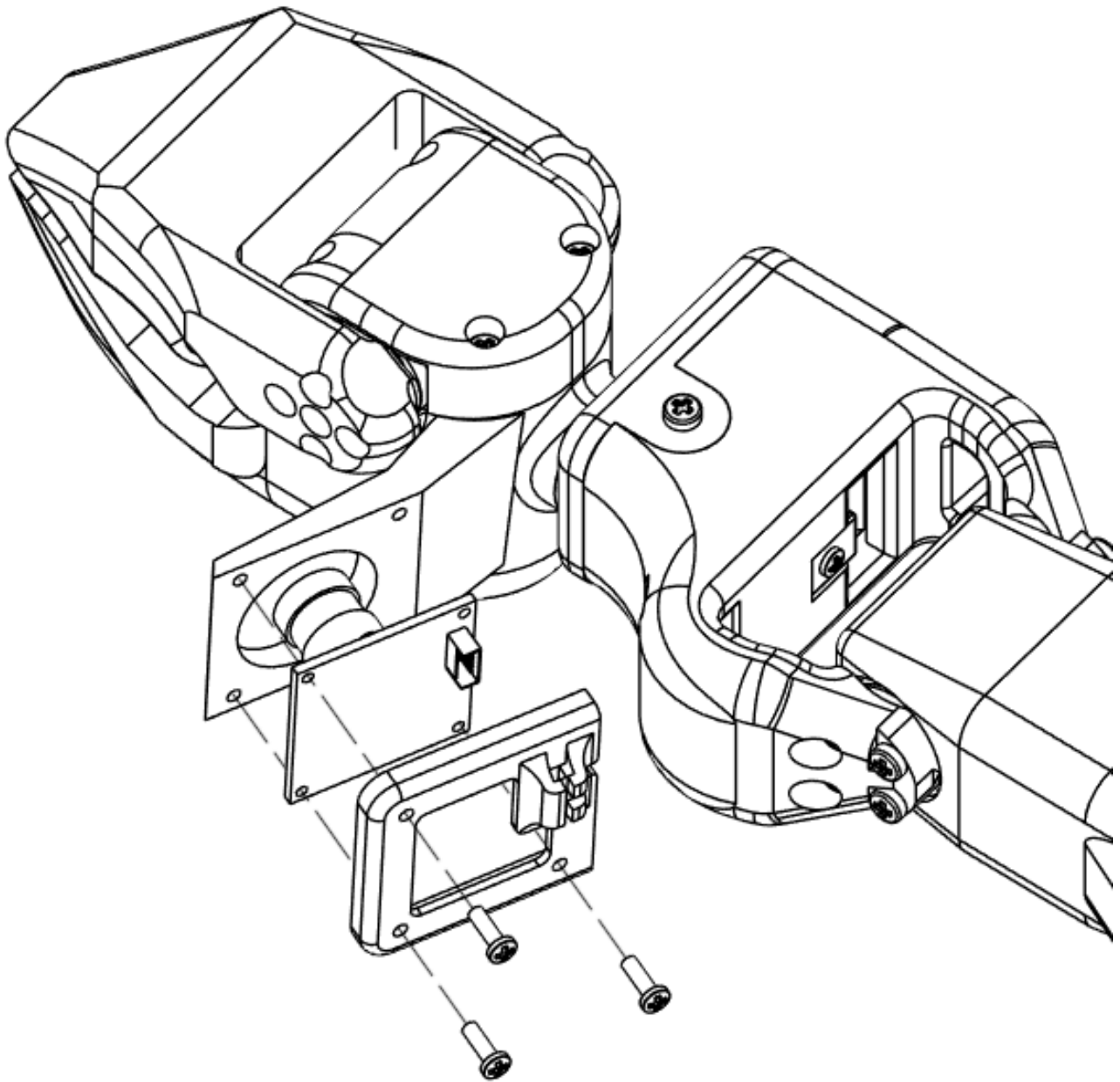


Figure 10.28 - Arm Camera & Cover Assembly

Stage 11: Dome Assembly

Dome Assembly

Materials

Type	Name	Count
Sensor	Camera	2
Electronics	LCD Screen	1
Part	Camera Holder	1
Part	Eye Camera Housing	1
Part	Dome Front	1
Part	Dome Left	1
Part	Dome Right	1
Part	Dome Rear	1
Part	Dome Top	1
Part	Magnet Cap	1
Part	Magnet	1
Part	Grommet Strip Cover	2
Part	LCD Screen Holder	1
Fastener	Grommet Strip Screw	4
Fastener	M2.5x8mm	8
Fastener	M3x8mm	14
Fastener	M3x10mm	14
Fastener	M3x12mm	8
Fastener	M3x20mm	2
Fastener	M3x40mm	2

Procedure

1. Fasten (2) Cameras to Camera Holder on the side marked "Cameras" as shown in Figure 11.1
 - a. (8) M2.5x8mm

2. Fasten Camera Holder Assembly to Eye Camera Housing as shown in Figure 11.2
 - a. (2) M3x8mm
3. Fasten Eye Camera Housing Assembly to Dome Front as shown in Figure 11.3
 - a. (2) M3x20mm
4. Fasten Dome Front, Dome Left, Dome Right, and Dome Rear to Dome Top as shown in Figure 11.4
 - a. (12) M3x10mm
5. Fasten Magnet Cap and Magnet to Dome Top as shown in Figure 11.4
 - a. (4) M3x12mm
6. Fasten Dome Assembly to Dome Level Base Plates as shown in Figure 11.5
 - a. (12) M3x8mm
7. Fasten Grommet Strip Cover to the top of Grommet Strip as shown in Figure 11.6
 - a. (2) Grommet Strip Screws
 - b. Repeat for other Grommet Strip
8. Fasten LCD Screen to LCD Screen Holder as shown in Figure 11.7
 - a. (4) M3x12mm
9. Fasten LCD Screen Holder Assembly to Dome Rear as shown in Figure 11.8
 - a. (2) M3x10mm
 - b. (2) M3x40mm

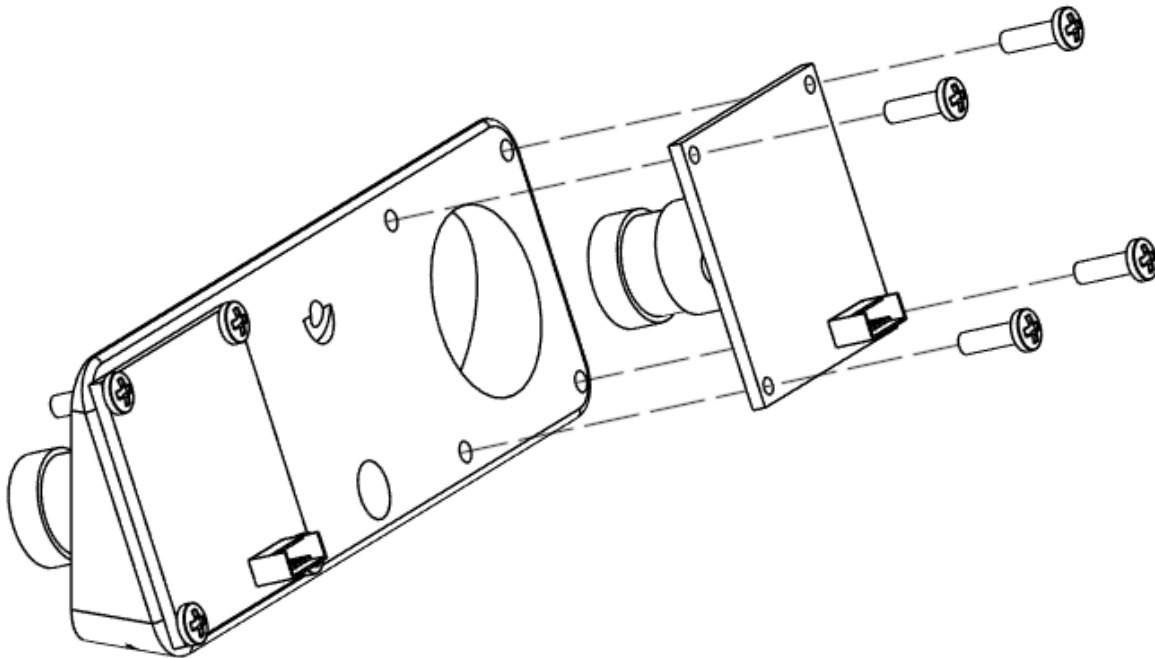


Figure 11.1 - Eye Camera Harness Assembly

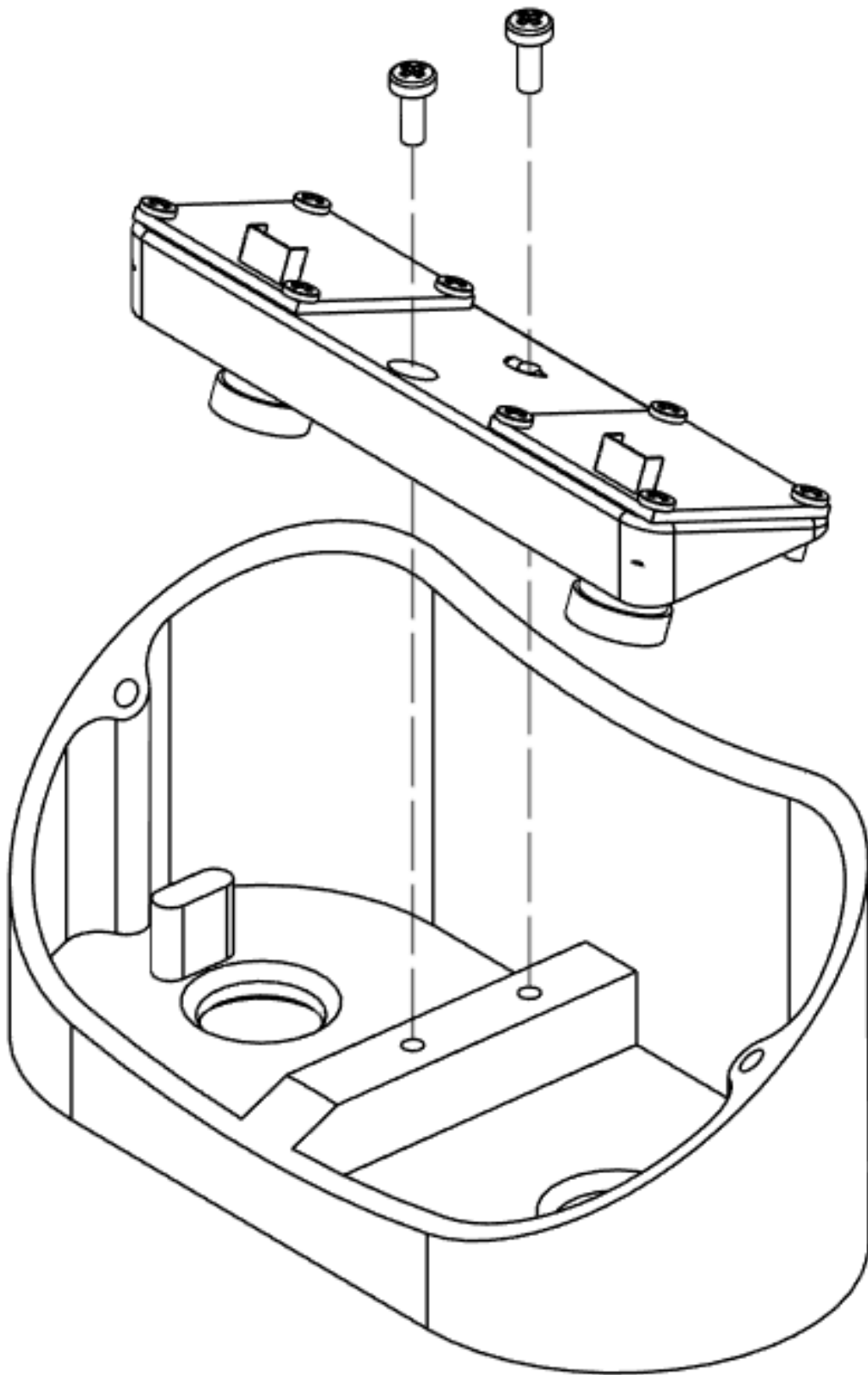


Figure 11.2 - Eye Camera Housing Assembly

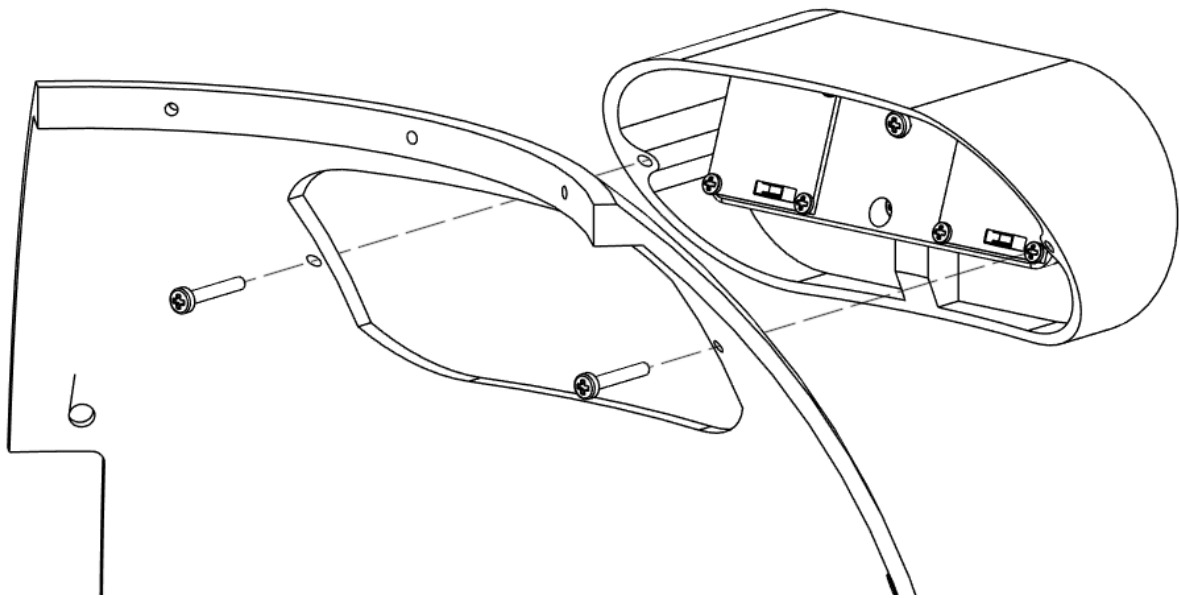


Figure 11.3 - Dome Front Assembly

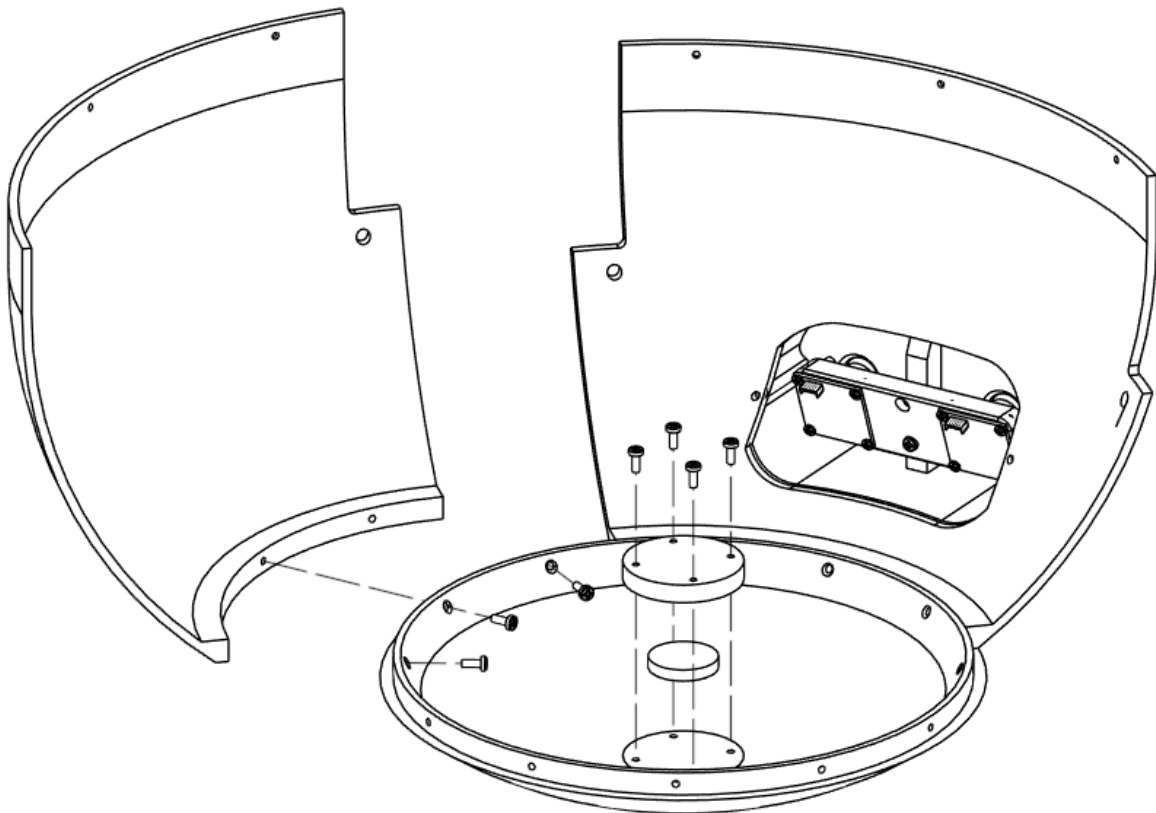


Figure 11.4 - Dome Assembly

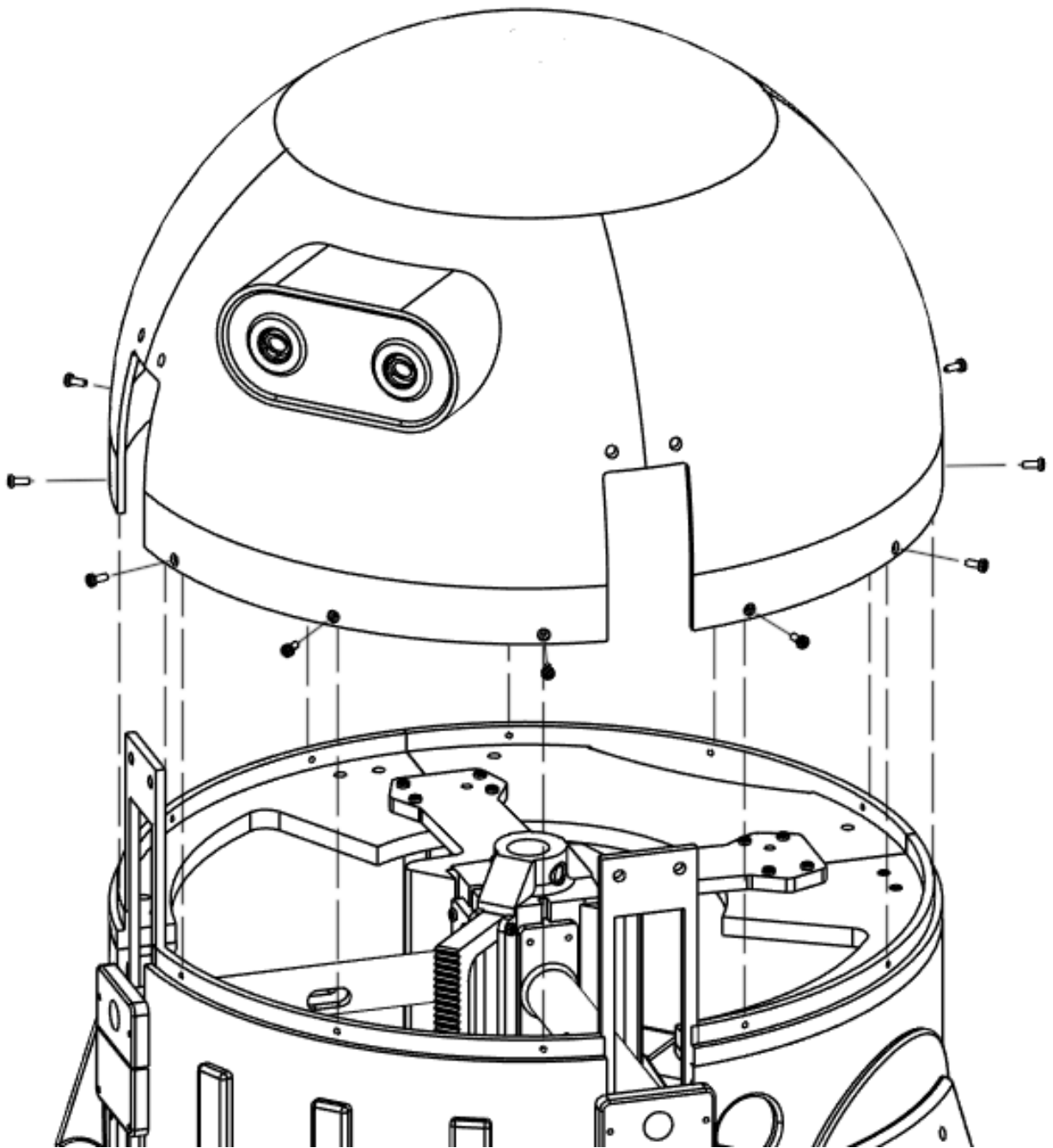


Figure 11.5 - Dome and Robot Assembly

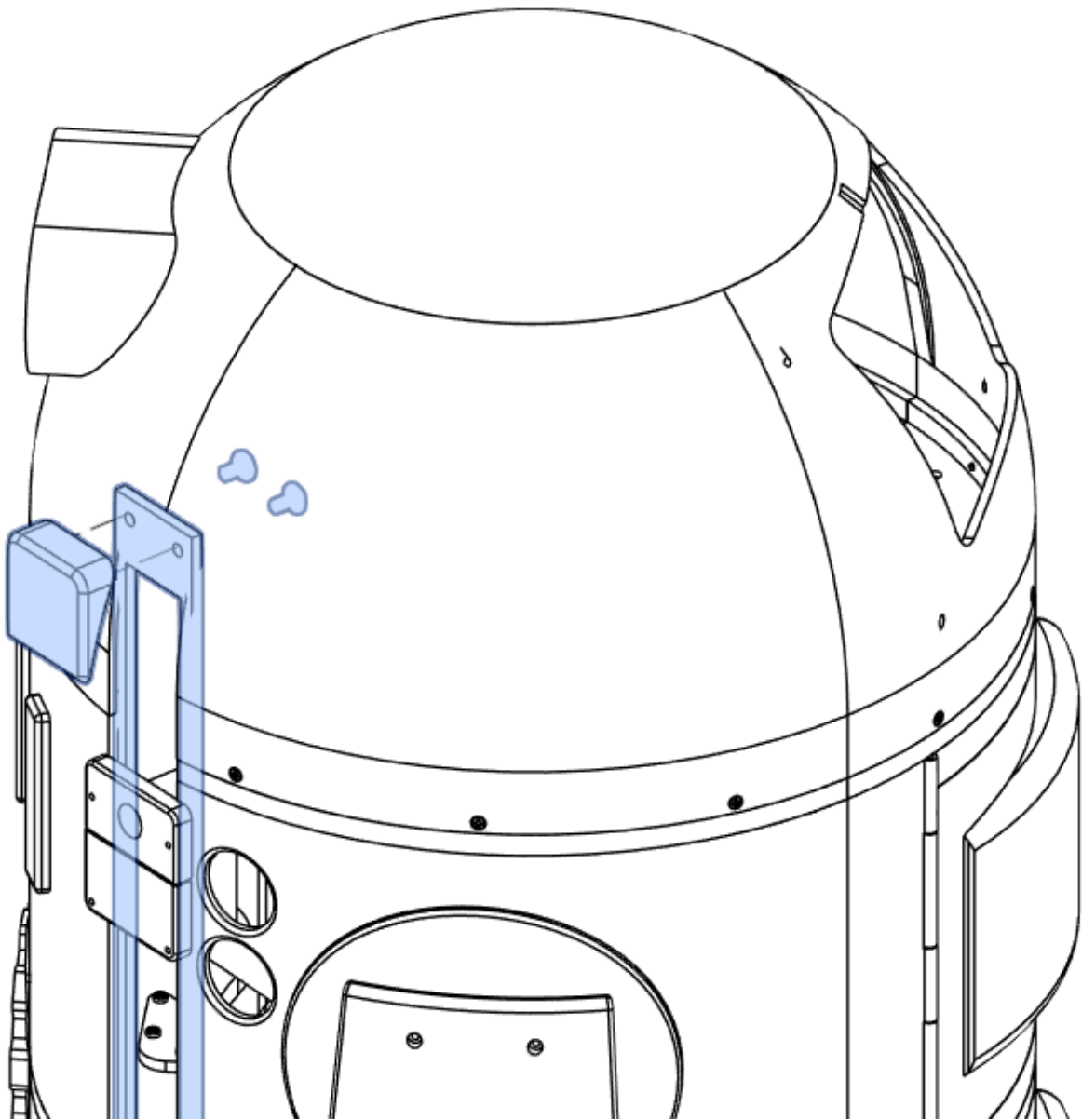


Figure 11.6 - Grommet Strip Cover Assembly

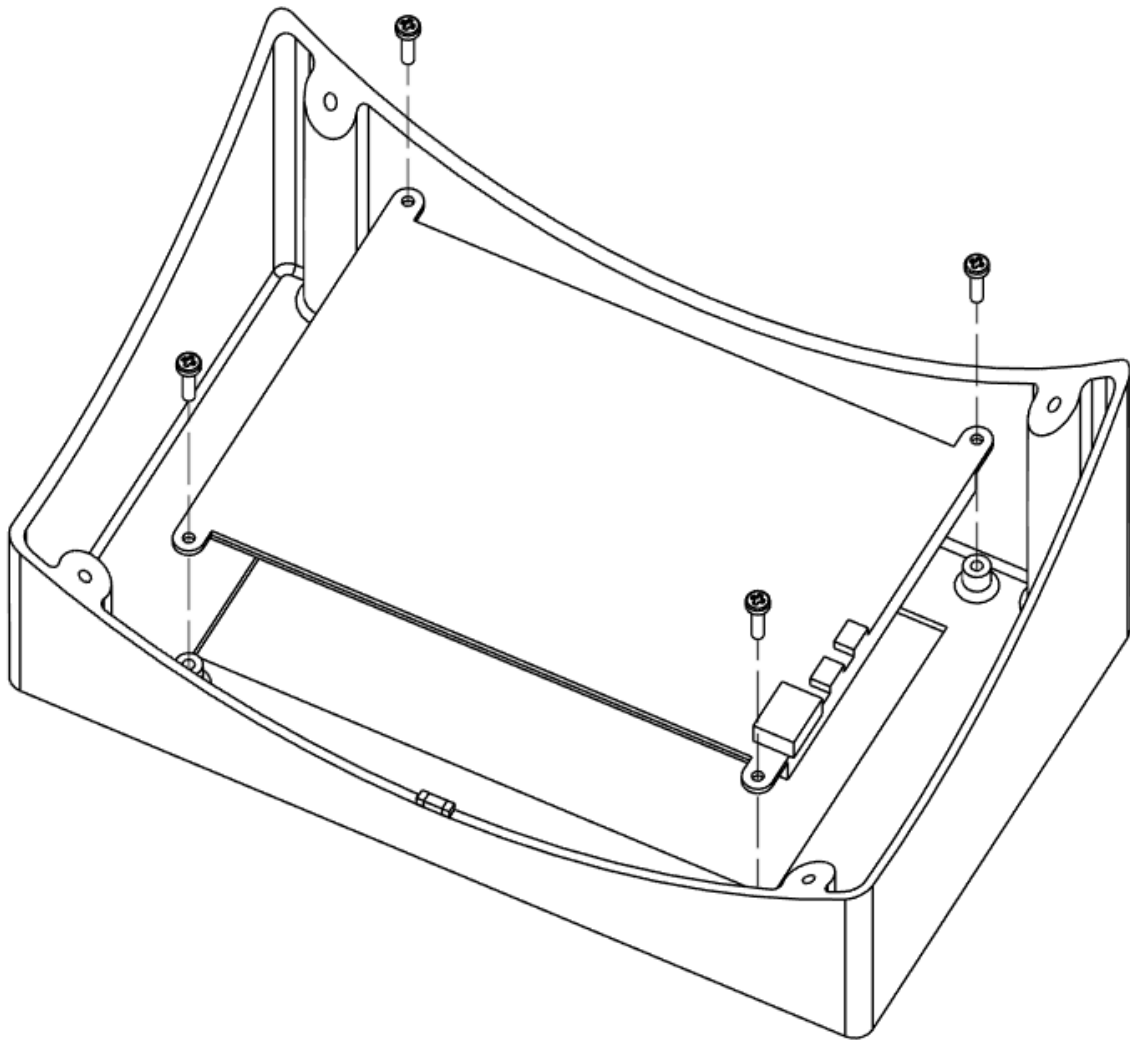


Figure 11.7 - LCD Screen & LCD Screen Holder Assembly

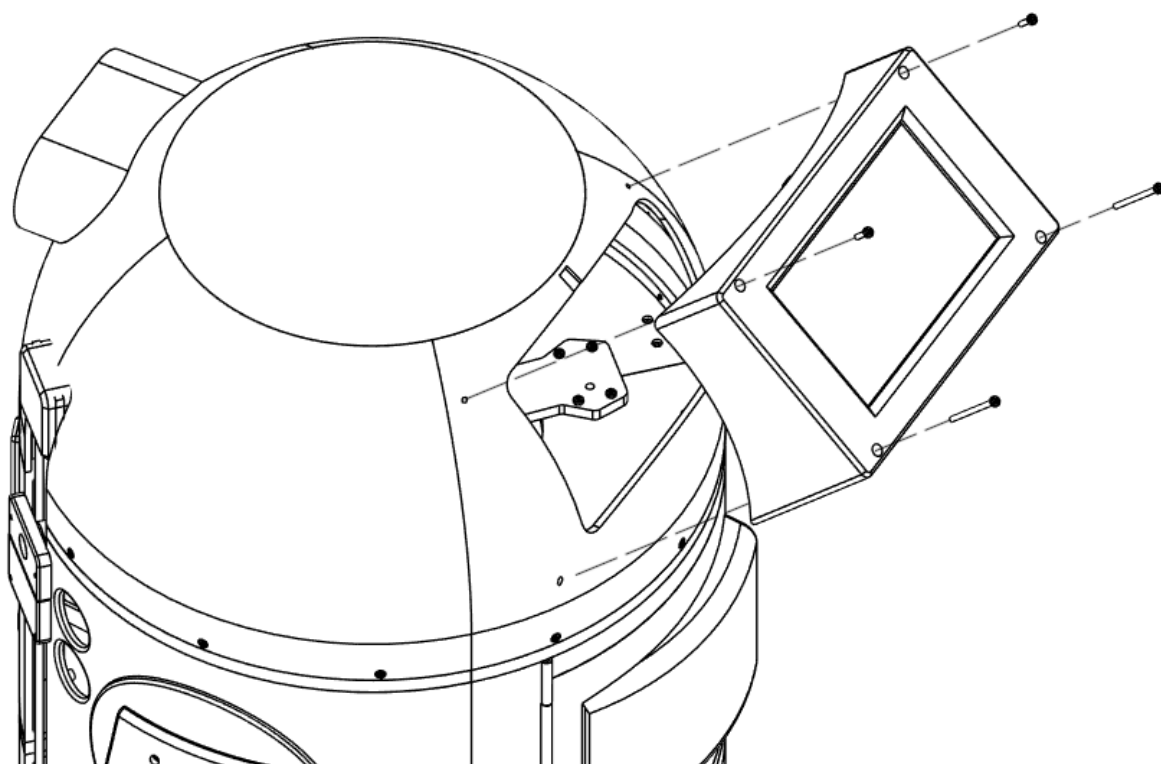


Figure 11.8 - LCD Screen & Dome Rear Assembly